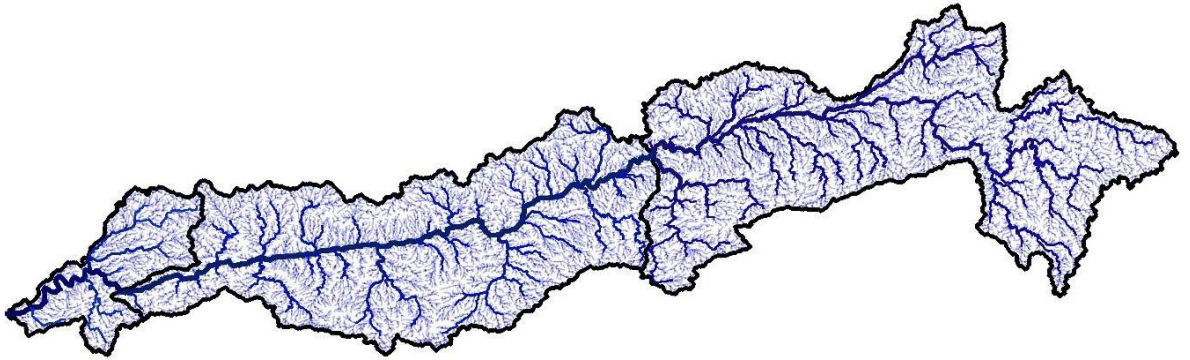




National River Conservation Directorate
Ministry of Jal Shakti,
Department of Water Resources,
River Development & Ganga Rejuvenation
Government of India

Biological Profile Report on Narmada River Basin



December 2025



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National River Conservation Directorate (NRCD)

The National River Conservation Directorate, functioning under the Department of Water Resources, River Development & Ganga Rejuvenation, and Ministry of Jal Shakti providing financial assistance to the State Government for conservation of rivers under the Centrally Sponsored Schemes of 'National River Conservation Plan (NRCP)'. National River Conservation Plan to the State Governments/ local bodies to set up infrastructure for pollution abatement of rivers in identified polluted river stretches based on proposals received from the State Governments/ local bodies.

www.nrcd.nic.in

Centres for Narmada River Basin Management and Studies (cNarmada)

The Center for Narmada River Basin Management and Studies (cNarmada) is a Brain Trust dedicated to River Science and River Basin Management. Established in 2024 by IIT Gandhinagar and IIT Indore, under the supervision of cGanga at IIT Kanpur, the center serves as a knowledge wing of the National River Conservation Directorate (NRCD). cNarmada is committed to restoring and conserving the Narmada River and its resources through the collation of information and knowledge, research and development, planning, monitoring, education, advocacy, and stakeholder engagement.

www.cnarmada.org

Centres for Ganga River Basin Management and Studies (cGanga)

cGanga is a think tank formed under the aegis of NMCG, and one of its stated objectives is to make India a world leader in river and water science. The Centre is headquartered at IIT Kanpur and has representation from most leading science and technological institutes of the country. cGanga's mandate is to serve as think-tank in implementation and dynamic evolution of Ganga River Basin Management Plan (GRBMP) prepared by the Consortium of 7 IITs. In addition to this, it is also responsible for introducing new technologies, innovations, and solutions into India.

www.cganga.org

Acknowledgment

This report is a comprehensive outcome of the project jointly executed by IIT Gandhinagar (Lead Institute) and IIT Indore (Fellow Institute) under the supervision of cGanga at IIT Kanpur. It was submitted to the National River Conservation Directorate (NRCD) in 2024. We gratefully acknowledge the individuals who provided information and photographs for this report.

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PREFACE

The Narmada River, often referred to as the lifeline of central and western India, holds unparalleled significance for the region's water resources, ecosystems, and communities. Its hydrological complexity, combined with its ecological and cultural heritage, underscores the critical need for its preservation and sustainable management.

This report, focusing on biological diversity studies of the Narmada River Basin, aims to provide an in-depth analysis of the flora and fauna of the Narmada River. The collection and compilation of data on the rich biological diversity of Upper, Middle and Lower basins will provide valuable insights into the ecological status of the Narmada Basin.

The findings and recommendations presented in this report are intended to serve as a guiding framework for policymakers, researchers, and practitioners committed to sustainable water management in the Narmada Basin. By identifying trends and challenges, it aspires to support the development of policies that balance ecological preservation with socioeconomic progress.

We are deeply grateful to all individuals, organizations, and institutions that contributed to the preparation of this report. Their dedication, expertise, and support have been instrumental in shaping this study. It is our hope that this report will inspire collaborative efforts and meaningful action toward the long-term health and sustainability of the Narmada River and its Basin.

Centre for Narmada River Basin
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ABBREVIATIONS AND ACRONYMS

cGanga	Centres for Ganga River Basin Management and Studies
cNarmada	Centres for Narmada River Basin Management and Studies
CWC	Central Water Commission
NRCD	National River Conservation Directorate
NRCP	National River Conservation Plan
NMCG	National Mission for Clean Ganga
GRBMP	Ganga River Basin Management Plan
IIT	Indian Institute of Technology
IMC	Indian Major Carps
EN	Endangered
VU	Vulnerable
NT	Near Threatened
CIFRI	Central Inland Fisheries Research Institute
WII	Wildlife Institute of India
ZSI	Zoological Survey of India
UZ	Upper Zone
MZ	Middle Zone
WB	Water Birds
WA	Water Associated Birds
WD	Water-dependent Birds

Chapter 1: Introduction

The Narmada River Basin extends from 21°40'12" to 23°41'24" N latitudes and 72°48'36" to 81°45'36" E longitudes, encompassing a total area of 97,560.80 sq. km, which constitutes about 3% of India's total geographical area (3,297,427.32 sq. km). Bounded by the Vindhyan ranges to the north, the Maikala hills to the east, the Satpura ranges to the south, and the Arabian Sea to the west, the basin lies at the northern extremity of the Deccan Plateau and covers significant portions of Madhya Pradesh and Gujarat, with smaller parts extending into Chhattisgarh and Maharashtra. The basin spans 39 districts, comprising 27 districts of Madhya Pradesh, six districts of Gujarat, four districts of Chhattisgarh, and two districts of Maharashtra. The Narmada Basin exhibits an elongated shape, stretching about 915.65 km east to west and about 236 km north to south. The basin experiences a tropical monsoon climate characterized by distinct wet and dry seasons, with an average annual rainfall range of 800 to 1200 mm. The Narmada River, the longest west-flowing river of India, drains into the Arabian Sea near Bharuch in Gujarat and is further subdivided into Upper, Middle, and Lower Narmada River Basins. (Figure 1)

In this report, we have collected and analyzed data on various categories of flora and fauna in the Narmada River and its tributaries. Flora data includes phytoplankton, periphyton, and information on riparian and floodplain vegetation. Data on fauna include zooplankton, zoobenthos, fish, higher aquatic vertebrates, and aquatic birds. Data have been collected from various government agencies, online research papers, and offline sources, including books.

The report includes compiled data for all the categories mentioned above, organized by year and sub-zone for the Upper, Middle, and Lower Basins, providing information on the biological status of the Narmada River based on the collected data. Biodiversity data may vary location to location, as some data may be missing or not collected.

The Narmada River is characterized by its geographical area, Population, Major Rivers, and Water resources. The details are provided in Table 1.

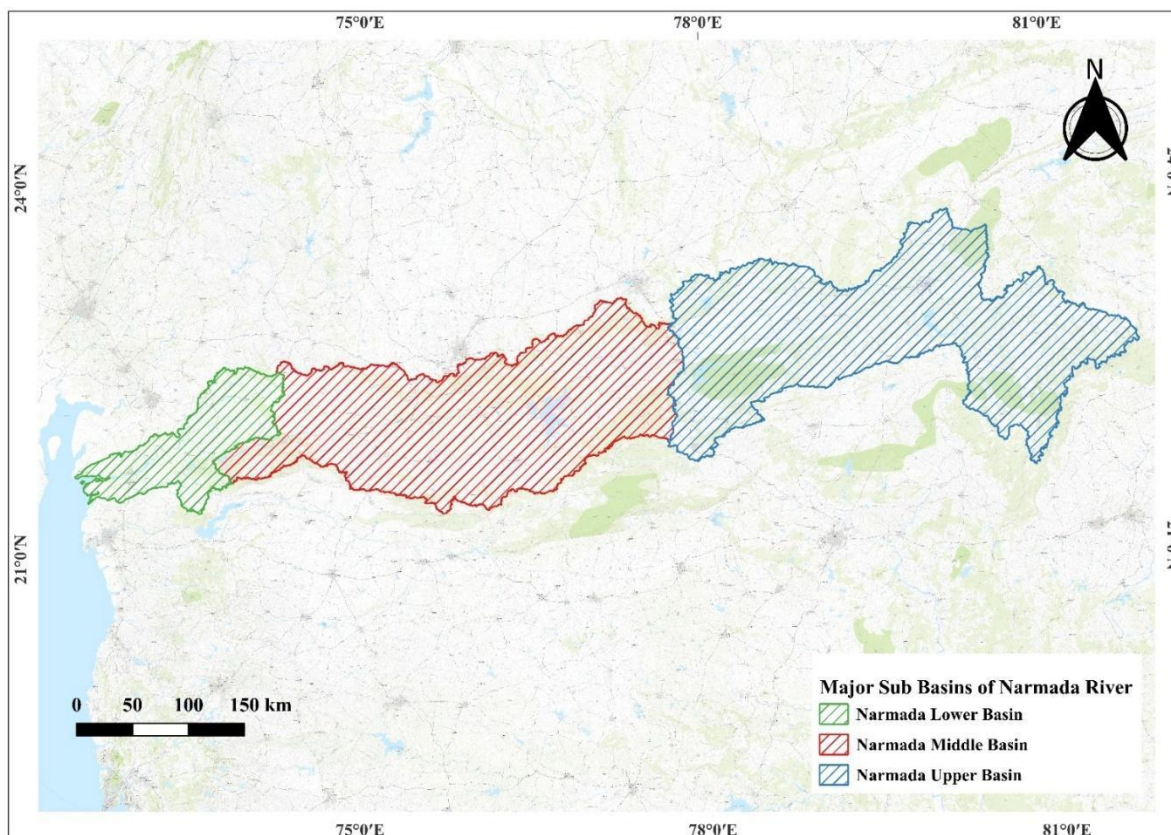


Figure 1: Sub-basin of the Narmada Basin

Table 1: Salient features of the Narmada River

S. No.	Particulars	Details
1	Basin Extent	Latitude: 21° 40' 12" to 23° 41' 24" N Longitude: 72° 48' 36" to 81° 45' 36" E
2	Total Geographical Area	97560.80 sq. km
3	States and their % area contribution in the basin:	Madhya Pradesh (87.82%), Gujarat (9.72%), Chhattisgarh (0.72%), Maharashtra (1.73%)
4	Spatial Variation of Elevation	<5 m: 2.42 %; 5-10 m: 0.15%; 10-50 m: 1.12%. 50-100m: 2.42%; 100-200 m: 11.68%; 200-300 m: 27.26%; 300-400 m: 20.30%; 400-500m: 12.86%; 500-750m: 18.40%; 750-1000m: 3.25%; 1000-1261 m: 0.14%

Biological Profile of Narmada Basin

5	Population	20,799,195 (Census, 2011)
6	Average Temperature Range (°C)	24.92 - 27.39
7	Mean Maximum Temperature (°C) (1970-2023)	32.47
8	Mean Minimum Temperature (°C) (1970-2023)	19.01
9	Mean Annual Rainfall (mm) (1970-2023)	1110.83
10	Highest Elevation (m)	1261
11	Major Rivers	Hiran, Tendon, Barna, Kolar, Man, Uri, Harni, Orsang, Burhner, Banjar, Sher, Shakkar, Dudhi, Tawa, Ganjal, Chhota Tawa
12	Districts	39
13	Number of Water Resources Structures	Dams-277; Barrages-2; Weirs-2; Lifts-4; Power Houses-9
14	Highest Dam	Sardar Sarovar dam-163 m
15	Longest Dam	Rani Avanti Bai Sagar -Bargi (NVDA) Dam-5.357 km
16	Highest Barrage	Kolar barrage-15.5 m
17	Longest Barrage	Kolar barrage-1.195 km
18	Number of Irrigation Projects	Major-21; Medium-23; ERM-1
19	Number of HE projects	6
20	Number of Groundwater Observation Wells	393

Biological Profile of Narmada Basin

21	Number of Hydro-Observation sites	29
22	Number of Flood Forecasting Sites	4
23	Water Tourism and Sport Sites	27

Source: Central Water Commission and National Remote Sensing Centre, Ministry of Jal Shakti, Government of India (2014a). Basin Report of the Narmada Basin.

Chapter 2: **Narmada River**

The Narmada River, once referred to as Narbada and anglicized as Nerbudda, ranks as the fifth-longest river in India and is the longest river flowing westward in the country. It is also the largest river in the state of Madhya Pradesh. The Narmada traverses the states of Madhya Pradesh and Gujarat and is often referred to as the "Lifeline of Madhya Pradesh and Gujarat" due to its substantial contributions to these states.

Originating from the Amarkantak Plateau in the Anuppur district of Madhya Pradesh, the Narmada forms the traditional boundary between North and South India. It flows westward for 1,312 km before emptying into the Arabian Sea through the Gulf of Khambhat, situated 30 km west of Bharuch city in Gujarat. The Narmada basin extends from 21° 40' 12" to 23° 41' 24" N latitudes and 72° 48' 36" to 81° 45' 36" E longitudes, covering a total area of 95,959.70 sq. km, which is nearly 3% of the total geographical area of India (3,297,427.32 sq. km)

2.1 TRIBUTARIES OF THE NARMADA RIVER BASIN

The important tributaries of the Narmada river are the Burhner, the Banjar, the Sher, the Shakkar, the Dudhi, the Tawa, the Ganjal, the Kundi, the Goi, and the Karjan, which join from left, whereas the Hiran, the Tendon, the Barna, the Kolar, the Man, the Uri, the Hatni, and the Orsang join from right. Narmada drains into the Arabian Sea through the Gulf of Khambhat. Details of the tributaries are provided in the flowchart Figure 2. Checklist of streams flowing through the comprising districts of Upper, Middle, and the Lower Narmada Basin is listed in the Annexures (2.1.1-2.1.3)

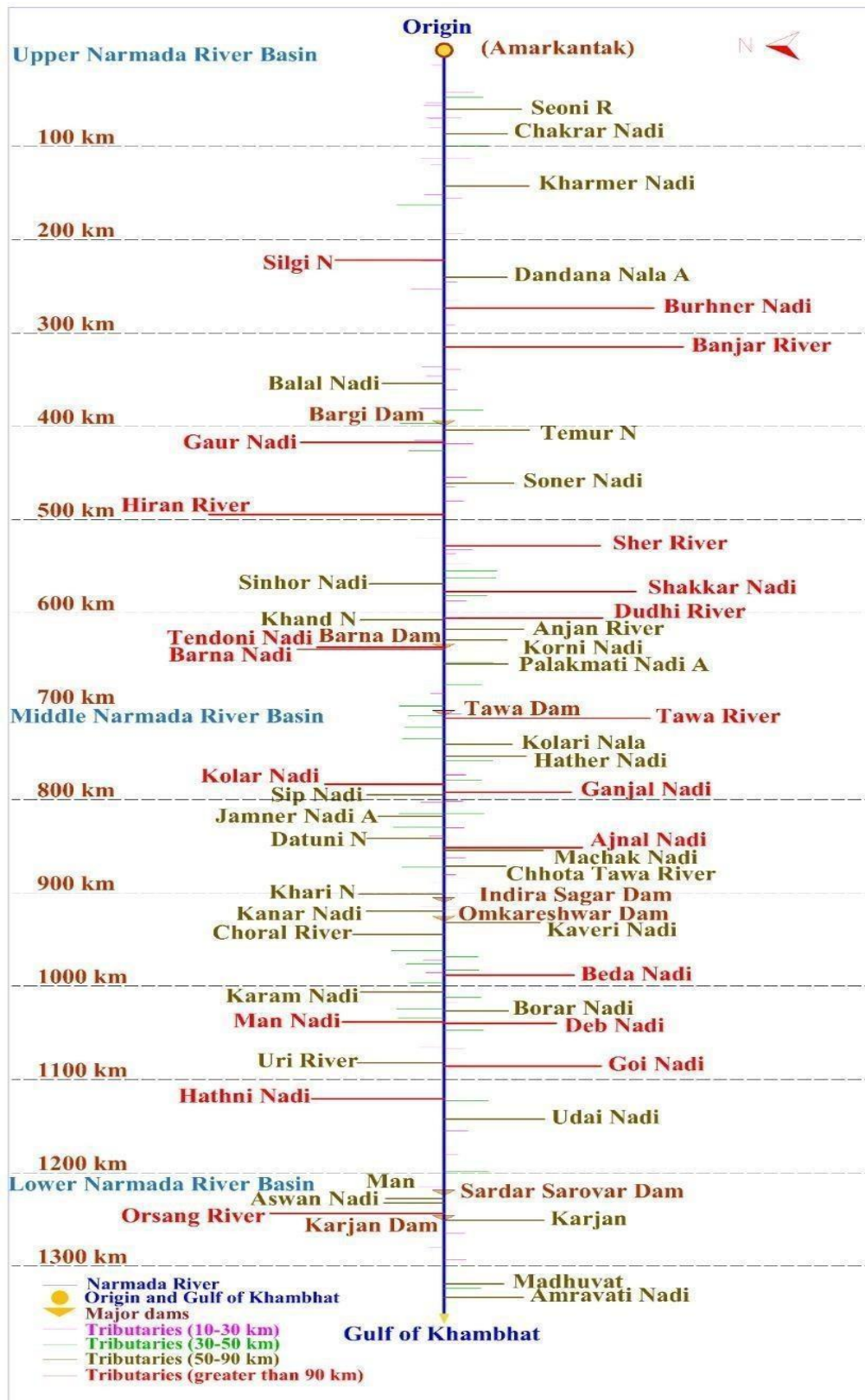


Figure 2: Flow Diagram: Narmada River and its Tributaries

Source: River Atlas, 2025 (cNarmada) IIT Indore.

2.2 RIVER ZONING

As per the WII report, the Narmada River is classified into three major Zones

- a) Upper Zone (UZ): Amarkantak and Bheraghat
- b) Middle Zone (MZ): Bheraghat and Chandod
- c) Lower Zone (LZ): Chandod in the Vadodara district to the Gulf of Khambhat

2.3 BIOLOGICAL DIVERSITY OF THE NARMADA RIVER

The time period for assessing the Biological Status of the Narmada River and its potentially perennial tributaries is divided into three zones: Reference Status (1960-1980), Intermediate Status (1981-2000), and Current Status (2001-present). In this report, we have compiled the data on the flora and fauna of the Narmada River and its tributaries. Flora includes Phytoplankton, Periphyton, Macrophytes, Riparian, and Floodplain Vegetation, and Fauna includes Zooplankton, Macro-benthos, Fishes, Higher Aquatic Vertebrates, and Aquatic Aves as shown in Figure 3.

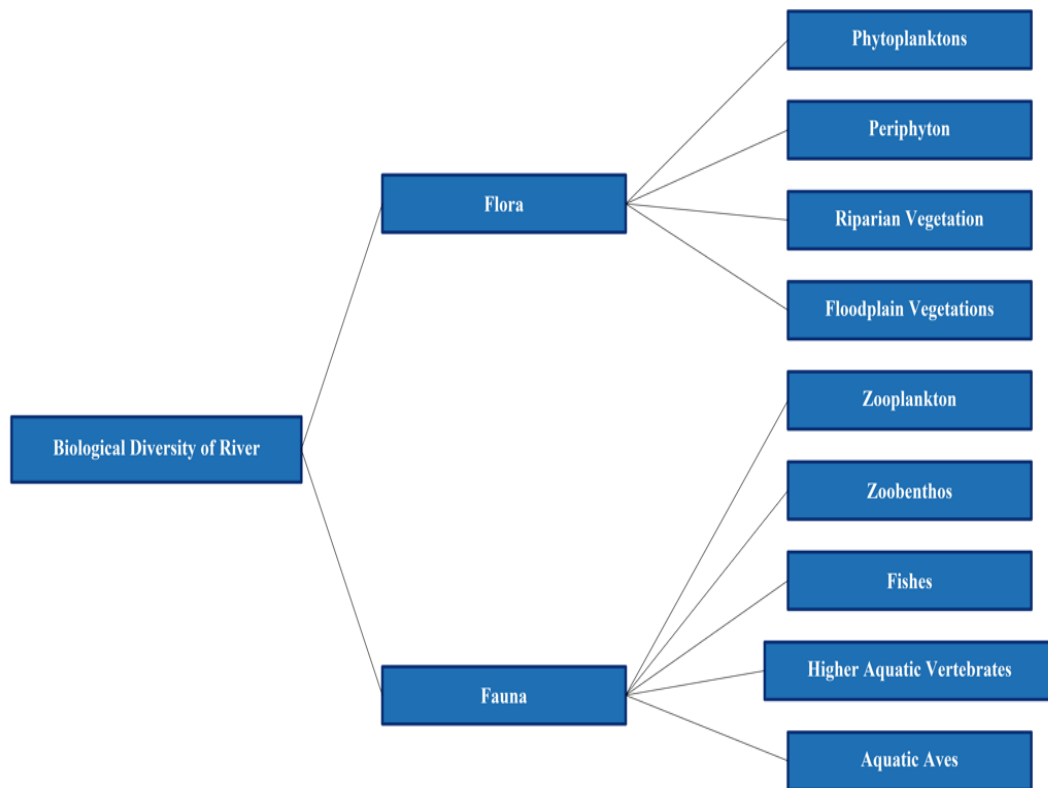


Figure 3: Tree diagram of the Biological Diversity of the Narmada Basin.

Chapter 3: Flora

3.1 PHYTOPLANKTON

According to the data collected for the Upper Narmada Basin and the Middle Narmada Basin, a total of four classes of phytoplankton (organisms that live in the water, unattached to surfaces) are present in the current time zone, totaling 60 species in the Upper Narmada Basin and 33 species in the Middle Narmada Basin. Some of the species of Phytoplankton that are common in both the Narmada Basin are *Fragilaria*, *Synedra ulna*, *Protococcus Hormidium*, *Anabaena*, and *Phormidium*.

According to Figure 4, the number of species in all classes of phytoplankton is greater than in the Middle Basin.

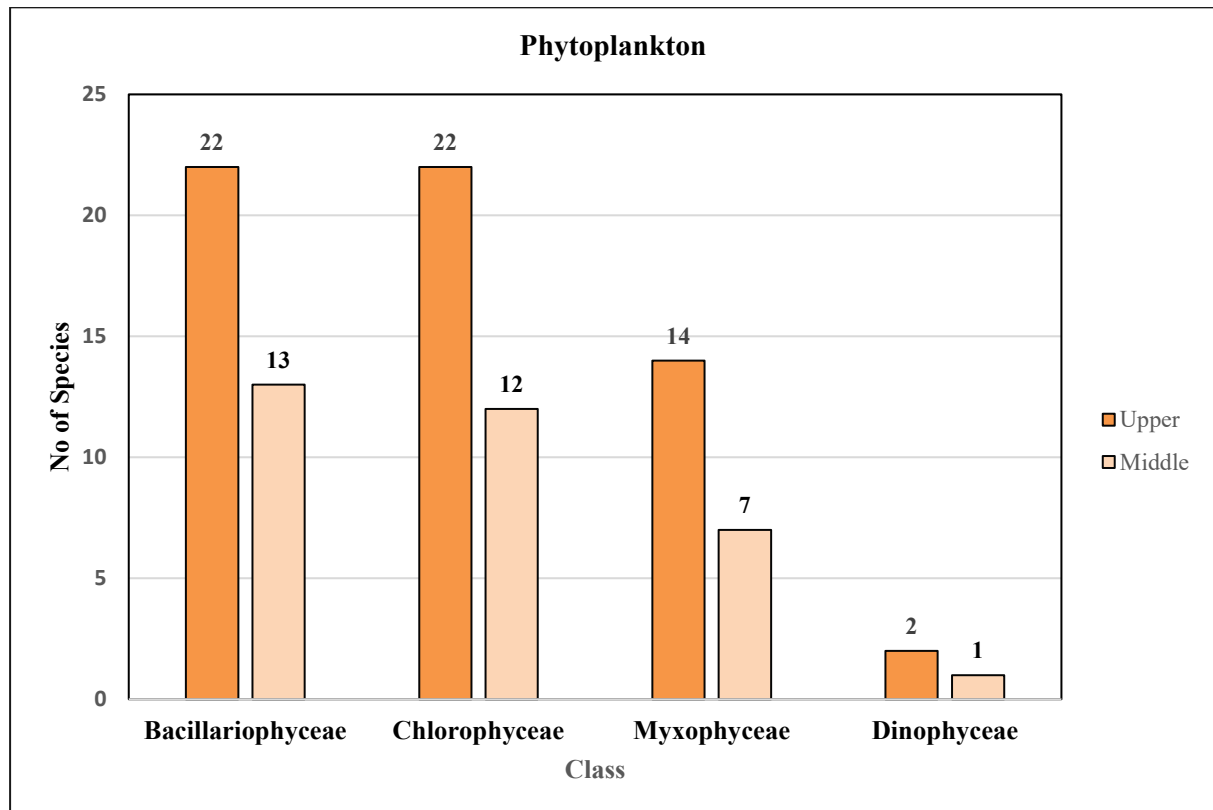


Figure 4: Different classes and species of Phytoplankton in the Upper and Middle Narmada Basin

Most of the species are in the Bargi Reservoir, situated in Jabalpur, the Barna Reservoir, located in the Raisen district, Sarni Reservoir on Tawa, the main tributary of Narmada near Tawa dam, and Sampna Reservoir on Manchna river to join river Tawa, the tributary of river Narmada in Betul district of the Narmada Basin, Madhya Pradesh.

Common species in the Upper Narmada Basin from the intermediate and present time zones are *Gyrosigma*, *Navicula*, *Surirella*, *Fragilaria*, *Gomphonema*, *Tabellaria*, and *Synedra ulna*. In the Middle Narmada Basin, intermediate and current zones, central locations where the species are found include Kolar Reservoir in the district of Sehore and the Narmada River stretch from Gadarwada to Harsud. Some species, like *Cymbella*, *Gyrosigma*, *Synedra Ulna*, *Gonatozygon*, *Lobocystis dichotoma*, *Pediastrum Duplex*, *Hormidium*, *Coelosphaerium*, are common in both the intermediate and current time zones. This data is combined for the Narmada River and its tributaries.

Details of the classes and species names are listed in Tables 2 and 3 for the Upper and Middle Basins.

Table 2: Details of the Phytoplankton found in the Upper Narmada Basin

Class			
Bacillariophyceae	Chlorophyceae	Myxophyceae	Dinophyceae
Species			
<i>Fragilaria</i>	<i>Pediastrum</i>	<i>Surirella</i>	<i>Ceratium hirundinella</i>
<i>Synedra ulna</i>	<i>Protococcus</i>	<i>Gonatozygon</i>	<i>Gonyaulax apiculata</i>
<i>Cymbella</i>	<i>Ulothrix</i>	<i>Pediastrum simplex</i>	
<i>Diatoma</i>	<i>Spirogyra</i>	<i>Amphithrix janthina</i>	
<i>Gomphonema</i>	<i>Surirella</i>	<i>Anabaena</i>	
<i>Navicula radiosa</i>	<i>Gonatozygon</i>	<i>Anacystis cyanea</i>	
<i>Melosira</i>	<i>Pediastrum simplex</i>	<i>Coelosphaerium</i>	
<i>Asterionella</i>	<i>Cerasterias irregulare</i>	<i>Oscillatoria tenuis</i>	
<i>Tabellaria</i>	<i>Closterium</i>	<i>Phormidium</i>	
<i>Surirella</i>	<i>Eudorina</i>	<i>Stichosiphon</i>	
<i>Meridion</i>	<i>Genicularia</i>	<i>Anacystis incerta</i>	
<i>Gyrosigma</i>	<i>Hormidium subtile</i>	<i>Oscillatoria</i>	
<i>Pinnularia</i>	<i>Pediastrum duplex</i>	<i>Lyngbya</i>	
<i>Navicula</i>	<i>Ankistrodesmus</i>	<i>Spirulina</i>	
<i>Synedra</i>	<i>Hormidium</i>		
<i>Mastogloia danseii</i>	<i>Palmella</i>		
<i>Nitzschia</i>	<i>Phacus suecica</i>		
<i>Rhopalodia gibba</i>	<i>Volvox aureus</i>		
<i>Amphora</i>	<i>Characium</i>		
<i>Bacillaria paradoxa</i>	<i>Coccomyxa dispar</i>		
<i>Epithemia turgida</i>	<i>Euglena</i>		

<i>Rhopalodia</i>	<i>Phacus</i>		
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Source: Data extracted from references 1 and 2 (CIFRI)

Table 3: Details of the Phytoplankton found in the Middle Narmada Basin.

Class			
Chlorophyceae	Bacillariophyceae	Myxophyceae	Dinophyceae
Species			
<i>Spirogyra</i>	<i>Fragilaria</i>	<i>Spirulina</i>	<i>Staurostrum leunocladium</i>
<i>Protococcus</i>	<i>Navicula</i>	<i>Anabaena</i>	
<i>Hormidium</i>	<i>Synedra</i>	<i>Oscillatoria</i>	
<i>Volvox</i>	<i>Surirella</i>	<i>Nostoc</i>	
<i>Pediastrum</i>	<i>Tabellaria</i>	<i>Phormidium</i>	
<i>Microspora</i>	<i>Nitzschia</i>	<i>Coelosphaerium</i>	
<i>Mougetia</i>	<i>Gyrosigma</i>	<i>Merismopedia</i>	
<i>Botryococcus</i>	<i>Rhopalodia</i>		
<i>Characium</i>	<i>Cymbella</i>		
<i>Gonatozygon</i>	<i>Asterionella</i>		
<i>Lobocystis dichotoma</i>	<i>Diatoma</i>		
<i>Pediastrum duplex</i>	<i>Synedra ulna</i>		
<i>Pediastrum simplex</i>			

Source: Data extracted from references 1 and 2 (CIFRI)

The stretch of the River Narmada below Surpan up to the Gulf of Cambay, including the estuarine part of the river, showed a high dominance of phytoplankton in the plankton community. The average total plankton population in this stretch varied from 63 to 1161 nos. l⁻¹. Among this total population, phytoplankton contributed a very large share, ranging from 70.42% to 98.70% (given in below **Table 4**), indicating that phytoplankton was the main biological component throughout this stretch.

The highest phytoplankton abundance was recorded at Jhanor, while the lowest abundance was observed at Ambetha. At all study sites, phytoplankton dominance followed a pattern similar to the upstream stretches of the river.

Qualitative composition of phytoplankton

The phytoplankton assemblage in this stretch was composed mainly of three groups:

- **Bacillariophyceae (diatoms)**

- **Myxophyceae (blue-green algae)**
- **Chlorophyceae (green algae)**

Bacillariophyceae (Diatoms)

Diatoms were the major and dominant phytoplankton group at most sites in this stretch. Their contribution ranged from 34.92% to 79.01%. Diatoms dominated the phytoplankton community at all sites except Sisodara and Jhanor. The high presence of diatoms indicates their strong adaptability to both riverine and estuarine conditions.

The important diatom taxa recorded in this stretch were:

- *Coscinodiscus grani*
- *Coscinodiscus subtilis*
- *Chaetoceros decipiens*
- *Biddulphia sinensis*
- *Amphora ovalis*
- *Rhizosolenia* sp.
- *Surirella elegans*
- *Anomoeoneis sphaerophora*
- *Synedra ulna*
- *Nitzschia* sp.
- *Gyrosigma* sp.
- *Asterionella* sp.
- *Pleurosigma angulatum*

These taxa together formed the main diatom assemblage in this stretch.

Myxophyceae (Blue-green algae)

Myxophyceae showed site-specific dominance. At Sisodara, blue-green algae were the dominant phytoplankton group, contributing 54.67% of the total phytoplankton population. At other sites, Myxophyceae occurred in lower proportions and did not dominate the assemblage.

Chlorophyceae (Green algae)

Chlorophyceae played a secondary role at most sites, except at Jhanor, where it became the dominant phytoplankton group, contributing 88.20%. This strong dominance of green algae at Jhanor resulted in the highest recorded phytoplankton abundance in this stretch.

The green algal assemblage was mainly represented by the following taxa:

- *Pediatrum simplex*
- *Spirogyra* sp.
- *Dictyosphaerium* sp.
- *Westella* sp.
- *Ankistrodesmus* sp.
- *Eudorina* sp.

These taxa were commonly observed at sites where Chlorophyceae showed moderate to high contribution.

Overall, phytoplankton pattern showed clear dominance across all sites, with percentage contribution ranging from 70.42% to 98.70%. Diatoms dominated most locations, while blue-green algae dominated at Sisodara and green algae dominated at Jhanor. Chlorophyceae remained the second most important group at the remaining sites of this stretch.

Table 4 Phytoplankton abundance and qualitative composition (%) (*Stretch below Surpan to Gulf of Cambay*)

Site	Myxophyceae (%)	Bacillariophyceae (%)	Chlorophyceae (%)	Total Phytoplankton (Average nos. l ⁻¹)	Phytoplankton (%)
Vedgam	0.93	76.85	12.96	98	90.74
Poicha	5.51	70.87	14.96	116	91.34
Sisodara	54.67	31.05	8.38	494	94.10
Velugam	4.94	79.01	9.88	152	93.83

Jhanor	5.68	4.82	88.20	1146	98.70
Ankalesh war	18.75	41.25	27.50	70	87.50
Shakkarpur a	22.22	34.92	33.33	57	90.47
Bhadbhut	8.46	43.85	20.00	94	72.31
Ambetha	7.04	52.11	11.27	50	70.42

Source:- CIFRI Report data

Phytoplankton abundance during monsoon season

During the monsoon season, the total phytoplankton population varied widely among different sites. The total phytoplankton ranged from 9 to 212 nos., and their percentage contribution ranged from 69.23% to 98.60% (presented in below **Table 5**). This shows that Phytoplankton remained the dominant planktonic group at almost all sites during monsoon.

Total phytoplankton

The highest total phytoplankton abundance was recorded at Poicha (212 nos., 97.24%), followed by Bharuch (148 nos., 97.36%) and Kevadia (141 nos., 98.60%). The lowest total phytoplankton abundance was observed at Ambetha (11 nos., 69.23%). At Bhadbhut, total phytoplankton was 137 nos. (90.73%), while at Shakkarpura, it was 139 nos. (98.58%).

Myxophyceae

Myxophyceae showed strong dominance during monsoon. Its abundance ranged from 6 to 144 nos., with percentage contribution varying from 13.98% to 77.31%. The highest Myxophyceae abundance was recorded at Poicha (144 nos., 66.05%), followed by Shakkarpura (109 nos., 77.31%) and Bhadbhut (102 nos., 67.55%). The lowest abundance was seen at Ambetha (6 nos., 46.16%).

Bacillariophyceae

Bacillariophyceae abundance during monsoon ranged from 2 to 112 nos., and its percentage contribution varied from 15.38% to 78.33%. The highest dominance of Bacillariophyceae was

recorded at Kevadia (112 nos., 78.33%). Lower values were observed at Shakkarpura (23 nos., 16.31%) and Ambetha (2 nos., 15.38%).

Chlorophyceae

Chlorophyceae showed the lowest contribution among phytoplankton groups during monsoon. Its abundance ranged from 1 to 13 nos., and percentage contribution varied from 1.69% to 8.55%. The highest value was recorded at Bharuch (13 nos., 8.55%), while Sisodara (1 nos., 1.69%) showed the lowest contribution.

Table 5 Phytoplankton abundance during monsoon season

Phytoplankton Group	Kevadia	Poicaha	Sisodara	Jhanor	Bharuch	Shakkarpura	Bhadrhut	Ambetha	Range
Myxophyceae	20 (13.98)	144 (66.05)	38 (64.41)	54 (60.67)	69 (45.39)	109 (77.31)	102 (67.55)	6 (46.16)	6 – 144 (13.98 – 77.31)
Bacillariophyceae	112 (78.33)	62 (28.44)	27 (27.12)	30 (30.34)	66 (43.42)	23 (16.31)	32 (21.19)	2 (15.38)	2 – 112 (15.38 – 78.33)
Chlorophyceae	9 (6.29)	6 (2.75)	1 (1.69)	2 (2.24)	13 (8.55)	7 (4.96)	3 (1.99)	1 (7.69)	1 – 13 (1.69 – 8.55)

									8.55)
Total Phytoplank ton	141 (98.6 0)	212 (97. 24)	55 (93.22)	83 (93.2 5)	148 (97.36)	139 (98.58)	137 (90.73)	11 (69.23)	9 – 212 (69. 23 – 98.6 0)

Source:- CIFRI Report data

Phytoplankton abundance during winter season

During the winter season, the total phytoplankton population increased compared to monsoon at many sites. The total phytoplankton ranged from 20 to 321 nos., and the percentage contribution ranged from 68.96% to 91.67% (presented in below **Table 6**), indicating continued dominance of phytoplankton during winter.

Total phytoplankton

The highest total phytoplankton abundance during winter was observed at Kevadia (321 nos., 88.19%), followed by Poicha (209 nos., 91.67%) and Jhanor (208 nos., 88.13%). The lowest abundance was recorded at Shakkarpura (20 nos., 68.96%). Moderate values were observed at Ambetha (126 nos., 89.36%) and Bhadbhut (96 nos., 91.43%).

Myxophyceae

Myxophyceae abundance during winter ranged from 9 to 162 nos., with percentage contribution varying from 8.57% to 44.54%. The highest abundance was recorded at Kevadia (162 nos., 44.54%), while the lowest was observed at Mahegam (9 nos., 8.57%). At other sites, Myxophyceae showed moderate contribution, such as Poicha (30.70%) and Sisodara (27.61%).

Bacillariophyceae

Bacillariophyceae showed strong presence during winter. Its abundance ranged from 2 to 129 nos., and percentage contribution varied from 6.89% to 54.66%. The highest dominance was observed at Jhanor (129 nos., 54.66%), followed by Ambetha (69 nos., 48.94%) and Bhadbhut (50 nos., 47.62%). The lowest value was recorded at Shakkarpura (2 nos., 6.89%).

Chlorophyceae

Chlorophyceae showed a clear increase during winter compared to monsoon. Its abundance ranged from 6 to 61 nos., and percentage contribution varied from 12.71% to 28.57%. The highest percentage contribution was recorded at Mahegam (28.57%), while the lowest was observed at Jhanor (12.71%). Other sites such as Bharuch (22.37%) and Bhadbhut (22.86%) also showed notable contribution.

Table 6 Phytoplankton abundance during winter season

Phytoplanktonic Group	Kevadia	Poicha	Sisodara	Jhanor	Bharuch	Shakkarpura	Bhadbhut	Mahegam	Ambetha	Range
Myxophyceae	162 (44.54)	70 (30.70)	66 (27.61)	49 (20.76)	25 (32.89)	12 (41.38)	22 (20.95)	9 (8.57)	24 (17.02)	9 – 162 (8.57 – 44.54)
Bacillariophyceae	98 (26.92)	97 (42.55)	72 (30.13)	129 (54.66)	17 (28.95)	2 (6.89)	50 (47.62)	42 (40.00)	69 (48.94)	2 – 129 (6.89 – 54.66)

Chlorophyll ceae	61 (16.7 6)	42 (18. 42)	51 (21.3 4)	30 (12. 71)	17 (22.3 7)	6 (20.69)	24 (22.86)	30 (28.57)	33 (23.4 0)	6 – 61 (12. 71 – 28. 57)
Total Phytoplankton	321 (88.1 9)	209 (91. 67)	189 (79.0 8)	208 (88. 13)	64 (84.2 1)	20 (68.96)	96 (91.43)	81 (77.14)	126 (89.3 6)	20 – 321 (68. 96 – 91. 67)

Source:- CIFRI Report data

3.2 PERIPHYTON

According to the data collected on periphyton (organisms that live attached to underwater surfaces) for the Upper and Middle Narmada Basin, a total of four classes of periphyton and 46 species are present in the intermediate time Zone (1980-2000) and the current time Zone (2001-present). Some species, such as *Cymbella*, *Fragilaria*, *Gomphonema*, *Navicula*, *Ankistrodesmus*, *Cladophora*, and *Phormidium*, are common in both the intermediate and current time zones.

Most of the species are located in the Bargi Reservoir, situated in Jabalpur, the Barna Reservoir, located in the Raisen district, the Sarni Reservoir on the Tawa, the main tributary of the Narmada river near Tawa dam, and the Sampna Reservoir on the Manchna river to join river Tawa, the tributary of river Narmada in Betul district of the Narmada Basin, Madhya Pradesh.

In the Middle Narmada Basin, there are four classes of periphyton and a total of 30 species. As we can see, the number of species is less compared to the Upper Narmada Basin. The dominant classes are Bacillariophyceae and Chlorophyceae, with a higher number of species in both the Basin, as shown in Figure 5.

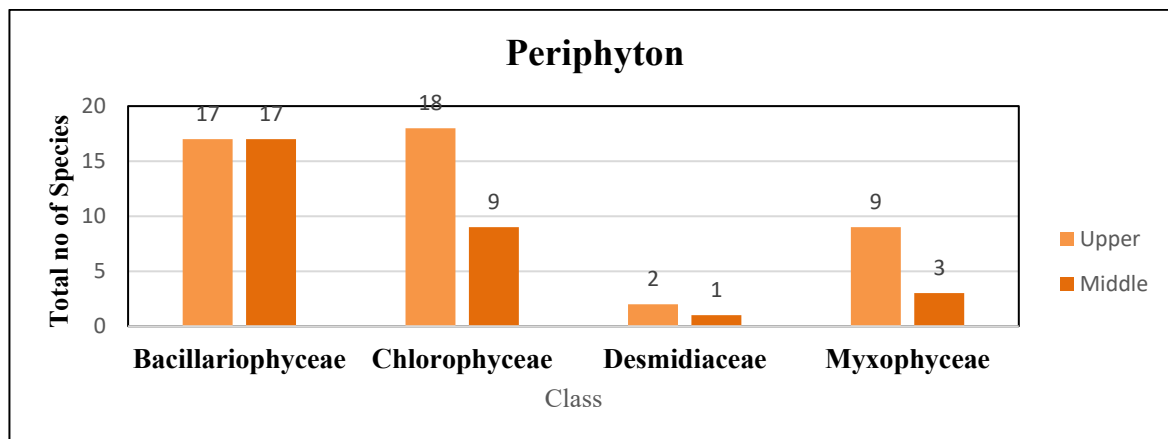


Figure 5: Different classes and numbers of species of Periphyton in the Upper and Middle Narmada Basin

Most of the species of the Middle Narmada Basin are in the stretch from Gadawada to Harsud, and specifically in the Kolar Reservoir area of Sehore district within the Narmada Basin, Madhya Pradesh. Some species, such as *Cocconeis*, *Cymbella*, *Gomphonema*, *Gyrosigma*, *Nitzschia*, *Surirella*, *Pediastrum*, and *Spirogyra*, are common in both the intermediate and current time zones. This data is combined for the Narmada River and its tributaries.

Details of the classes, species names, and locations are listed in Tables 7 and 8.

Table 7: Details of the Periphyton found in the Upper Narmada Basin

Class			
Bacillariophyceae	Chlorophyceae	Myxophyceae	Desmidiaceae
Species			
<i>Amphora</i>	<i>Binuclearia tatrana</i>	<i>Anabaena</i>	<i>Cosmarium</i>
<i>Asterionella</i>	<i>Cladophora</i>	<i>Phormidium</i>	<i>Staurastrum</i>
<i>Cymbella</i>	<i>Closterium</i>	<i>Amphithrix</i>	
<i>Diatoma</i>	<i>Gonatozygon</i>	<i>Characiopsis</i>	
<i>Gomphonema</i>	<i>Genicularia</i>	<i>Oscillatoria</i>	
<i>Navicula</i>	<i>Hormidium</i>	<i>Stichosiphon</i>	
<i>Nitzschia</i>	<i>Zygnemopsis</i>	<i>Rhizoclonium</i>	
<i>Rhopalodia</i>	<i>Ankistrodesmus</i>	<i>Spirogyra</i>	
<i>Synedra</i>	<i>Pediastrum simplex</i>	<i>Merismopedia</i>	
<i>Tabellaria</i>	<i>Meridion</i>		
<i>Cocconeis</i>	<i>Navicula</i>		

<i>Fragilaria</i>	<i>Nitzschia</i>		
<i>Surirella</i>	<i>Rhopalodia</i>		
<i>Bacillaria</i>	<i>Synedra</i>		
<i>Gyrosigma</i>	<i>Tabellaria</i>		
<i>Mastogloia</i>	<i>Scenedesmus</i>		
<i>Synedra ulna</i>	<i>Closteriopsis</i>		
	<i>Spirogyra</i>		

Source: Data extracted from references 1 and 2 (CIFRI)

Table 8: Different types of semi-aquatic grass in the Upper Zone of the Narmada Basin

Class			
Bacillariophyceae	Chlorophyceae	Myxophyceae	Desmidiaceae
Species			
<i>Amphora</i>	<i>Gonatozygon</i>	<i>Spirulina</i>	<i>Staurastrum</i>
<i>Cocconeis</i>	<i>Hormidium</i>	<i>Anabaena</i>	
<i>Cymbella</i>	<i>Pediastrum</i>	<i>Oscillatoria</i>	
<i>Diatoma</i>	<i>Spirogyra</i>		
<i>Gomphonema</i>	<i>Microspora</i>		
<i>Gyrosigma</i>	<i>Tribonema</i>		
<i>Mastogloia</i>	<i>Actinastrum</i>		
<i>Navicula</i>	<i>Ulothrix</i>		
<i>Nitzschia</i>	<i>Spirotaenia</i>		
<i>Pinnularia</i>			
<i>Rhopalodia</i>			
<i>Surirella</i>			
<i>Synedra</i>			
<i>Tabellaria</i>			
<i>Fragilaria</i>			
<i>Asterionella</i>			
<i>Diploneis</i>			

Source: Data extracted from references 1 and 2 (CIFRI)

In the stretch of the river from below Surpan to the Gulf of Cambay, the periphytic population showed clear variation. The number of periphyton ranged from 340 to 1130 numbers per cm². The highest periphytic growth was recorded at the Poicha site, while the lowest population was observed at Ankaleshwar. This variation may be due to changes in water flow, light availability, nutrients, and surface conditions at different locations.

Among all periphytic groups, Bacillariophyceae (diatoms) were the most dominant. They contributed about 63.85% to 87.51% of the total periphytic population in this stretch. The

dominance of diatoms shows that the river stretch provides suitable conditions such as moderate nutrients and good light penetration for their growth.

Several diatom species were commonly found and played an important role in the periphytic community. These included.

- Nitzschia
- Navicula sp.
- Surirella sp.
- Fragilaria sp.
- Asterionella sp.
- Diatoma sp.
- Synedra sp.
- Gomphonema sp.
- Gyrosigma sp.
- Tabellaria sp.
- Cymatopleura sp.

The presence of these species reflects a stable and productive periphytic ecosystem in this part of the river.

Overall, the periphyton in this stretch of the river shows healthy growth, with strong dominance of diatoms, which supports the aquatic food web and contributes to the ecological balance of the river system.

3.3 RIPARIAN AND FLOODPLAIN VEGETATION

In this report, we present the compiled data for the aquatic and semi-aquatic grasses and herbs categories of the vegetation in the Narmada Basin. The data has been categorized into the Upper and Middle Zones. In the Narmada Basin, grasses are the least abundant species, with four classes and 27 species of semi-aquatic species, as shown in Figure 6. There is only one aquatic species of grass, *Typha angustifolia* L. The classes that are dominant contributors are Poaceae and Cyperaceae. *Cyperus difformis* L. of semi-aquatic habitat is in the exotic category of the Cyperaceae family; the rest are native species. Many of the species are perennial and annual.

For Herbs, 60 families and 142 genera are found in the Narmada River, encompassing terrestrial, aquatic, and semi-aquatic species. *Marsilea quadrifolia* L., *Eichhornia crassipes* (Mart.) Solms, *Portulaca oleracea* L., is an exotic species, and the rest are native species. There are 24 aquatic, semi-aquatic herb families reported in the Upper Zone, with different species as shown in Figure 7 (Ref. 9). Checklist for grass and herbs is in the Annexure (3.3.1)

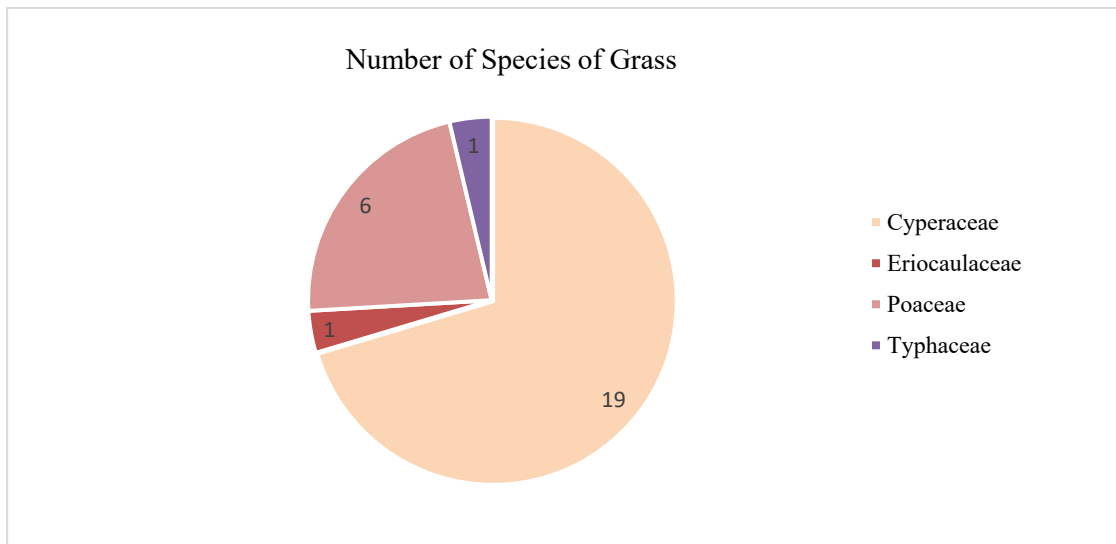


Figure 6: Different types of semi-aquatic grass in the Upper Zone of the Narmada Basin.

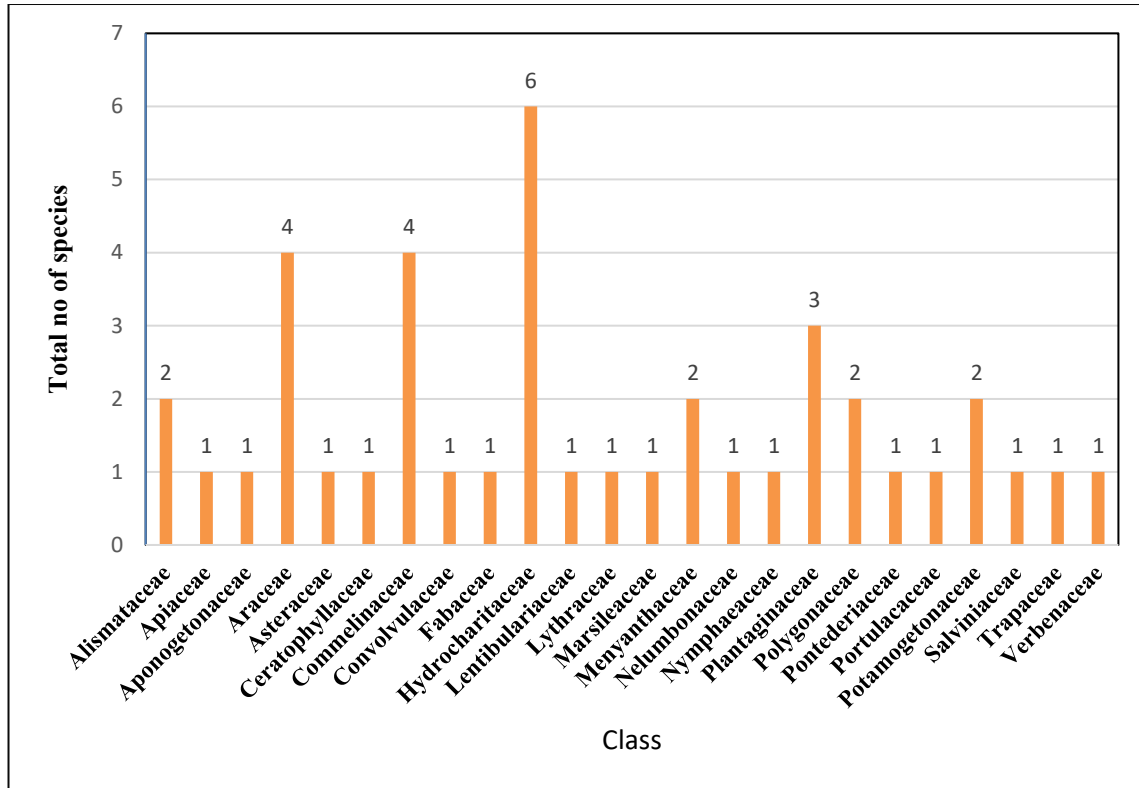


Figure 7: Different classes of aquatic and semi-aquatic Herbs in the Upper Zone of the Narmada Basin.

The riparian and floodplain vegetation of the lower Narmada River shows a disturbance-tolerant and simplified plant community. This condition is mainly due to changed river flow, agriculture, urban growth, grazing, and spread of invasive species. Native vegetation still exists, but it is under strong pressure from exotic plants.

Floristic Composition and Species Richness

The lower zone supports a total of 94 plant species, belonging to 38 families and 85 genera. Herbs form the largest group, followed by trees and grasses.

Life-form composition (Lower Zone)

Table 9 Life-form composition (Lower Zone)

Life Form	Number of Species
Trees	20

Shrubs	7
Herbs	42
Grasses	11
Sedges	4
Climbers	10
Total	94

Source:- NRCD Narmada Report

Out of these, 70 species are native, while 24 species (25.5%) are exotic. The high percentage of exotic species clearly shows strong invasion pressure in the lower riparian belt.

2. Tree Layer (Riparian Canopy)

A total of 20 tree species belonging to 11 families and 19 genera were recorded. The average tree density is 35.78 ± 3.44 individuals per hectare, but this does not represent a healthy forest. The density is high mainly because of dominance by invasive trees.

Table 10 Major tree species density and Importance Value Index (IVI)

Species	Density (Ind ha ⁻¹)	IVI
<i>Prosopis juliflora</i>	22.82 ± 4.75	97.64
<i>Ficus racemosa</i>	5.31 ± 1.53	27.28
<i>Pithecellobium dulce</i>	4.25 ± 1.40	19.14
<i>Cordia dichotoma</i>	2.12 ± 1.03	16.79
Other species (15 spp.)	≤ 1.06	< 15

Source:- NRCD Narmada Report

Prosopis juliflora alone contributes more than 63% of total trees. Its dense canopy, chemical effects on soil, and aggressive growth suppress native tree regeneration.

3. Shrub Layer

The shrub layer is poorly developed, with only 7 species. Total shrub density is 786.34 ± 135.94 Ind ha⁻¹.

Table 11 Dominant shrub species density

Species	Density (Ind ha ⁻¹)
<i>Lippia alba</i>	295.45 ± 99.27
<i>Alhagi maurorum</i>	168.00 ± 75.31
<i>Phyllanthus reticulatus</i>	144.83 ± 65.69
<i>Lantana camara</i>	92.69 ± 64.44

Source:- NRCD Narmada Report

The dominance of exotic shrubs like *Lippia alba* and *Lantana camara* shows that the plants growing below trees are degraded and disturbed.

Alhagi maurorum, a native shrub, survives mainly in dry and disturbed floodplain patches.

4. Herb Layer (Floodplain Ground Cover)

Herbs are the largest and most dominant life form, with 42 species and a total density of 7931.04 ± 262.03 Ind ha⁻¹.

- Exotic herbs: 14 species (about 33%)
- This is the highest invasion level among all life forms.

Table 12 Dominant herb species density

Species	Density (Ind ha ⁻¹)
<i>Alternanthera sessilis</i>	809.45 ± 294.18
<i>Solanum xanthocarpum</i>	703.29 ± 180.63
<i>Xanthium strumarium</i>	504.25 ± 191.37

<i>Parthenium hysterophorus</i>	464.44 ± 178.97
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Source:- NRCD Narmada Report

The herb layer is mainly dominated by Asteraceae and Amaranthaceae, which are typical of disturbed, nutrient-rich, and open floodplain soils.

5. Grasses (Floodplain Stabilizers)

Grasses are extremely abundant in the lower floodplain. A total of 11 grass species were recorded with a density of $16918.79 \pm 1198.06 \text{ Ind ha}^{-1}$.

Table 13 Major grass species density

Species	Density (Ind ha ⁻¹)
<i>Cynodon dactylon</i>	9315.29 ± 1163.55
<i>Paspalum distichum</i> (Exotic)	2003.72 ± 497.57
<i>Dactyloctenium aegyptium</i>	1791.40 ± 469.73

Source:- NRCD Narmada Report

Cynodon dactylon plays an important role in bank stabilization and soil binding. However, the spread of exotic *Paspalum distichum* indicates future invasion risk.

6. Sedges (Floodplain and Marsh Interface)

The lower zone supports 4 sedge species, making it the most sedge-rich stretch of the river.

Table 14 Sedge species density

Species	Density (Ind ha ⁻¹)
<i>Cyperus difformis</i>	490.98 ± 259.79
<i>Fimbristylis miliacea</i>	345.01 ± 172.60
<i>Cyperus haspan</i>	318.47 ± 214.62

<i>Cyperus nutans</i>	145.97 ± 145.66
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Source:- NRCD Narmada Report

Total sedge density: 1300.42 ± 399.41 Ind ha⁻¹

These sedges indicate seasonal or permanent waterlogging and are important for sediment trapping and floodplain stability.

7. Climbers (Riparian Structural Complexity)

The lower Narmada supports 10 climber species with a total density of 1393.31 ± 30.51 Ind ha⁻¹.

Table 15 Climbers species density

Species	Density (Ind ha ⁻¹)
<i>Momordica charantia</i>	305.20 ± 224.72
<i>Ampelocissus latifolia</i>	291.93 ± 225.97
<i>Cayratia trifolia</i>	212.31 ± 211.87

Source:- NRCD Narmada Report

Climbers add vertical structure and habitat diversity, but their abundance is lower than in upper river stretches.

3.4 AQUATIC VEGETATION

According to the data collected, there is only one reported study on aquatic vegetation, which was conducted at Bargi Dam in Jabalpur district, part of the Upper Narmada Basin (Ref. 5). Details of the studies, including the aquatic species present, their families, and orders, are provided in Tables 16 and 17.

Table 16 : Details of Aquatic vegetation in the Bargi Dam, Upper Narmada Basin

Category	Details
Study Referenced	Parte et al. (2018)

Study Location	Catchment area of Bargi Dam, Jabalpur
Total Aquatic Species	25 species
Species Composition	24 aquatic herbs, 1 grass species
Taxonomic Coverage	10 orders, 16 families, 22 genera
Dominant Families	Hydrocharitaceae (6 species), Araceae (3 species), Menyanthaceae (2 species)
Families Represented by a Single Species	12 families
Life Forms	4 annual species, 21 perennial species
Exotic Species	2 species exotics to the Narmada River

Source: Data extracted from the reference 5 (WII)

The study was conducted by Parte et al. in 2018 at Bargi dam in the Jabalpur district of the Upper Narmada Basin. According to the study, there are two exotic species in the Upper Narmada River, with the dominant families being Hydrocharitaceae (six species), Araceae (three species), and Menyanthaceae (two species).

Table 17: Number of species in various families of aquatic species found in the Upper Narmada Basin.

S.No	Family Name	Species
1	Mariliaceae	1
2	Aponogetonaceae	1
3	Typhaceae	1
4	Nelumbonaceae	1
5	Lentibulariaceae	1
6	Nymphaeaceae	1
7	Alismataceae	1
8	Pontederiaceae	1
9	Lythraceae	1

10	Salviniaceae	1
11	Ceratophyllaceae	1
12	Trapaceae	1
13	Menyanthaceae	2
14	Potamogetonaceae	2
15	Araceae	3
16	Hydrocharitaceae	6

Source: Data extracted from reference 5 (WII)

The lower zone of the Narmada River supports a small number of aquatic and semi-aquatic plant species. Most of these plants grow in shallow water, along river margins, or in areas with slow current. Some species grow fast and spread widely, especially under nutrient-rich conditions.

Key aquatic and semi-aquatic species recorded in the lower zone are given below table:

Table 18 Aquatic and semi-aquatic species recorded in the lower zone

Species	Density (ha ⁻¹)
<i>Grangea maderaspatana</i>	411.36 ± 170.45
<i>Alternanthera philoxeroides</i> (Exotic)	132.70 ± 81.35
<i>Pistia stratiotes</i> (Exotic)	119.43 ± 119.18
<i>Colocasia esculenta</i>	92.89 ± 92.69
<i>Caldesia oligococca</i>	79.62 ± 79.45

Source:- NRCD Narmada Report

Among these, *Grangea maderaspatana* shows the highest density and is commonly found in moist and shallow areas. Two species, *Alternanthera philoxeroides* and *Pistia stratiotes*, are exotic and invasive. Their presence indicates disturbed and nutrient-rich water conditions.

Chapter 4: Fauna

4.1 ZOOPLANKTONS

Data collected for zooplanktons in the Upper and Middle Narmada Basin is categorized into intermediate (1980-2000) and current zones (2000—present). According to the data, there are four classes present in the Upper and Middle Narmada Basin, comprising species that are mainly located in Barna, Bargi, Sampna, Sarni Reservoir, and the Amarkantak-Gadarwada stretch in the Upper Narmada Basin, as well as in the Kolar Reservoir and the Gadawada to Harsud stretch in the Middle Narmada Basin. As we examine the current status of these zooplankton classes and species, they are found in the same locations. A few species, like *Brachionus Bidentata*, *Brachionus Quadridentatus*, *Filinia longiseta*, *Hexarthra Mira*, *Brachionus Angularis*, *polyarthra* of class Rotifera, then *Cyclop Diaptomus* of class Copepoda are common in both time zones.

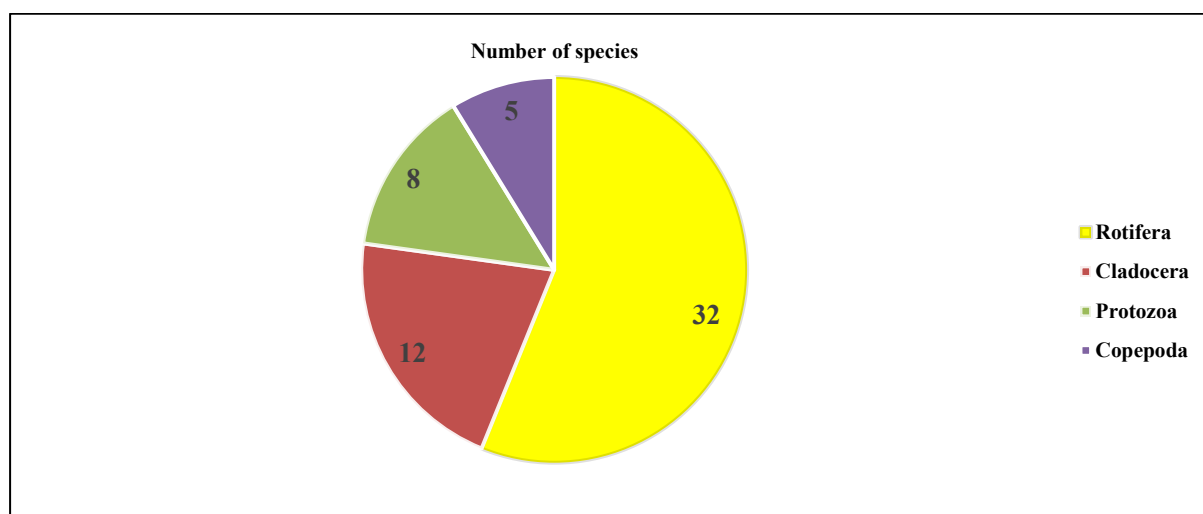


Figure 8: Different classes and number of species of Zooplankton in the Upper Narmada Basin

Data collected on zooplankton in the Middle Narmada Basin have been categorized into two periods: the intermediate (1980-2000) and current zones (2001—present). Classes present in the Middle Narmada Basin comprise species primarily found in the Kolar Reservoir, the Gadawada to Harsud stretch, in the Middle Narmada Basin. As we examine the current status of these zooplankton classes and species, they are found in the same locations. A few of the species, such as *Euglena*, *Arcella*, *Centrophyxis*, of class Protozoa, *Cyclopus* of Copepod, *Filinia*, *Notholca* from class Rotifera, etc., are common in both time zones found at Kolar Reservoir in Sehore district and Gadawada to Harsud.

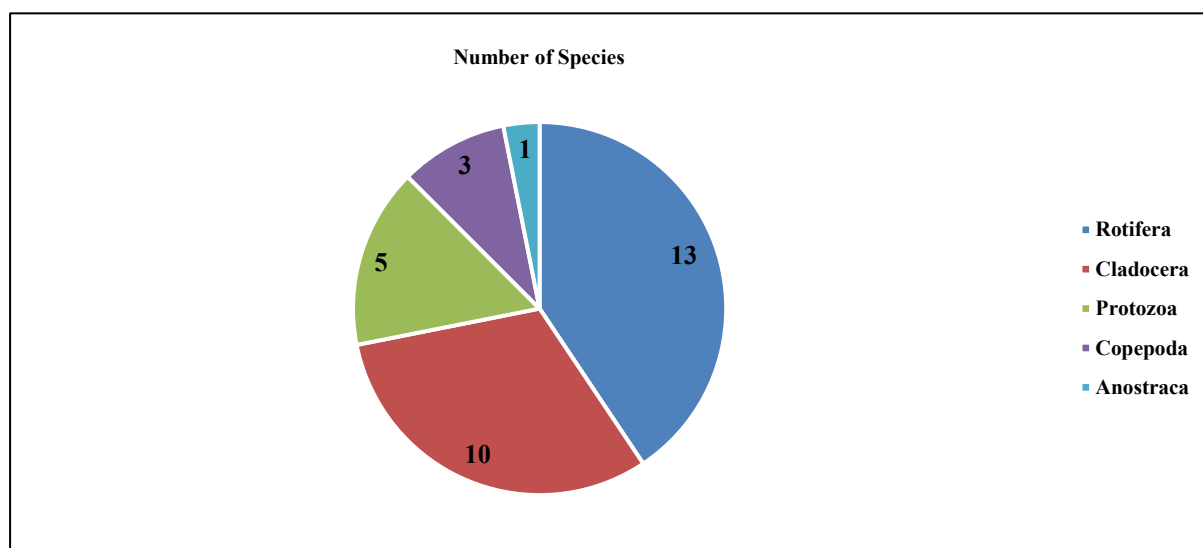


Figure 9: Different classes and number of species of Zooplankton in the Middle Narmada Basin

As shown in Figures 8 and 9, the dominant class is Rotifera, with 32 species, followed by Cladocera, with 12 species in the Upper Narmada Basin and 13 species (Rotifera) and 12 species (Cladocera) in the Middle Basin. Compared to the Zooplankton species numbers, there are more in the Upper Narmada Basin. This data is combined for the Narmada River and its tributaries. A detailed list of class and species names of the Upper and Middle Narmada Basin is in Tables 19 and 20.

Table 19: Details of different classes and species of Zooplanktons of the Upper Narmada Basin

Class			
Rotifera	Cladocera	Protozoa	Copepoda
Species			
<i>Anuraeopsis</i>	<i>Alona</i>	<i>Diffugia lobostoma</i>	<i>Cyclops</i>
<i>Brachionus angularis</i>	<i>Bosminopsis dietersi</i>	<i>Diffugia urceolata</i>	<i>Diaptomus</i>
<i>Brachionus bidentata</i>	<i>Ceriodaphnia cornuta</i>	<i>Diffugia</i>	<i>Nauplii</i>
<i>Brachionus havanaensis</i>	<i>Chydorus ovalis</i>	<i>Heterophrys myriopoda</i>	<i>Limnocalanus</i>
<i>Brachionus quadridentata</i>	<i>Diaphanosoma</i>	<i>Pompholyxophrys punicea</i>	<i>Mesocyclops</i>
<i>Filinia longiseta</i>	<i>Acroperus harpae</i>	<i>Vampyrella lateritia</i>	
<i>Hexarthra mira</i>	<i>Alonella</i>	<i>Plagiopyxis</i>	
<i>Keratella valga</i>	<i>Bosmina longirostris</i>	<i>Pompholyxophrys</i>	
<i>Polyarthra</i>	<i>Ilyocryptus spinifer</i>		
<i>Pompholyx sulcata</i>	<i>Bosmina</i>		
<i>Trichocerca longiseta</i>	<i>Chydorus</i>		
<i>Trichocerca multirinis</i>	<i>Ophryoxus gracilis</i>		
<i>Keratella</i>			
<i>Anuraeopsis fissa</i>			
<i>Asplanchna priodonta</i>			
<i>Elosa woralli</i>			
<i>Gastropus stylifer</i>			
<i>Keratella cochlearis</i>			
<i>Lecane aspasia</i>			
<i>Mytilina mucronata</i>			
<i>Polyarthra vulgaris</i>			
<i>Proales decipiens</i>			
<i>Proales fallaciosa</i>			
<i>Synchaeta pectinata</i>			
<i>Trichocerca capucina</i>			
<i>Trichocerca cylindrica</i>			
<i>Filinia brachiata</i>			
<i>Filinia minuta</i>			
<i>Gastropus</i>			
<i>Ploesoma</i>			
<i>Lecane</i>			
<i>Brachionus</i>			

Table 20: Details of different classes and species of Zooplankton in the Middle Narmada Basin.

Class				
Rotifera	Cladocera	Protozoa	Copepoda	Anostraca
Species				
<i>Asplanchna</i>	<i>Bosminopsis</i>	<i>Diffugia</i>	<i>Cyclops</i>	<i>Eubbranchipus</i>
<i>Brachionus angularis</i>	<i>Ceriodaphnia cornuta</i>	<i>Centropyxis</i>	<i>Diaptomus</i>	
<i>Brachionus bidentata</i>	<i>Chydorus ovalis</i>	<i>Euglena</i>	<i>Nauplii</i>	
<i>Gastropus</i>	<i>Diaphanosoma</i>	<i>Arcella</i>		
<i>Hexarthra mira</i>	<i>Sida</i>	<i>Oxytricha</i>		
<i>Keratella hiemalis</i>	<i>Moina</i>			
<i>Keratella cochlearis</i>	<i>Chydorus</i>			
<i>Brachionus</i>	<i>Alonella</i>			
<i>Filinia</i>	<i>Daphnia</i>			
<i>Polyarthra</i>	<i>Bosmina</i>			
<i>Trichocerca</i>				
<i>Lecane</i>				
<i>Epiphanes</i>				

Source: Data extracted from references 1 and 2 (CIFRI)

The lower stretch of the Narmada River, extending from Surpan to the Gulf of Cambay, including the estuarine region in Gujarat, exhibited a moderate but ecologically significant zooplankton population. Zooplankton formed a secondary yet vital component of the total plankton community in this stretch, contributing between 1.30% and 29.58% of the overall plankton abundance.

The zooplankton assemblage in the Lower Narmada was predominantly represented by Protozoa, Copepoda, and Rotifera, reflecting the transitional nature of this stretch from freshwater to estuarine conditions. Among these groups, protozoans (0.52–14.61%) showed considerable dominance, particularly in zones influenced by estuarine mixing and increased nutrient availability. Important protozoan taxa recorded included *Centropyxis* sp., *Arcella* sp., and *Diffugia* sp., which are characteristic of stable microhabitats and favorable organic matter conditions in these riverine environments.

Copepoda (0.26–12.68%) constituted the second major zooplankton group and played a crucial role in linking primary producers to higher trophic levels. Their abundance highlights the productive nature of the lower river stretch and its suitability for supporting fish larvae and

other higher aquatic organisms. Rotifera (0.43–5.63%), though contributing a smaller fraction, were consistently present across sites. Rotifers are known indicators of nutrient enrichment and moderate organic loading, reflecting the combined influence of freshwater inflow and estuarine processes.

Overall, the zooplankton composition demonstrates a mixed freshwater–estuarine character, shaped by tidal influence and upstream discharge regulation. These patterns indicate that the stretch provides favorable ecological conditions for secondary productivity, essential for sustaining the river’s fisheries.

Table 21 Zooplankton Statistical Table

Zooplankton Group	Mean Value (%)	Key Representative Taxa
Protozoa	7.57%	Centropyxis sp., Arcella sp., Diffugia sp.
Copepoda	6.47%	Cyclops sp., Mesocyclops sp., Nauplii
Rotifera	3.03%	Brachionus sp., Keratella sp.

Source : Singh, S. N. (2009). River Narmada: Its Environment and Fisheries.

4.1.1 Macrobenthos

According to the data collected on macrobenthos, the three major phyla identified are Annelida, Arthropoda, and Mollusca, with a total of 8 classes in the Upper and 4 classes in the Middle Narmada Basin during the intermediate and current time zones.

In the present analysis, 31 species of the Upper Narmada Basin were identified in Bargi, Barna, Sarni, Sampna Reservoir, and the Amarkantak to Gadarwada Narmada stretch. Common species observed in both time zones include *Tubifex* (annelid), *Chironomus*, *Chaoborus*, *Culicoides*, *Anox* (Arthropoda), *Pisidium*, *Corbicula*, *Parreysia*, *Thiara*, *Bellantya*, *Gyraulus*, and *Lymnaea* (Mollusca).

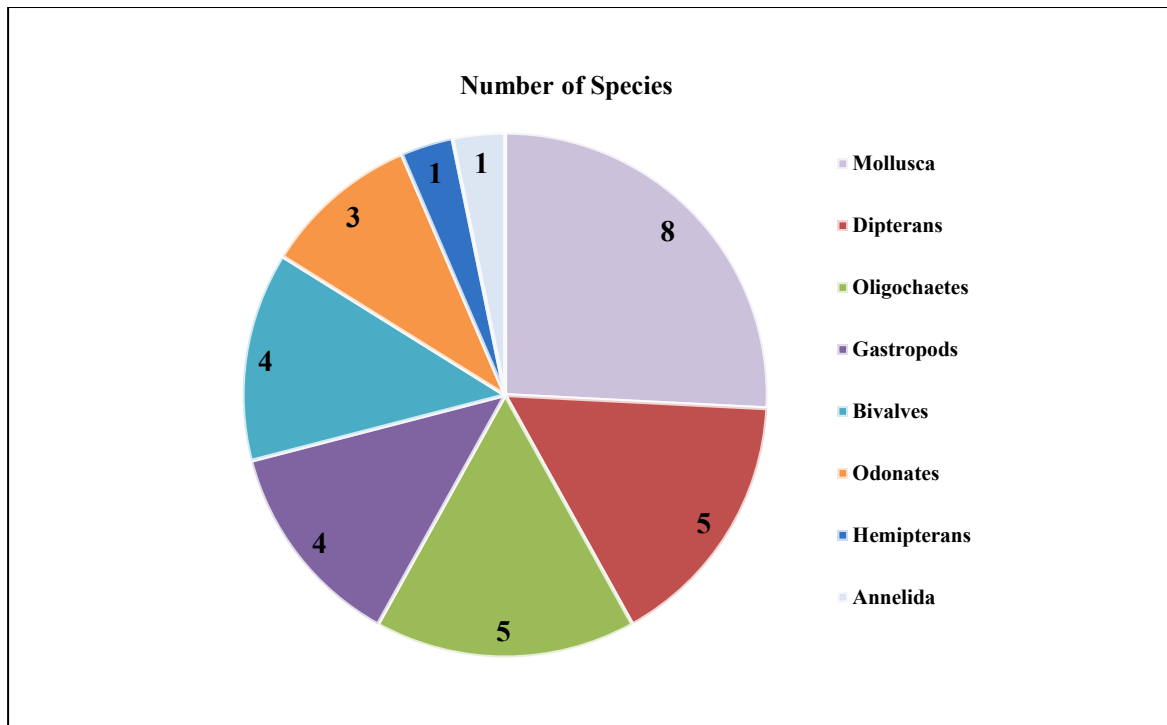


Figure 10: Different classes and species numbers of the Macrobenthos of the Upper Narmada Basin.

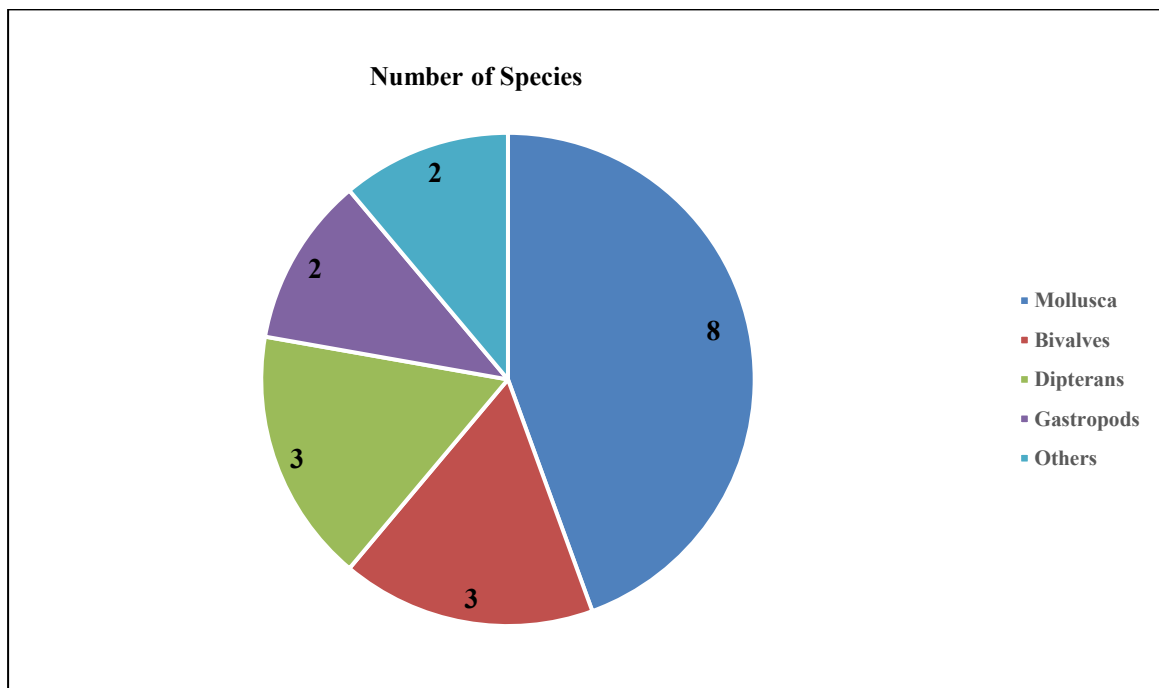


Figure 11: Different classes and species numbers of the Macrobenthos of the Middle Narmada Basin.

The macrobenthic communities of the Kolar reservoir in Sehore district, which is part of the Middle Narmada Basin during the 1980-2000 period, consist of Dipterans (53.0%), Gastropods (24.4%), Oligochaetes (12.2%), and Bivalves (10.4%). Major forms recorded with the species include *Chironomus*, *Chaoborus*, *Thiara*, *Bellamya*, *Parreysia*, *Corbicula*, and *Lamellidens*. The Middle Narmada Basin comprises the same Phyla (Annelida, Arthropoda, and Mollusca), classes, and a total of 18 species, which are found mainly in the Middle Narmada Basin. This data is combined for the Narmada River and its tributaries. Details of the macrobenthos class and species are provided in Tables 22 and 23 for the Upper and the Middle Basin.

Table 22: Different classes and species names of the Macrobenthos of the Upper Narmada Basin.

Class						
Mollusca	Dipterans	Oligochaetes	Bivalves	Odonates	Hemipterans	Annelida
Species						
<i>Thiara scabra</i>	<i>Chironomus</i>	<i>Tubifex</i>	<i>Corbicula</i>	<i>Chironomus</i>	<i>Laccotrephes</i>	<i>Tubifex</i>
<i>Thiara tuberculata</i>	<i>Chaoborus</i>	<i>Stylaria</i>	<i>Parreysia</i>	<i>Chaoborus</i>		
<i>Bellamya bengalensis</i>	<i>Culicoides</i>	<i>Nais</i>	<i>Pisidium</i>	<i>Anax</i>		
<i>Gyraulus</i>	<i>Lumbricillus</i>	<i>Chaetogaster</i>	<i>Lamellidens</i>			
<i>Lymnaea acuminata</i>	<i>Chironomid larvae (Chironomus)</i>	<i>Dero</i>				
<i>Corbicula striatella</i>						
<i>Lamellidens marginalis</i>						
<i>Sphaerium</i>						

Source: Data extracted from references 1 and 2 (CIFRI)

Table 23: Different classes and species of Macrobenthos of the Middle Narmada Basin

Class				
Mollusca	Bivalves	Dipterans	Gastropods	Others
Species				
<i>Thiara</i>	<i>Parreysia</i>	<i>Chironomus</i>	<i>Thiara</i>	<i>Trichoptera</i>
<i>Viviparus</i>	<i>Corbicula</i>	<i>Chaoborus</i>	<i>Bellamya</i>	<i>Odonata</i>
<i>Lymnaea</i>	<i>Lamellidens</i>	<i>Culicoides</i>		
<i>Gyraulus</i>				
<i>Corbicula</i>				
<i>Parreysia</i>				
<i>Pisidium</i>				
<i>Lamellidens</i>				

Source: Data extracted from references 1 and 2 (CIFRI)

The lower stretch of the Narmada River, extending from Surpan to the Gulf of Cambay, exhibits a highly variable and diverse macro-benthic population. This diversity reflects the transition from freshwater to estuarine conditions along the Gujarat reach. The total average macro-benthos abundance in this stretch varies significantly, ranging from 108 to 13,739 nos. m⁻², indicating strong spatial heterogeneity in benthic productivity.

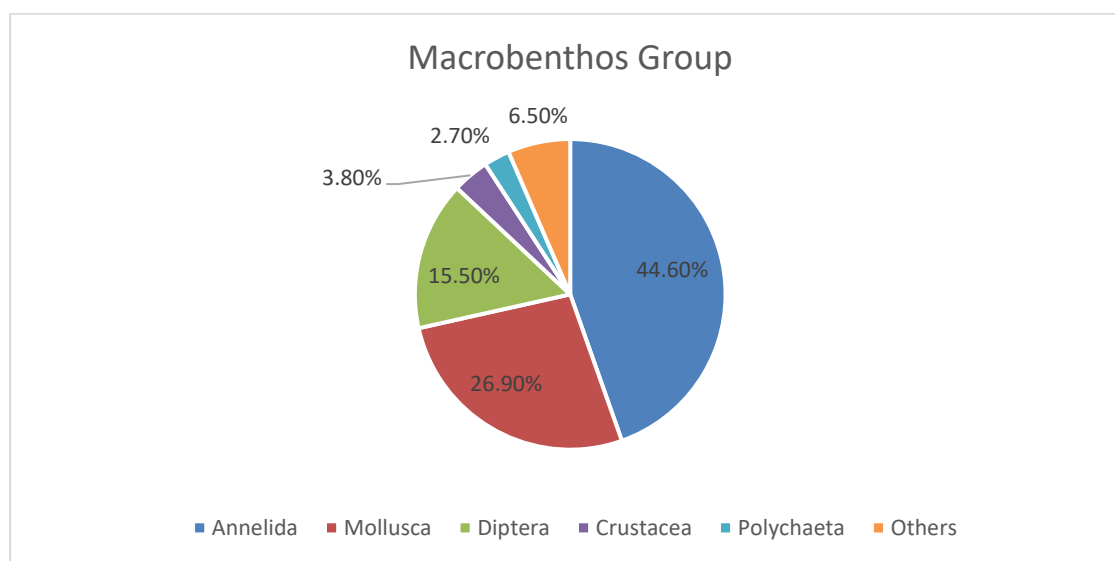
The macro-benthic community composition varies distinctly based on the local substrate and salinity levels. Annelids emerged as the most dominant group at sites like Ambetha, Bhadbhut, Ankaleshwar, and Jhanor, where they contributed between 65.84% and 93.81% of the population. These areas are characterized by organic-rich sediments and stable depositional environments. The population is primarily represented by *Tubifex tubifex*, *Nais* sp., and *Neanthes* sp.

At the Sisodara and Poicha sites, Diptera (represented by *Chironomus* sp., *Culicoides* sp., and *Chaoborus* sp.) replaced annelids as the dominant group, with dominance ranging from 47.36% to 86.46%. This shift indicates fine sediment conditions and high tolerance to varying environmental stresses. In contrast, Molluscs were the principal component at Shakkarpura, Velugam, and Vedgam, ranging from 57.06% to 95.15%. This group included *Thiara tuberculata*, *Bellamya bengalensis*, and *Corbicula* sp., reflecting stable substrates.

Overall, the macro-benthic community of the Lower Narmada is a vital indicator of the river's health, playing a crucial role in nutrient cycling and supporting the food web for commercial fisheries like the Hilsa and Giant Prawn.

Table 24 Macroenthos Group Table

Macroenthos Group	Mean Composition (%)	Dominant Sites
Annelida	44.6%	Ambetha, Bhadbhut, Ankaleshwar
Mollusca	26.9%	Shakkarpura, Velugam, Vedgam
Diptera	15.5%	Sisodara, Poicha
Crustacea	3.8%	Estuarine transition zones
Polychaeta	2.7%	Near the Gulf of Cambay
Others	6.5%	Scattered across all sites



Source: Singh, S. N. (2009). River Narmada: Its Environment and Fisheries

4.1.2 Macrophytes

Macrophyte species such as *Vallisneria*, *Hydrilla*, *Ceratophyllum*, *Potamogeton*, *Chara*, *Najas*, *Nymphaea*, *Eichhornia crassipes*, *Ipomoea*, and *Typha* are reported in the Barna, Bargi, Sarni Reservoir, and Amarkantak to Gadarwada stretch in the Upper Basin.

Macrophyte species such as *Hydrilla*, *Vallisneria*, *Potamogeton*, *Ceratophyllum*, *Chara*, and *Najas* are found in the Kolar reservoir, Gadarwada to Harsud stretch in the Middle Basin (Ref. 1, 2).

The lower stretch of the Narmada River, extending from Surpan to the Gulf of Cambay, represents a critical ecological transition zone where freshwater flow meets tidal influence. In this region, macrophyte distribution is not continuous but occurs in isolated patches, reflecting a moderate yet ecologically significant abundance. Detailed surveys within Gujarat show that macrophytic biomass in this stretch fluctuates between 1.875 and 4.825 kg m⁻². The spatial distribution is highly site-specific: the highest densities are consistently recorded at Sisodara, where favorable localized conditions support thicker growth, while the lowest densities are found at Ankleshwar.

Ecologically, the growth of these aquatic plants is strictly confined to marginal areas and sheltered coves where the water remains relatively stagnant, protecting the flora from the swifter mid-stream currents. The macrophytic assemblage in this Gujarat stretch is dominated by a mix of submerged and emergent species, including *Hydrilla verticillata*, *Potamogeton crispus*, *Vallisneria spiralis*, *Chara* sp., and *Ceratophyllum demersum*. However, as the river nears the Gulf of Cambay, a sharp decline is observed. The lower estuarine sites are practically devoid of macrophytes, a phenomenon attributed to the high salinity levels and the physical stress of intense tidal action, which prevents freshwater aquatic plants from establishing root systems.

Table 25 Macrophyte Distribution in the Lower Narmada (Gujarat)

Feature	Data Detail	Reference Source
Geographic Range	Surpan to Gulf of Cambay (Gujarat)	CIFRI Bulletin No. 157
Biomass Range	1.875 – 4.825 kg m ⁻²	Sardar Sarovar Dam Report
Peak Abundance Site	Sisodara	CIFRI Bulletin No. 157

Lowest Abundance Site	Ankleshwar	CIFRI Bulletin No. 157
Dominant Species	H. verticillata, P. crispus, V. spiralis, Chara sp.	GSBB Fish & Habitat Study
Estuarine Status	Devoid (due to salinity/tides)	CIFRI Bulletin No. 157

Source: Singh, S. N. (2009). *River Narmada: Its Environment and Fisheries*

4.1.3 Fishes

In the Narmada Basin, a diverse array of fish species is found in different zones, ranging from the Upper to the Lower Zone. Different studies conducted by the Central Inland Fisheries Research Institute (CIFRI), the Zoological Survey of India (ZSI), and the Wildlife Institute of India (WII) at various time zones indicate that most species are common.

In the reference time zone, as per the data, a total of 77 species are reported. Most of the species belonged to the order Cypriniformes and the families Cyprinidae, Bagridae, and Sisoridae, which were the majorly dominant groups (Ref. .3). Among these 77 species, 40 species are from Hoshangabad district, which is part of the Upper Narmada Basin. The checklist of species in the Annexure (4.4.1)

In the Intermediate Zone status of fishes as per CIFRI (1998-2003), is that in the Upper Narmada Basin, the major carps were *C.catla* (17.2%), *L.rohita* (14.6 %), *C.mirgala*(13.4 %), *Tortor* (0.5%), As per the data in the last five years (1996-97 to 2000-01) these major carps declined to 45.7 %. in the Barna reservoir at Jabalpur district. Local major and minor groups included species of fish named *W. attu*, *A.aor*, *A. seenghala*, *N.chitala*, *N. notopterus*, *L. gonius*, *L. bata*, *C. reba*, *Mastacembelus*, *Chanlla*, *L. Mystus* and *L. calbasu* (0.5 kg). Similarly, in the Bargi reservoir, major carps (83.0%) formed the most important part of the fishery. Major contributors are: *C. catla*(77.3%), *C.mirgala*(2.8 %), *L.rohita* (2.7%) and *T.tor*(0.2%). (Ref 1)

In the current Zone, data collected from CIFRI (2009, 2013), ZSI (2007, 2010), and WII (2022) for the Narmada Basin.

The data collected from CIFRI (2009) focused on the findings of the fish in a stretch-wise manner.

In this, the Narmada River is divided into four stretches:

- (a) Amarkantak to Gadarwada was represented by six (06) sites, namely Amarkantak, Mandla, Jabalpur, Narsinghpur, Kareli, and Gadarwada.
- (b) Below Gadarwada to Harsud encompasses seven (07) sites, viz. Sandia, Shahganj, Hoshangabad, Dongarwada, Gondagaon, Handia and Mola.
- (c) Below Harsud to Surpan, the lowermost expanse of the Narmada River was represented by six (06) sites, namely Maheshwar, Khalghat, Hanfeshwar, Charbara, Gadher, and Surpan.
- (d) Below Surpan to the Gulf of Cambay

The stretch from Amarkantak to Gadarwada is considered the uppermost stretch and is part of the Upper Basin. The area from Gadarwada to Harsud and from Harsud to Surpan is considered part of the Middle Narmada Basin, as Maheshwar and Khalghat are located within this region. A total of 64 species were reported in all three stretches of the Narmada River. Out of 64 species, 19 species are common found in all three stretches mentioned, i.e, from the uppermost stretch to the lowermost stretch of the Narmada River.

Two species, i.e, *Clarias batrachus* (Linnaeus, 1758), *Heteropneustes fossilis* (Bloch, 1794), are specifically found only in the stretch Amarkantak to Gadarwada, the Upper Narmada Basin. Figure 12 shows the species number and family of the common fishes found in all three stretches of the Narmada Basin (Ref 2). Checklist of the fishes is in the Annexures (4.4.2).

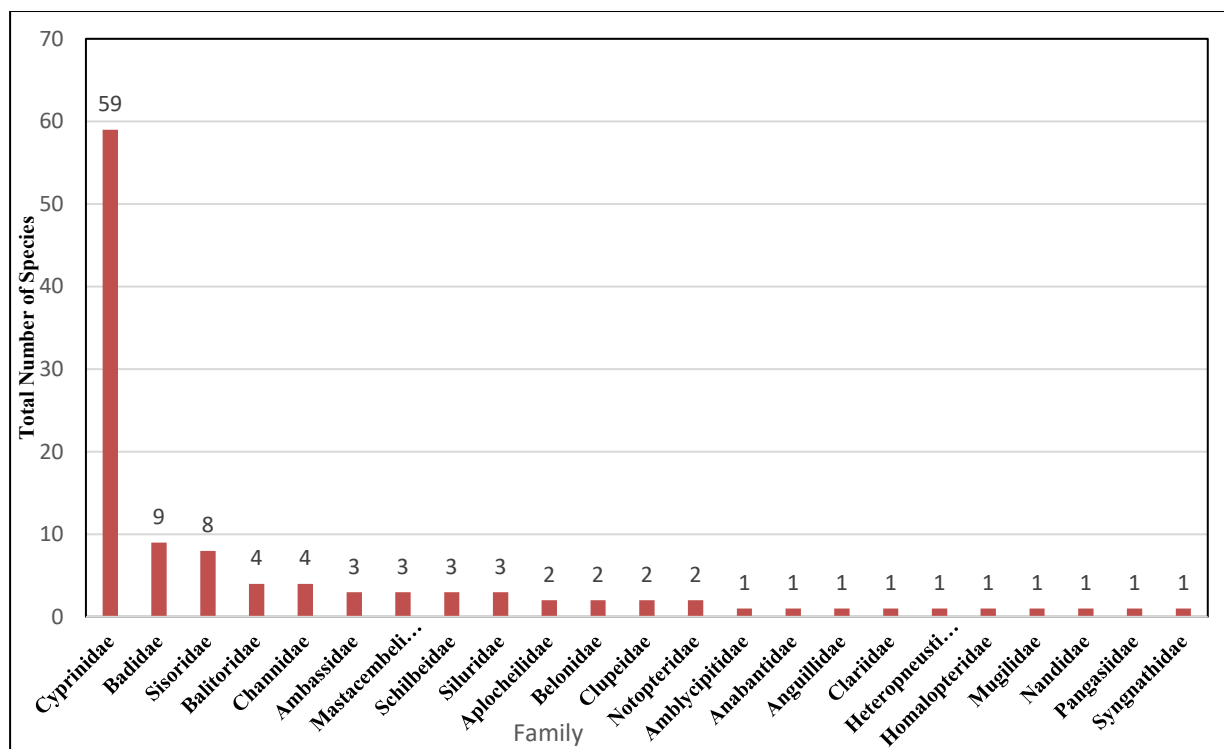


Figure 12: Class and species number of the fish found in all three stretches of the Upper and Middle Narmada Basin.

There are 25 species found in the below Gadarwada to Harsud stretch only. Six (6) species *Rita rita* (Hamilton, 1822), *Cyprinus carpio* (Linnaeus, 1758), *Puntius sarana* (Hamilton, 1822), *Chitala chitala* (Hamilton, 1822), *Pangasius pangasius* (Hamilton, 1822), *Ailia coila* (Hamilton, 1822) are found only in the Harsud to Surpan stretch of the Narmada river, in which some part of the Middle Basin contributes. The detailed list of fish names and their families, common to all three stretches, is presented in Table 26.

Table 26: Names of the fish present in all three stretches of the Narmada River

Family	Fish Taxa
Bagridae	<i>Aorichthys aor</i> (Hamilton, 1822)
Bagridae	<i>Aorichthys seenghala</i> (Sykes, 1839)
Bagridae	<i>Rita pavimentata</i> (Valenciennes, 1840)
Cypridae	<i>Catla catla</i> , <i>Cirrhinus mrigala</i> , <i>Cirrhinus reba</i> , <i>Labeo bata</i> , <i>Labeo calbasu</i> , <i>Labeo gonius</i> , <i>Labeo rohita</i> , <i>Pethia sophore</i> , <i>Salmostoma bacaila</i> , <i>Tor tor</i> (Hamilton, 1822), <i>Labeo fimbriatus</i> (Bloch, 1797)
Channidae	<i>Channa punctata</i> (Hamilton, 1822)
Mastacembelidae	<i>Mastacembelus armatus</i> (Lacepède, 1800)
Notopteridae	<i>Notopterus notopterus</i> (Pallas, 1769)
Schilbeidae	<i>Clupisoma garua</i> (Hamilton, 1822)
Siluridae	<i>Wallago attu</i> (Schneider, 1801)

Source: Data extracted from reference 2 (CIFRI)

Another analysis has been conducted using data collected from CIFRI (1966, 2009, 2013) and WII (2022), and it has been concluded that 31 species are common across all these years. Checklist of the fishes (WII) is in Annexure 4.4.3

A detailed list of the Fish common in reference and the present time zone is given in Table 27.

Table 27: Fishes common in Reference and Present Time Zone (1966-2022).

<u>S. No.</u>	<i>Fish Taxa</i>	<u>S. No</u>	<i>Fish taxa</i>
1	<i>Xenentodon cancila</i>	17	<i>Labeo dyocheilus</i>
2	<i>Wallago attu</i>	18	<i>Labeo calbasu</i>
3	<i>Tor tor</i>	19	<i>Labeo boggut</i>
4	<i>Rasbora daniconius</i>	20	<i>Labeo bata</i>
5	<i>Puntius sophore</i>	21	<i>Heteropneustes fossilis</i>

6	<i>Osteobrama cotio</i>	22	<i>Glossogobius giuris</i>
7	<i>Ompok bimaculatus</i>	23	<i>Garra gotyla</i>
8	<i>Nandus nandus</i>	24	<i>Clupisoma garua</i>
9	<i>Mystus vittatus</i>	25	<i>Cirrhinus reba</i>
10	<i>Mystus cavasius</i>	26	<i>Cirrhinus mrigala</i>
11	<i>Mystus bleekeri</i>	27	<i>Channa marulius</i>
12	<i>Mastacembelus armatus</i>	28	<i>Channa gachua</i>
13	<i>Lepidocephalichthys guntea</i>	29	<i>Chanda nama</i>
14	<i>Labeo rohita</i>	30	<i>Barilius barila</i>
15	<i>Labeo gonius</i>	31	<i>Amblypharyngodon mola</i>
16	<i>Labeo fimbriatus</i>		

Source: Data extracted from references 1, 2,3, and 5 (CIFRI&WII)

Biodiversity of Fishes in the Upper and Middle Narmada Basin.

Data collected, analysed, and separated into the Upper and Middle Basin locations. There are 9 orders in which a total of 167 species of fish are reported in the Upper Basin of the Narmada River. The major order dominating fish diversity is Cypriniformes, with a total of 93 species, followed by Siluriformes with 34 species, and Perciformes with 17 species. The major locations from which sampling was conducted, along with the fish species recorded, are Mandla, Tilwaraghat (Jabalpur), Kareli (Narsinghpur), Jabalpur, Dindori (shared between Narsinghpur and Betul), Bargi (located on the Bargi River), and Sagar.

Similarly, in the Middle Basin, a total of 157 species are conserved in 9 orders. Here, the dominating order of fish diversity is Cypriniformes, with a total of 90 fish species, followed by Siluriformes with 34 species, and Perciformes with 16 species of fish. (Ref 4)

In both the Upper and Middle Basins, there are 114 species in common, listed in Table 28.

Table 28: Common fishes in the Upper and Middle Narmada Basin

Family	Order	Zoological Name	Family	Order	Zoological Name
Ambassidae	Perciformes	<i>Chanda baculis</i>	Cyprinidae	Cypriniformes	<i>Labeo boggut</i>
Ambassidae	Perciformes	<i>Chanda nama</i>	Cyprinidae	Cypriniformes	<i>Labeo calbasu</i>
Ambassidae	Perciformes	<i>Chanda ranga</i>	Cyprinidae	Cypriniformes	<i>Labeo dyocheilus</i>
Amblycipitidae	Siluriformes	<i>Amblyceps mangois</i>	Cyprinidae	Cypriniformes	<i>Labeo fimbriatus</i>
Anabantidae	Perciformes	<i>Anabas testudineus</i>	Cyprinidae	Cypriniformes	<i>Labeo gonius</i>
Anguillidae	Anguilliformes	<i>Anguilla bengalensis bengalensis</i>	Cyprinidae	Cypriniformes	<i>Labeo pangusia</i>
Aplocheilidae	Cyprinodontiformes	<i>Aplocheilus lineatus</i>	Cyprinidae	Cypriniformes	<i>Labeo rohita</i>
Aplocheilidae	Cyprinodontiformes	<i>Aplocheilus panchax</i>	Cyprinidae	Cypriniformes	<i>Nemacheilus beavani</i>
Badidae	Perciformes	<i>Badis badis</i>	Cyprinidae	Cypriniformes	<i>Nemacheilus botia</i>
Bagridae	Siluriformes	<i>Aorichthys seenghala</i>	Cyprinidae	Cypriniformes	<i>Nemacheilus denisoni denisoni</i>
Bagridae	Siluriformes	<i>Mystus bleekeri</i>	Cyprinidae	Cypriniformes	<i>Nemacheilus sikmaiensis</i>
Bagridae	Siluriformes	<i>Mystus cavasius</i>	Cyprinidae	Cypriniformes	<i>Oreichthys cosuatis</i>
Bagridae	Siluriformes	<i>Mystus vittatus</i>	Cyprinidae	Cypriniformes	<i>Osteobrama vigorsii</i>
Bagridae	Siluriformes	<i>Rita gogra</i>	Cyprinidae	Cypriniformes	<i>Parapsilorhynchus tentaculatus</i>
Bagridae	Siluriformes	<i>Rita rita</i>	Cyprinidae	Cypriniformes	<i>Puntius chola</i>
Bagridae	Siluriformes	<i>Sperata aor</i>	Cyprinidae	Cypriniformes	<i>Puntius conchoni</i>
Bagridae	Siluriformes	<i>Sperata seenghala</i>	Cyprinidae	Cypriniformes	<i>Puntius curmuca</i>
Balitoridae	Cypriniformes	<i>Acanthocobitis botia</i>	Cyprinidae	Cypriniformes	<i>Puntius dorsalis</i>
Balitoridae	Cypriniformes	<i>Schistura beavani</i>	Cyprinidae	Cypriniformes	<i>Puntius guganio</i>
Balitoridae	Cypriniformes	<i>Schistura dayi</i>	Cyprinidae	Cypriniformes	<i>Puntius sarana</i>
Balitoridae	Cypriniformes	<i>Schistura denisonii</i>	Cyprinidae	Cypriniformes	<i>Puntius sarana sarana</i>
Belonidae	Beloniformes	<i>Xenentodon cancila</i>	Cyprinidae	Cypriniformes	<i>Puntius sophore</i>
Belontiidae	Perciformes	<i>Colisa fasciatus</i>	Cyprinidae	Cypriniformes	<i>Puntius ticto</i>
Channidae	Perciformes	<i>Channa marulius</i>	Cyprinidae	Cypriniformes	<i>Raiamas bola</i>
Channidae	Perciformes	<i>Channa orientalis</i>	Cyprinidae	Cypriniformes	<i>Rasbora daniconius daniconius</i>

Biological Profile of Narmada Basin

Channidae	Perciformes	<i>Channa punctatus</i>	Cyprinidae	Cypriniformes	<i>Salmophasia balookee</i>
Channidae	Perciformes	<i>Channa striatus</i>	Cyprinidae	Cypriniformes	<i>Salmostoma bacaila</i>
Clariidae	Siluriformes	<i>Clarias batrachus</i>	Cyprinidae	Cypriniformes	<i>Securicula gora</i>
Clupeidae	Clupeiformes	<i>Goniolosa manmina</i>	Cyprinidae	Cypriniformes	<i>Sinilabeo dero</i>
Clupeidae	Clupeiformes	<i>Tenualosa ilisha</i>	Cyprinidae	Cypriniformes	<i>Tor khudree</i>
Cyprinidae	Cypriniformes	<i>Amblypharyngodon mola</i>	Cyprinidae	Cypriniformes	<i>Tor putitora</i>
Cyprinidae	Cypriniformes	<i>Aspidoparia morar</i>	Cyprinidae	Cypriniformes	<i>Tor tor</i>
Cyprinidae	Cypriniformes	<i>Barilius barila</i>	Heteropneustidae	Siluriformes	<i>Heteropneust es fossilis</i>
Cyprinidae	Cypriniformes	<i>Barilius bendelisis</i>	Homalopteridae	Cypriniformes	<i>Indoreonectes evezardi</i>
Cyprinidae	Cypriniformes	<i>Barilius evezardi</i>	Mastacembelidae	Synbranchiformes	<i>Macragnathus aculeatus</i>
Cyprinidae	Cypriniformes	<i>Barilius radiolatus</i>	Mastacembelidae	Synbranchiformes	<i>Macragnathus pancalus</i>
Cyprinidae	Cypriniformes	<i>Barilius shacra</i>	Mastacembelidae	Synbranchiformes	<i>Mastacembelus armatus</i>
Cyprinidae	Cypriniformes	<i>Barilius vagra</i>	Mugilidae	Perciformes	<i>Rhinomugil corsula</i>
Cyprinidae	Cypriniformes	<i>Catla catla</i>	Nandidae	Perciformes	<i>Nandus nandus</i>
Cyprinidae	Cypriniformes	<i>Chagunius chagunio</i>	Notopteridae	Osteoglossiformes	<i>Chitala chitala</i>
Cyprinidae	Cypriniformes	<i>Chela laubuca</i>	Notopteridae	Osteoglossiformes	<i>Notopterus notopterus</i>
Cyprinidae	Cypriniformes	<i>Cirrhinus cirrhosus</i>	Pangasiidae	Siluriformes	<i>Pangasius pangasius</i>
Cyprinidae	Cypriniformes	<i>Cirrhinus mrigala</i>	Schilbeidae	Siluriformes	<i>Clupisoma garua</i>
Cyprinidae	Cypriniformes	<i>Cirrhinus reba</i>	Schilbeidae	Siluriformes	<i>Eutropiichthys vacha</i>
Cyprinidae	Cypriniformes	<i>Crossocheilus latius latius</i>	Schilbeidae	Siluriformes	<i>Silonia silondia</i>
Cyprinidae	Cypriniformes	<i>Cyprinus carpio</i>	Siluridae	Siluriformes	<i>Ompok bimaculatus</i>
Cyprinidae	Cypriniformes	<i>Danio rerio</i>	Siluridae	Siluriformes	<i>Ompok pabda</i>
Cyprinidae	Cypriniformes	<i>Devario devario</i>	Siluridae	Siluriformes	<i>Wallago attu</i>
Cyprinidae	Cypriniformes	<i>Esomus danricus</i>	Sisoridae	Siluriformes	<i>Bagarius bagarius</i>
Cyprinidae	Cypriniformes	<i>Garra gotyla</i>	Sisoridae	Siluriformes	<i>Gagata cenia</i>
Cyprinidae	Cypriniformes	<i>Garra lamta</i>	Sisoridae	Siluriformes	<i>Gagata itchkeea</i>
Cyprinidae	Cypriniformes	<i>Garra mullya</i>	Sisoridae	Siluriformes	<i>Glyptothorax lonah</i>
Cyprinidae	Cypriniformes	<i>Hypophthalmichthys molitrix</i>	Sisoridae	Siluriformes	<i>Glyptothorax telchitta</i>
Cyprinidae	Cypriniformes	<i>Hypselobarbus curmuca</i>	Sisoridae	Siluriformes	<i>Gogangra viridescens</i>

Cyprinidae	Cypriniformes	<i>Labeo ariza</i>	Sisoridae	Siluriformes	<i>Laguvia ribeiroi ribeiroi</i>
Cyprinidae	Cypriniformes	<i>Labeo bata</i>	Sisoridae	Siluriformes	<i>Pseudolaguvi a ribeiroi</i>
Cyprinidae	Cypriniformes	<i>Labeo boga</i>	Syngnathidae	Anguilliformes	<i>Microphis cuncalus</i>

Source: Data extracted from reference 4 (CIFRI)

There are also species that are specific to the Basin, including 53 fish species in the Upper Narmada Basin and 43 in the Middle Narmada Basin. Checklist is provided in the Annexure (4.4.4 and 4.4.5)

Figure 13 shows the common species of both the Upper and Middle Basins and species specifically present or located in the Upper or Middle Basin.

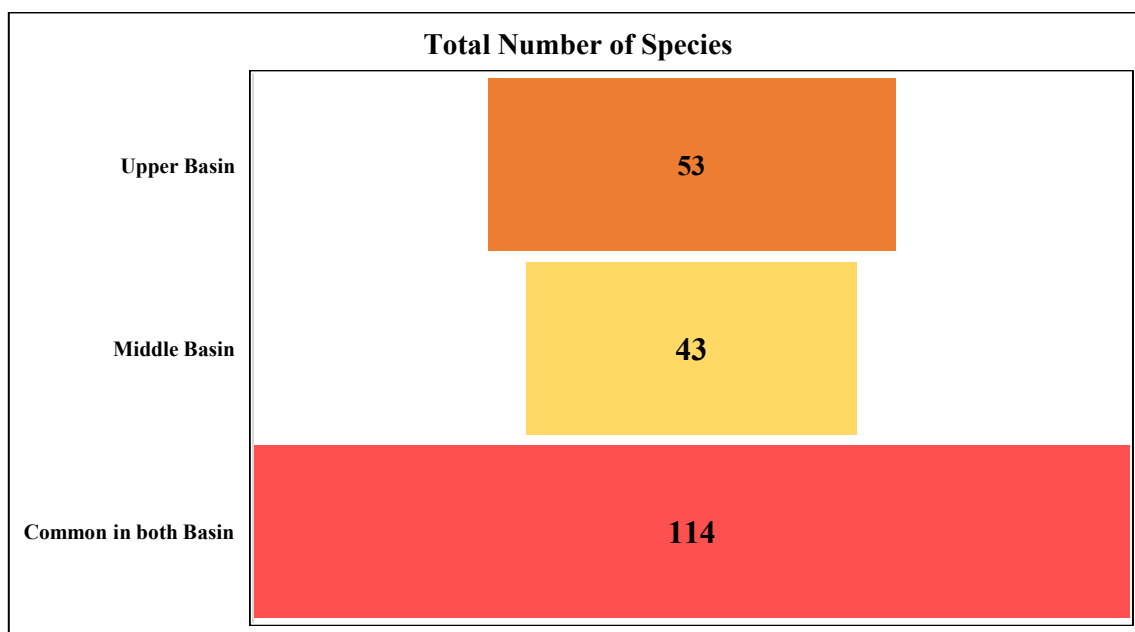


Figure 13: Number of species present in the Upper, Middle, and both the Narmada Basins.

4.1.4 Fish and Fisheries of lower narmada basin

Faunal Composition (Below Surpan to Gulf of Cambay)

The lower stretch of the Narmada River, extending from Surpan to the Gulf of Cambay, serves as a vital freshwater–estuarine transition zone. This region supports the highest taxonomic diversity of the entire river, characterized by a mix of freshwater, brackish, and marine migrant species.

Freshwater Component (Cyprinids & Catfishes): Dominant major carps include *Catla catla*, *Cirrhinus cirrhosus* (formerly *C. mrigala*), and *Labeo rohita*. Minor carps and minnows are represented by *Bangana ariza* (formerly *Cirrhinus reba*), *Puntius sarana*, and *Salmophasia bacaila*. The catfish assemblage is particularly diverse, featuring *Rita rita*, *Wallago attu*, *Sperata aor* (formerly *Aorichthys aor*), and *Sperata seenghala*.

Estuarine and Marine Migrants: The influence of tidal ingress is evident through the presence of *Tenualosa ilisha* (Hilsa), *Mugil cephalus* (Flathead mullet), *Liza parsia*, and *Lates calcarifer* (Barramundi).

Shellfish: The estuarine zone is a major hub for shellfish, primarily the giant freshwater prawn *Macrobrachium rosenbergii* and the Indian white prawn *Penaeus indicus*.

Fisheries and Market Landings

Fisheries data from key Gujarat markets (Rajpipla, Katupura, and Dandia Bazar) highlight distinct spatial shifts in catch composition.

Rajpipla Market (Upper-Lower Stretch): This market reflects a riverine fishery dominated by carps. Major carps contribute 27.76–48.23% (Avg. 37.99%) and minor carps 15.84–27.15% (Avg. 21.50%). Catfishes account for 12.00–19.60% of the landings.

Katupura Market (Estuarine Zone): Located near the estuary, this market is dominated by the Hilsa fishery, with *Tenualosa ilisha* contributing 20.43–66.74% of total landings. Shellfishes are also significant, comprising 17.38–39.53% of the catch.

Dandia Bazar (Bharuch): This market shows the strongest estuarine influence, with *T. ilisha* dominating at 32.88–69.45% and shellfishes contributing 9.53–21.07%.

Table 29: Fish Catch Composition (%) – Lower Narmada Basin

Fish Group	Percentage Contribution	Average (%)	Market Dominance
Major Carps	27.76 – 48.23	37.99	Rajpipla
Minor Carps	15.84 – 27.15	21.5	Rajpipla
Catfishes	12.00 – 19.60	15.8	Rajpipla

Tenulosa ilisha	20.43 – 69.45	34.73	Katupura / Dandia Bazar
Shellfishes	9.53 – 39.53	24.53	Katupura / Dandia Bazar
Mahseer (Tor tor)	2.53 – 7.40	4.97	Rajpipla

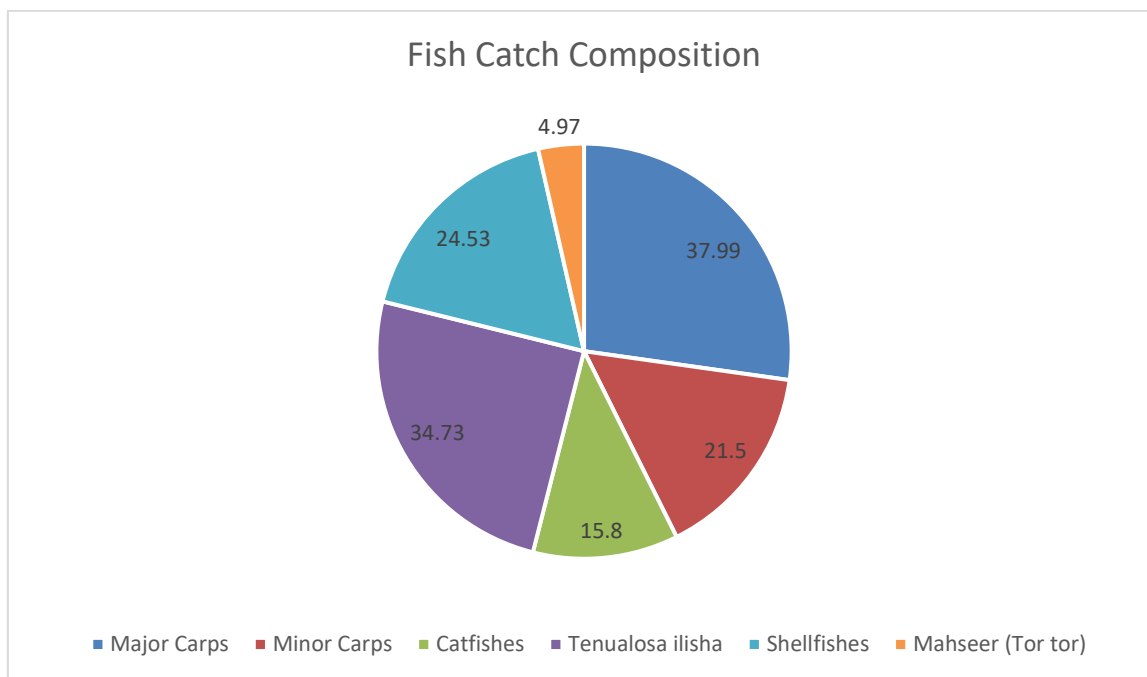


Figure 14 Fish Catch Composition

Source : Singh, S. N. (2009). *River Narmada: Its Environment and Fisheries*

Table 30 Fish Habitat

No.	Scientific Name	Former Name / Synonym	Habitat
1	Tenulosa ilisha	Hilsa ilisha	Estuarine / Anadromous
2	Cirrhinus cirrhosus	Cirrhinus mrigala	Freshwater
3	Bangana ariza	Cirrhinus reba	Freshwater
4	Sperata seenghala	Aorichthys seenghala	Freshwater / Benthic

5	Sperata aor	Aorichthys aor	Freshwater / Benthic
6	Salmophasia bacaila	Salmostoma bacaila	Freshwater
7	Laubuka laubuca	Chela laubuca	Freshwater
8	Leptomelanosoma indicum	Polynemus indicus	Marine Migrant
9	Otolithoides biauritus	Sciaenoides biauritus	Marine Migrant
10	Parambassis ranga	Chanda ranga	Freshwater

Source: Taxonomy based on Mankodi, P. C., Documentation of Inland Fishes of Gujarat, Gujarat State Biodiversity Board (GSBB).

4.2 AQUATIC AVES

The Narmada Basin supports a diverse assemblage of threatened and near-threatened waterbirds and water-associated bird species, reflecting its ecological importance as a key habitat.

The avifaunal diversity of the Narmada Basin exhibits considerable variation across its Upper, Middle, and Lower zones, collectively supporting 78 families, 250 genera, and 379 species. Species richness is highest in the Middle Zone (315 species) and lowest in the Lower Zone (43 species). Terrestrial birds dominate the assemblage, accounting for 263 species basin-wide, while 91 species are waterbirds and 14 are water-associated. Water-dependent species number 11 across the Basin. Among the Endangered species recorded are the Black-bellied Tern (*Sterna acuticauda*), Indian Skimmer (*Rynchops albicollis*), and the Great Adjutant (*Leptoptilos dubius*). Several species fall under the Vulnerable category, including the Common Pochard (*Aythya ferina*), Sarus Crane (*Grus antigone*), River Tern (*Sterna aurantia*), Lesser Adjutant (*Leptoptilos javanicus*), and Painted Stork (*Mycteria leucocephala*). The basin also harbors a significant number of Near Threatened species, such as the Oriental Darter (*Anhinga melanogaster*), Black-necked Stork (*Ephippiorhynchus asiaticus*), Asian Woollyneck (*Ciconia episcopus*), Curlew Sandpiper (*Calidris ferruginea*), River Lapwing (*Vanellus duvaucelii*), Great Thick-knee (*Esacus recurvirostris*), Common Coot (*Fulica atra*), Black-headed Ibis (*Threskiornis melanocephalus*), Grey-headed Fish-Eagle (*Ichthyophaga ichthyaetus*), and Eurasian Curlew (*Numenius arquata*). The presence of these species highlights the conservation significance of the Narmada Basin and underscores the need for continued protection of its wetland and riverine habitats. (Ref. 5).

Figure 15 shows the different species numbers in each category of Water Birds (WB), Water-associated (WA), and Water-dependent Birds (WB) in the Upper and Middle zones of the Narmada Basin. Details about the Water Birds are listed in Tables 31-35.

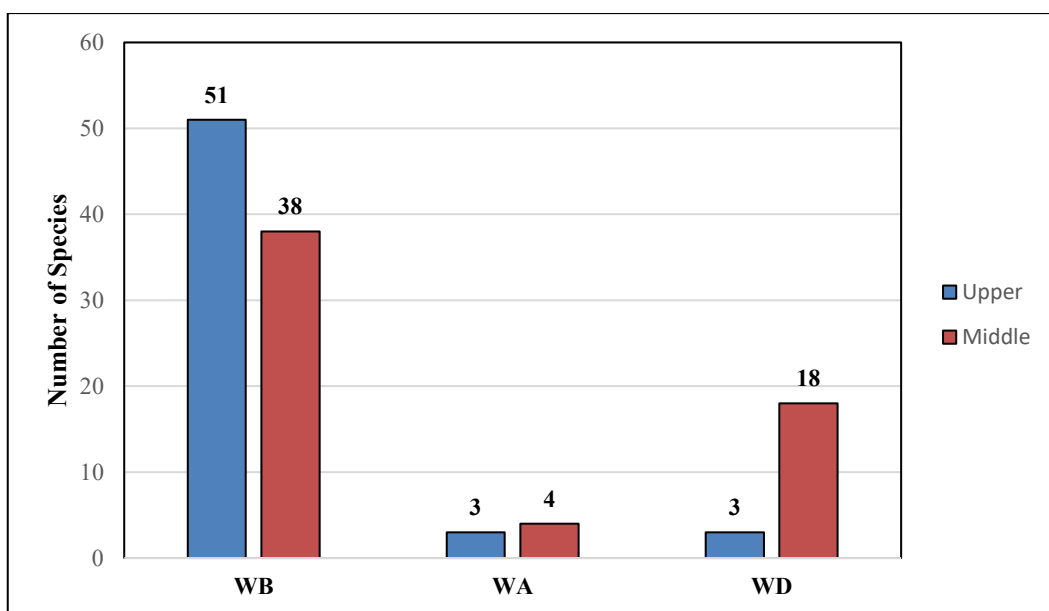


Figure 15: Different species numbers in three categories of aquatic birds. (WB, WA, WD)

Table 31: Details of WA birds of the Upper and Middle Zone.

Order	Family	Birds	Scientific Name
Anseriformes (UZ)	Anatidae	Gadwall	<i>Mareca strepera</i> (Linnaeus, 1758)
Pelicaniformes (UZ)	Ardeidae	Green-backed Heron	<i>Butorides striata</i> (Linnaeus, 1758)
Pelicaniformes (UZ)	Ardeidae	Intermediate Egret	<i>Ardea intermedia</i> (Wagler, 1829)
Anseriformes (MZ)	Anatidae	Gadwall	<i>Mareca strepera</i> (Linnaeus, 1758)
Pelicaniformes (MZ)	Ardeidae	Green-backed Heron	<i>Butorides striata</i> (Linnaeus, 1758)
Pelicaniformes (MZ)	Ardeidae	Intermediate Egret	<i>Ardea intermedia</i> (Wagler, 1829)
Charadriiformes (MZ)	Glareolidae	Indian Courser	<i>Cursorius caromandelicus</i> (Gmelin, 1789)

Source: Data extracted from reference 5(WII)

Table 32: Details of WB birds of the Upper Zone.

Order	Family	Birds	Scientific Name
Pelicaniformes	Threskiornithidae	Red naped Ibis	<i>Pseudibis papillosa</i> (Temminck, 1824)
Gruiformes	Rallidae	White-breasted Waterhen	<i>Amaurornis phoenicurus</i> (Pennant, 1769)
Charadriiformes	Charadriidae	Red-wattled Lapwing	<i>Vanellus indicus</i> (Boddaert, 1783)

Anseriformes	Anatidae	Ruddy Shelduck	<i>Tadorna ferruginea</i> (Pallas, 1764)
Anseriformes	Anatidae	Indian Spot-billed Duck	<i>Anas poecilorhyncha</i> (J.R. Forster, 1781)
Ciconiiformes	Ciconiidae	Black-necked Stork	<i>Ephippiorhynchus asiaticus</i> (Latham, 1790)
Ciconiiformes	Ciconiidae	Woolly-necked stork	<i>Ciconia episcopus</i> (Boddaert, 1783)
Pelicaniformes	Threskiornithidae	Eurasian Spoonbill	<i>Platalea leucorodia</i> (Linnaeus, 1758)
Pelicaniformes	Ardeidae	Purple Heron	<i>Ardea purpurea</i> (Linnaeus, 1766)
Pelicaniformes	Ardeidae	Cattle Egret	<i>Bubulcus ibis</i> (Linnaeus, 1758)
Pelicaniformes	Ardeidae	Little Egret	<i>Egretta garzetta</i> (Linnaeus, 1766)
Pelicaniformes	Ardeidae	Black-crowned Night-heron	<i>Nycticorax nycticorax</i> (Linnaeus, 1758)
Pelicaniformes	Ardeidae	Gray Heron	<i>Ardea cinerea</i> (Linnaeus, 1758)
Pelicaniformes	Phalacrocoraci Dae	Great Cormorant	<i>Phalacrocorax carbo</i> (Linnaeus, 1758)
Pelicaniformes	Phalacrocoraci Dae	Indian Shag	<i>Phalacrocorax fuscicollis</i> (Stephens, 1826)
Pelicaniformes	Phalacrocoraci Dae	Oriental Darter	<i>Anhinga melanogaster</i> (Pennant, 1769)
Gruiformes	Gruidae	Sarus Crane	<i>Grus antigone</i> (Linnaeus, 1758)
Charadriiformes	Charadriidae	Little Ringed Plover	<i>Charadrius dubius</i> (Scopoli, 1786)
Charadriiformes	Charadriidae	Kentish Plover	<i>Charadrius alexandrinus</i> (Linnaeus, 1758)
Charadriiformes	Scolopacidae	Common Sandpiper	<i>Actitis hypoleucos</i> (Linnaeus, 1758)
Charadriiformes	Scolopacidae	Little Stint	<i>Calidris minuta</i> (Leisler, 1812)
Charadriiformes	Laridae	River Tern	<i>Sterna aurantia</i> (J.E. Gray, 1831)
Pelicaniformes	Ardeidae	Indian Pond Heron	<i>Ardeola grayii</i> (Sykes, 1832)
Anseriformes	Anatidae	Greylag Goose	<i>Anser anser</i> (Linnaeus, 1758)
Anseriformes	Anatidae	Barheaded Goose	<i>Anser indicus</i> (Latham, 1790)
Anseriformes	Anatidae	Common Shelduck	<i>Tadorna tadorna</i> (Linnaeus, 1758)
Anseriformes	Anatidae	Comb Duck	<i>Sarkidiornis melanotos</i> (Pennant, 1769)
Anseriformes	Anatidae	Northern Pintail	<i>Anas acuta</i> (Linnaeus, 1758)
Anseriformes	Anatidae	Common Teal	<i>Anas crecca</i> (Linnaeus, 1758)
Anseriformes	Anatidae	Eurasian Wigeon	<i>Mareca Penelope</i> (Linnaeus, 1758)
Anseriformes	Anatidae	Northern Shoveler	<i>Spatula clypeata</i> (Linnaeus, 1758)
Anseriformes	Anatidae	Red-crested Pochard	<i>Netta rufina</i> (Pallas, 1773)

Anseriformes	Anatidae	Common Pochard	<i>Aythya ferina</i> (Linnaeus, 1758)
Anseriformes	Anatidae	Tufted Pochard	<i>Aythya fuligula</i> (Linnaeus, 1758)
Anseriformes	Anatidae	Cotton Pygmy-goose	<i>Nettapus coromandelianus</i> (J.F. Gmelin, 1789)
Podicipediform Es	Podicipedidae	Little Grebe	<i>Tachybaptus ruficollis</i> (Pallas, 1764)
Podicipediform Es	Podicipedidae	Great Crested Grebe	<i>Podiceps cristatus</i> (Linnaeus, 1758)
Pelicaniformes	Threskiornithidae	Black-headed ibis	<i>Threskiornis melanocephalus</i> (Latham, 1790)
Pelicaniformes	Ardeidae	Great Egret	<i>Ardea alba</i> (Linnaeus, 1758)
Gruiformes	Rallidae	Watercock	<i>Gallicrex cinerea</i> (J.F. Gmelin, 1789)
Gruiformes	Rallidae	Common Moorhen	<i>Gallinula chloropus</i> (Linnaeus, 1758)
Gruiformes	Rallidae	Purple Moorhen	<i>Porphyrio porphyrio</i> (Linnaeus, 1758)
Gruiformes	Rallidae	Common Coot	<i>Fulica atra</i> (Linnaeus, 1758)
Charadriiformes	Charadriidae	River Lapwing	<i>Vanellus duvaucelii</i> (Lesson, 1826)
Charadriiformes	Scolopacidae	Marsh Sandpiper	<i>Tringa stagnatilis</i> (Bechstein, 1803)
Charadriiformes	Scolopacidae	Wood Sandpiper	<i>Tringa glareola</i> (Linnaeus, 1758)
Charadriiformes	Scolopacidae	Curlew Sandpiper	<i>Calidris ferruginea</i> (Pontoppidan, 1763)
Charadriiformes	Scolopacidae	Dunlin	<i>Calidris alpina</i> (Linnaeus, 1758)
Charadriiformes	Laridae	Little Tern	<i>Sterna albifrons</i> (Pallas, 1764)
Charadriiformes	Laridae	Brown-headed Gull	<i>Chroicocephalus brunnicephalus</i> (Jerdon, 1840)
Anseriformes	Anatidae	Lesser Whistling Duck	<i>Dendrocygna javanica</i> (Horsfield, 1821)

Source: Data extracted from reference 5(WII)

Table 33: Details of WD birds of the Upper Zone.

Order	Family	Birds	Scientific Name
Coraciiformes	Alcedinidae	White-throated Kingfisher	<i>Halcyon smyrnensis</i> (Linnaeus, 1758)
Coraciiformes	Alcedinidae	Pied Kingfisher	<i>Ceryle rudis</i> (Linnaeus, 1758)
Coraciiformes	Alcedinidae	Common Kingfisher	<i>Alcedo atthis</i> (Linnaeus, 1758)

Source: Data extracted from reference 5(WII)

Table 34: Details of WB birds of the Middle Zone.

Order	Family	Birds	Scientific Name
Pelicaniformes	Threskiornithidae	Glossy Ibis	<i>Plegadis falcinellus</i> (Linnaeus, 1766)
Pelicaniformes	Ardeidae	Chestnut Bittern	<i>Ixobrychus cinnamomeus</i> (J.F. Gmelin, 1789)
Pelicaniformes	Phalacrocoraci Dae	Little Cormorant	<i>Microcarbo niger</i> (Vieillot, 1817)
Charadriiformes	Burhinidae	Great Thickknee	<i>Esacus recurvirostris</i> (Cuvier, 1829)
Charadriiformes	Charadriidae	Yellow-wattled Lapwing	<i>Vanellus malabaricus</i> (Boddaert, 1783)
Charadriiformes	Charadriidae	Lesser Sand Plover	<i>Charadrius mongolus</i> (Pallas, 1776)
Charadriiformes	Charadriidae	Grey Plover	<i>Pluvialis squatarola</i> (Linnaeus, 1758)
Charadriiformes	Recurvirostridae	Black-winged Stilt	<i>Himantopus himantopus</i> (Linnaeus, 1758)
Charadriiformes	Rostratulidae	Greater painted-snipe	<i>Rostratula benghalensis</i> (Linnaeus, 1758)
Charadriiformes	Jacaniidae	Pheasant-tailed Jacana	<i>Hydrophasianus chirurgus</i> (Scopoli, 1786)
Charadriiformes	Scolopacidae	Common Greenshank	<i>Tringa nebularia</i> (Gunnerus, 1767)
Charadriiformes	Scolopacidae	Common Redshank	<i>Tringa tetanus</i> (Linnaeus, 1758)
Charadriiformes	Scolopacidae	Green Sandpiper	<i>Tringa ochropus</i> (Linnaeus, 1758)
Charadriiformes	Glareolidae	Small Pratincole	<i>Glareola lactea</i> (Temminck, 1820)
Charadriiformes	Laridae	Black-bellied Tern	<i>Sterna acuticauda</i> (Gray, 1832)
Pelicaniformes	Threskiornithidae	Red naped Ibis	<i>Pseudibis papillosa</i> (Temminck, 1824)
Gruiformes	Rallidae	White-breasted Waterhen	<i>Amaurornis phoenicurus</i> (Pennant, 1769)
Charadriiformes	Charadriidae	Red-wattled Lapwing	<i>Vanellus indicus</i> (Boddaert, 1783)
Anseriformes	Anatidae	Ruddy Shelduck	<i>Tadorna ferruginea</i> (Pallas, 1764)
Anseriformes	Anatidae	Indian Spot-billed Duck	<i>Anas poecilorhyncha</i> (J.R. Forster, 1781)
Ciconiiformes	Ciconiidae	Black-necked Stork	<i>Ephippiorhynchus asiaticus</i> (Latham, 1790)
Ciconiiformes	Ciconiidae	Woolly-necked stork	<i>Ciconia episcopus</i> (Boddaert, 1783)
Pelicaniformes	Threskiornithidae	Eurasian Spoonbill	<i>Platalea leucorodia</i> (Linnaeus, 1758)
Pelicaniformes	Ardeidae	Purple Heron	<i>Ardea purpurea</i> (Linnaeus, 1766)
Pelicaniformes	Ardeidae	Cattle Egret	<i>Bubulcus ibis</i> (Linnaeus, 1758)
Pelicaniformes	Ardeidae	Little Egret	<i>Egretta garzetta</i> (Linnaeus, 1766)

Pelicaniformes	Ardeidae	Black-crowned Night-heron	<i>Nycticorax nycticorax</i> (Linnaeus, 1758)
Pelicaniformes	Ardeidae	Gray Heron	<i>Ardea cinerea</i> (Linnaeus, 1758)
Pelicaniformes	Phalacrocoraci Dae	Great Cormorant	<i>Phalacrocorax carbo</i> (Linnaeus, 1758)
Pelicaniformes	Phalacrocoraci Dae	Indian Shag	<i>Phalacrocorax fuscicollis</i> (Stephens, 1826)
Pelicaniformes	Phalacrocoraci Dae	Oriental Darter	<i>Anhinga melanogaster</i> (Pennant, 1769)
Gruiformes	Gruidae	Sarus Crane	<i>Grus antigone</i> (Linnaeus, 1758)
Charadriiformes	Charadriidae	Little Ringed Plover	<i>Charadrius dubius</i> (Scopoli, 1786)
Charadriiformes	Charadriidae	Kentish Plover	<i>Charadrius alexandrinus</i> (Linnaeus, 1758)
Charadriiformes	Scolopacidae	Common Sandpiper	<i>Actitis hypoleucos</i> (Linnaeus, 1758)
Charadriiformes	Scolopacidae	Little Stint	<i>Calidris minuta</i> (Leisler, 1812)
Charadriiformes	Laridae	River Tern	<i>Sterna aurantia</i> (J.E. Gray, 1831)
Pelicaniformes	Ardeidae	Indian Pond Heron	<i>Ardeola grayii</i> (Sykes, 1832)

Source: Data extracted from reference 5(WII)

Table 35: Details of WD birds of the Middle Zone.

Order	Family	Birds	Scientific Name
Accipitriiformes	Accipitridae	Crested Hawk Eagle	<i>Nisaetus cirrhatus</i> (Gmelin, 1788)
Accipitriiformes	Accipitridae	Grey-headed Fish Eagle	<i>Haliaeetus ichthyaetus</i> (Horsfield, 1821)
Strigiformes	Strigidae	Brown Fish Owl	<i>Ketupa zeylonensis</i> (Gmelin, 1788)
Coraciiformes	Alcedinidae	Stork-billed Kingfisher	<i>Pelargopsis capensis</i> (Linnaeus, 1766)
Coraciiformes	Meropidae	Asian Green Bee-Eater	<i>Merops orientalis</i> (Latham, 1801)
Coraciiformes	Meropidae	Blue-tailed Bee eater	<i>Merops philippinus</i> (Linnaeus, 1766)
Passeriformes	Dicruridae	Black Drongo	<i>Dicrurus macrocercus</i> (Vieillot, 1817)
Passeriformes	Rhipiduridae	White-browed Fantail	<i>Rhipidura aureola</i> (Lesson, 1830)
Passeriformes	Turdidae	Tickell's Thrush	<i>Tardus unicolor</i> (Tickell, 1833)
Passeriformes	Muscicapidae	Blue Rock-thrush	<i>Monticola solitarius</i> (Linnaeus, 1758)
Passeriformes	Motacillidae	Citrine Wagtail	<i>Motacilla citreola</i> (Pallas, 1776)

Passeriformes	Motacillidae	Grey Wagtail	<i>Motacilla cinerea</i> (Tunstall, 1771)
Passeriformes	Motacillidae	White Wagtail	<i>Motacilla alba</i> (Linnaeus, 1758)
Passeriformes	Motacillidae	White browed Wagtail	<i>Motacilla maderaspatensis</i> (Gmelin, 1789)
Coraciiformes	Coraciidae	Indian Roller	<i>Coracias benghalensis</i> (Linnaeus, 1758)
Coraciiformes	Alcedinidae	White-throated Kingfisher	<i>Halcyon smyrnensis</i> (Linnaeus, 1758)
Coraciiformes	Alcedinidae	Pied Kingfisher	<i>Ceryle rudis</i> (Linnaeus, 1758)
Coraciiformes	Alcedinidae	Common Kingfisher	<i>Alcedo atthis</i> (Linnaeus, 1758)

Source: Data extracted from reference 5(WII)

4.2.1 Higher Vertebrates

Amphibians

A total of 14 species of amphibians has been reported from the Upper Narmada Basin, listed in Table 36 with their respective categories and locations. (Ref. 6,7)

Figure 16 shows the number of species of amphibians and reptiles in the Upper and Middle Narmada Basin.

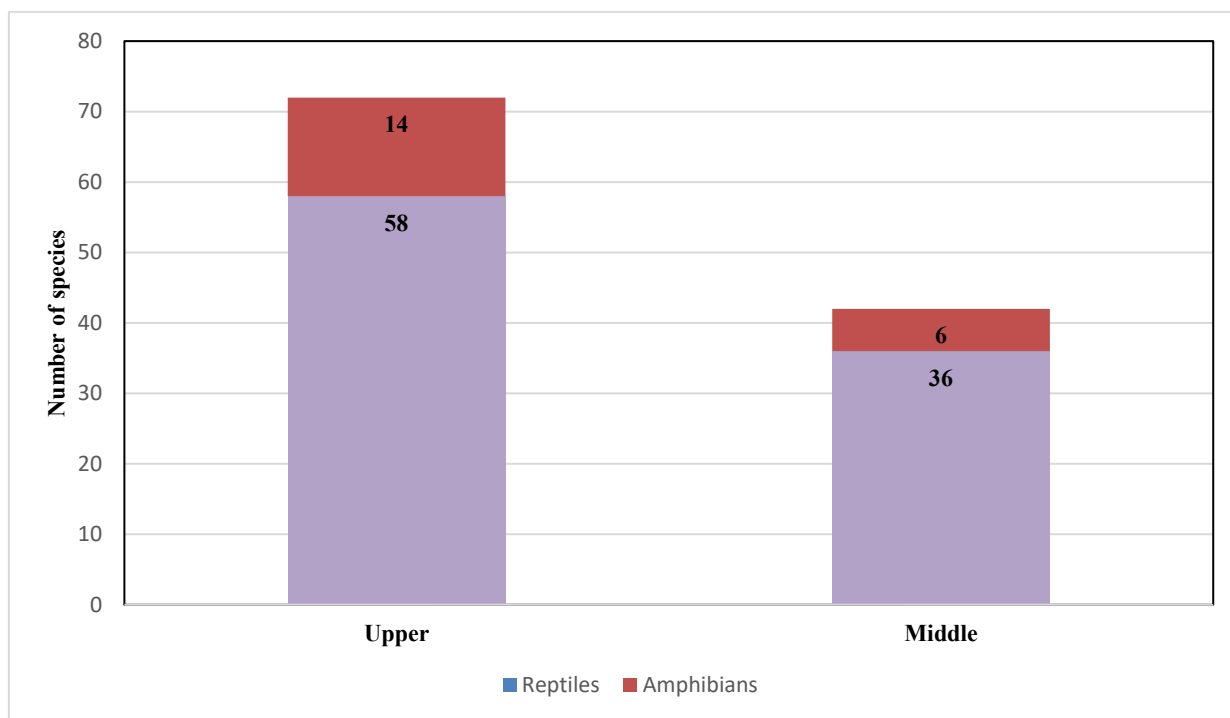


Figure 16: Number of Amphibians and Reptiles in the Upper and Middle Basin

Table 36: Details of different Amphibians found in the Upper Narmada Basin

Local Name	Zoological Name	Order	Family	Location
Common Indian Toad	<i>Bufo melanostictus</i> <i>Schneider</i>	Anura	Bufonidae	Chhindwara, Dindori, Jabalpur, Mandla, Seoni, Common throughout the plains
Marbled Toad	<i>Bufo stomaticus</i> <i>Lutken</i>	Anura	Bufonidae	Jabalpur
Gunther's Warty frog	<i>Fejervarya keralensis</i> <i>(Dubois)</i>	Anura	Ranidae	Chhindwara
Cricket frog	<i>Fejervarya limnocharis</i> <i>(Gravenhorst)</i>	Anura	Ranidae	Jabalpur, Mandla, Balaghat, Amarkantak, Seoni

Skipping frog	<i>Hoplobatrachus cyanophlyctis</i> (Schneider)	Anura	Ranidae	Jabalpur, Mandla, Damoh, Balaghat, Amarkantak, Seoni
Indian bull frog	<i>Hoplobatrachus tigerinus</i> (Daudin)	Anura	Ranidae	Dindori, Jabalpur, Mandla, Seoni, Chhindwara
Painted frog	<i>Kaloula pulchra</i> Gray	Anura	Microhylidae	Jabalpur, Balaghat, Chhindwara, Damoh, Seoni,
Ornate Microhylid frog	<i>Microhyla ornata</i> (Dumeril & Bibron)	Anura	Microhylidae	Chhindwara, Damoh, Jabalpur, Mandla, Seoni, Common throughout the plains
Sacred grove bush frog	<i>Philautus sanctisilvaticus</i> Das & Chanda	Anura	Rhacophoridae	Chhindwara
Tree frog	<i>Polypedates maculatus</i> (Gray)	Anura	Rhacophoridae	Jabalpur, Seoni, Mandla, Raisen
Fungoid frog	<i>Rana malabarica</i> (Tschudi)	Anura	Ranidae	Mandla
Burrowing frog	<i>Sphaerotheca breviceps</i> (Schneider)	Anura	Ranidae	Jabalpur, Seoni, Damoh, Amarkantak, Chhindwara
Balloon frog	<i>Uperodon globulosum</i> (Gunther)	Anura	Microhylidae	Jabalpur
Marbled Baloon frog	<i>Uperodon systoma</i> (Schneider)	Anura	Microhylidae	Seoni

Source: Data extracted from reference 6,7(ZSI)

A total of 6 species of Amphibians are reported from the Middle Narmada Basin, listed in Table 37.

Table 37: Different species of the Amphibians of the Middle Narmada Basin

Local Name	Zoological Name	Order	Family	Location
Common Indian Toad	<i>Bufo melanostictus</i> Schneider	Anura	Bufonidae	Hoshangabad, Common throughout the plains
Skipping frog	<i>Hoplobatrachus cyanophlyctis</i> (Schneider)	Anura	Ranidae	Hoshangabad, Dhar, Indore

Indian bull frog	<i>Hoplobatrachus tigerinus</i> (Daudin)	Anura	Ranidae	Hoshangabad, Dhar
Cricket frog	<i>Fejervarya limnocharis</i> (Gravenhorst)	Anura	Ranidae	Hoshangabad
Sacred grove bush frog	<i>Philautus sanctisilvaticus</i> Das & Chanda	Anura	Rhacophoridae	Hoshangabad
Tree frog	<i>Polypedates maculatus</i> (Gray)	Anura	Rhacophoridae	Hoshangabad

Source: Data extracted from reference 6,7(ZSI)

4.2.2 Reptiles

As per ZSI (Ref. 6,7), many different species are reported in the entire Narmada Basin. A total of 58 species of Reptiles are reported from the Upper Narmada Basin, listed in Table 38.

Table 38: Different species of Reptiles in the Upper Narmada Basin

Local Name	Zoological Name	Order	Family	Location
Common vine snake	<i>Ahaetulla nasutus</i> (Lacepede)	Serpentes	Colubridae	Seoni
Striped keelback	<i>Amphiesma stolatum</i> (Linnaeus)	Serpentes	Colubridae	Jabalpur, Dindori, Seoni
Banded racer	<i>Argyrogena fasciolatus</i> (Shaw)	Serpentes	Colubridae	Jabalpur, Mandla
Indian softshell turtle	<i>Aspideretes gangeticus</i> (Cuvier)	Testudines	Trionychidae	Seoni
Olivaceous keelback	<i>Atretium schistosum</i> (Daudin)	Serpentes	Colubridae	Seoni
Cat snake	<i>Boiga trigonata</i> (Schneider)	Serpentes	Colubridae	Seoni,
Common Indian krait	<i>Bungarus caeruleus</i> (Schneider)	Serpentes	Elapidae	Jabalpur, Mandla, Dindori, Seoni
Emma Gray's Forest Lizards	<i>Calotes emma</i> Gray	Testudines	Agamidae	Mandla
Common garden lizard	<i>Calotes versicolor</i> (Daudin)	Testudines	Agamidae	Balaghat, Dindori, Jabalpur, Seoni,

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Indian Chamaeleon	<i>Chamaeleo zeylanicus Laurenti</i>	Testudines	Chamaeleonidae	Chhindwara, Jabalpur, Narsinhapur, Seoni
Mugger or Marsh Crocodile	<i>Crocodylus palustris Lesson</i>	Crocodylia	Crocodylidae	Jabalpur, Seoni
Collegal Ground gecko	<i>Cyrtodactylus collegalensis (Beddome)</i>	Testudines	Gekkonidae	Mandla
Clouded Ground gecko	<i>Cyrtodactylus nebulosus (Beddome)</i>	Testudines	Gekkonidae	Mandla, Balaghat
Russell's viper	<i>Daboia russellii (Shaw)</i>	Serpentes	Viperidae	Jabalpur, Anuppur
Common bronzeback tree snake	<i>Dendrelaphis tristis (Daudin)</i>	Serpentes	Colubridae	Dindori, Jabalpur, Seoni
Saw scaled viper	<i>Echis carinatus (Schneider)</i>	Serpentes	Viperidae	Jabalpur, Seoni
Indian Trinket snake	<i>Elaphe helena (Daudin)</i>	Serpentes	Colubridae	Jabalpur, Seoni
Copper-headed Trinket Snake	<i>Elaphe radiata (Schegel)</i>	Serpentes	Colubridae	Balaghat, Mandla
Common sand boa	<i>Eryx conicus (Schneider)</i>	Serpentes	Boidae	Jabalpur, Seoni
Red sand boa	<i>Eryx johani (Russell)</i>	Serpentes	Boidae	Seoni
Common Fat-tailed gecko	<i>Eublepharis hardwickii Gray</i>	Testudines	Eublepharidae	Hoshangabad, Umari
Kollegal ground gecko	<i>Geckoella collegalensis (Beddome)</i>	Testudines	Gekkonidae	Mandla
Clouded ground gecko	<i>Geckoella nabulosus (Beddome)</i>	Testudines	Gekkonidae	Mandla
Indian star tortoise	<i>Geochelone elegans (Schoepff)</i>	Testudines	Testudinidae	Central India
Brook's House gecko	<i>Hemidactylus brooki Gray</i>	Testudines	Gekkonidae	Balaghat, Dindori, Jabalpur, Seoni
Yellow-Green House gecko	<i>Hemidactylus flaviviridis Ruppell</i>	Testudines	Gekkonidae	Balaghat, Jabalpur
Slender gecko	<i>Hemidactylus gracilis Blandford</i>	Testudines	Gekkonidae	Mandla
Bark gecko	<i>Hemidactylus leschenaulti Dumeril & Bibron</i>	Testudines	Gekkonidae	Jabalpur

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Indian termite hill gecko, Giant leaf-toed gecko	<i>Hemidactylus subtriedrus</i> (Daudin)	Testudines	Gekkonidae	Seoni
Three-striped roofed turtle	<i>Kachuga dhongoka</i> (Gray)	Testudines	Bataguridae	Entire Basin
Lesser Agama	<i>Laudakia minor</i> (Hardwicke & Gray)	Testudines	Agamidae	Jabalpur, Sagar
Indian flapshell turtle	<i>Lissemys punctata punctata</i> (Lacepede)	Testudines	Trionychidae	Narmadapuram, Seoni
Common Wolf snake	<i>Lycodon aulicus</i> (Linnaeus)	Serpentes	Colubridae	Balaghat, Dindori, Seoni
Travancore wolf snake	<i>Lycodon tranvancoricus</i> (Beddome)	Serpentes	Colubridae	Jabalpur
Snake skink	<i>Lygosoma albopunctata</i> (Gray)	Testudines	Scincidae	Balaghat
Lined supple skink	<i>Lygosoma lineata</i> (Gray)	Testudines	Scincidae	Jabalpur
Spotted supple skink	<i>Lygosoma punctatus</i> (Gmelin)	Testudines	Scincidae	Jabalpur
Common skink	<i>Mabuya carinata</i> (Schneider)	Testudines	Scincidae	Dindori, Jabalpur, Mandla, Seoni
Striped Grass skink	<i>Mabuya dissimilis</i> (Hallowell)	Testudines	Scincidae	Jabalpur, Katni
Bronze Grass skink	<i>Mabuya macularia</i> (Blyth)	Testudines	Scincidae	Jabalpur, Mandla
Green keelback	<i>Macropisthodon plumbicolor</i> (Cantor)	Serpentes	Colubridae	Seoni
Peninsular black turtle	<i>Melanochelys trijuga trijuga</i> (Schweigger)	Testudines	Bataguridae	Seoni
Common cobra	<i>Naja naja</i> (Linnaeus)	Serpentes	Elapidae	Jabalpur, Mandla, Seoni
Banded kukri snake	<i>Oligodon arnensis</i> (Shaw)	Serpentes	Colubridae	Jabalpur
Streaked Kukri snake	<i>Oligodon taeniolatus</i> (Jerdon)	Serpentes	Colubridae	Jabalpur, Mandla
Snake eyed lacerta	<i>Ophisops jerdoni</i> Blyth	Testudines	Lacertidae	Jabalpur
Blanford's Agama	<i>Psammophilus blanfordanus</i> (Stoliczka)	Testudines	Agamidae	Balaghat, Dindori, Jabalpur, Seoni
Rat snake	<i>Ptyas mucosus</i> (Linnaeus)	Serpentes	Colubridae	Dindori, Jabalpur, Seoni, Mandla

Indian python	<i>Python molurus (Linnaeus)</i>	Serpentes	Boidae/ Phythonidae	Dindori, Mandla, Seoni
Brahminy worm snake	<i>Ramphotyphlops braminus (Daudin)</i>	Serpentes	Typhlopidae	Balaghat, Dindori, Seoni, Jabalpur
White-spotted supple shink	<i>Riopa albopunctata Gray</i>	Testudines	Scincidae	Balaghat
Cantor's Black- headed snake	<i>Sibynophis sagittarius Cantor</i>	Serpentes	Colubridae	Mandla
Fan throated lizard	<i>Sitana ponticeriana Cuvier</i>	Testudines	Agamidae	Balaghat, Dindori, Jabalpur, Seoni
Bamboo Pit Viper	<i>Trimeresurus gramineus (Shaw)</i>	Serpentes	Viperidae	Jabalpur
Beaked worm snake	<i>Typhlops acutus (Dumeril & Bibron)</i>	Serpentes	Typhlopidae	Seoni
Common Indian Monitor	<i>Varanus bengalensis (Daudin)</i>	Testudines	Varanidae	Mandla, Seoni, Dindori, Jabalpur
Russell's viper	<i>Vipera russelli (Shaw)</i>	Serpentes	Viperidae	Jabalpur, Mandla Dindori, Seoni
Checkered keelback water snake	<i>Xenochrophis piscator (Schneider)</i>	Serpentes	Colubridae	Jabalpur, Seoni

Source: Data extracted from references 6,7, (ZSI)

A total of 36 species of Reptiles are reported from the Middle Narmada Basin, listed in Table 39.

Table 39: Different species of Reptiles in the Middle Narmada Basin

Local Name	Zoological Name	Order	Family	Location
Common vine snake	<i>Ahaetulla nasutus (Lacepede)</i>	Serpentes	Colubridae	Dewas, Indore
Striped keelback	<i>Amphiesma stolatum (Linnaeus)</i>	Serpentes	Colubridae	Dewas, Dhar, Indore
Banded racer	<i>Argyrogena fasciolatus (Shaw)</i>	Serpentes	Colubridae	Dewas, Indore
Olivaceous keelback	<i>Atretium schistosum (Daudin)</i>	Serpentes	Colubridae	Dewas, Dhar, Indore

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Cat snake	<i>Boiga trigonata</i> (Schneider)	Serpentes	Colubridae	Dewas, Dhar, Indore
Common Indian krait	<i>Bungarus caeruleus</i> (Schneider)	Serpentes	Elapidae	Dewas, Dhar, Indore
Slender coral snake	<i>Calliophis melanurus</i> (Shaw)	Serpentes	Elapidae	Dhar
Common garden lizard	<i>Calotes versicolor</i> (Daudin)	Testudines	Agamidae	Hoshangabad
Painted bronzeback tree snake	<i>Dendrelaphis pictus</i> (Gmelin)	Serpentes	Colubridae	Dewas
Saw scaled viper	<i>Echis carinatus</i> (Schneider)	Serpentes	Viperidae	Dhar
Indian Trinket snake	<i>Elaphe helena</i> (Daudin)	Serpentes	Colubridae	Dewas, Dhar, Indore
Common sand boa	<i>Eryx conicus</i> (Schneider)	Serpentes	Boidae	Dewas, Dhar, Indore
Red sand boa	<i>Eryx johani</i> (Russell)	Serpentes	Boidae	Dewas, Dhar, Indore
Common Fat-tailed gecko	<i>Eublepharis hardwickii</i> Gray	Testudines	Eublepharidae	Hoshangabad
Indian termite hill gecko	<i>Hemidactylus subtriadrus</i> (Daudin)	Testudines	Gekkonidae	Indore
Three-striped roofed turtle	<i>Kachuga dhongoka</i> (Gray)	Testudines	Bataguridae	Entire Basin
Indian flapshell turtle	<i>Lissemys punctata punctata</i> (Lacepede)	Testudines	Trionychidae	Hoshangabad
Common Wolf snake	<i>Lycodon aulicus</i> (Linnaeus)	Serpentes	Colubridae	Dewas, Dhar, Indore
Shaw's Wolf Snake	<i>Lycodon striatus</i> (Shaw)	Serpentes	Colubridae	Dewas
Common skink	<i>Mabuya carinata</i> (Schneider)	Testudines	Scincidae	Hoshangabad
Bronze Grass skink	<i>Mabuya macularia</i> (Blyth)	Testudines	Scincidae	Hoshangabad
Green keelback	<i>Macropisthodon plumbicolor</i> (Cantor)	Serpentes	Colubridae	Dewas, Indore
Common cobra	<i>Naja naja</i> (Linnaeus)	Serpentes	Elapidae	Dewas, Dhar, Indore
Black cobra	<i>Naja naja oxiana</i> (Eichwald)	Serpentes	Elapidae	Dewas, Dhar, Indore

Banded kukri snake	<i>Oligodon arnensis</i> (Shaw)	Serpentes	Colubridae	Dewas, Dhar, Indore
Rat snake	<i>Ptyas mucosus</i> (Linnaeus)	Serpentes	Colubridae	Dewas, Dhar
Indian python	<i>Python molurus</i> (Linnaeus)	Serpentes	Boidae	Dewas, Dhar, Indore
Brahminy worm snake	<i>Ramphotyphlops braminus</i> (Daudin)	Serpentes	Typhlopidae	Dewas, Indore, Dhar, Hoshangabad
-	<i>Sibynophis subpunctatus</i> (Dumerill & Bibron)	Serpentes	Colubridae	Dhar
Fan throated lizard	<i>Sitana ponticeriana</i> Cuvier	Testudines	Agamidae	Hoshangabad
Bamboo Pit Viper	<i>Trimeresurus gramineus</i> (Shaw)	Serpentes	Viperidae	Hoshangabad
Diard's Blind snake	<i>Typhlops diardi</i> (Schlegel)	Serpentes	Typhlopidae	Dewas, Dhar, Indore
Hardwick's Spiny-tailed Lizard	<i>Uromastix hardwickii</i> Gray	Testudines	Agamidae	Hoshangabad
Ocellate shield tail	<i>Uropeltis ocellatus</i> (Beddome)	Serpentes	Uropeltidae	Hoshangabad
Russell's viper	<i>Vipera russelli</i> (Shaw)	Serpentes	Viperidae	Dewas, Dhar,
Checkered keelback water snake	<i>Xenochrophis piscator</i> (Schneider)	Serpentes	Colubridae	Dewas, Dhar, Indore

Source: Data extracted from references 6,7(ZSI)

A total of 14 species of Reptiles are reported from the Lower Narmada Basin, listed in Table 40.

Table 40: Detail of Reptiles in the Lower Narmada Basin:

Local Name	Zoological Name	Order	Family	Location
Striped keelback	<i>Amphiesma stolatum</i> (Linnaeus)	Serpentes	Colubridae	Jhabua
Common Indian krait	<i>Bungarus caeruleus</i> (Schneider)	Serpentes	Elapidae	Jhabua
Saw scaled viper	<i>Echis carinatus</i> (Schneider)	Serpentes	Viperidae	Jhabua
Common sand boa	<i>Eryx conicus</i> (Schneider)	Serpentes	Boidae	Jhabua
Three-striped roofed turtle	<i>Kachuga dhongoka</i> (Gray)	Testudines	Bataguridae	Entire Basin

Common Wolf snake	<i>Lycodon aulicus (Linnaeus)</i>	Serpentes	Colubridae	Jhabua
Common cobra	<i>Naja naja (Linnaeus)</i>	Serpentes	Elapidae	Jhabua
Black cobra	<i>Naja naja oxiana (Eichwald)</i>	Serpentes	Elapidae	Jhabua
Rat snake	<i>Ptyas mucosus (Linnaeus)</i>	Serpentes	Colubridae	Jhabua
Indian python	<i>Python molurus (Linnaeus)</i>	Serpentes	Boidae	Jhabua
Brahminy worm snake	<i>Ramphotyphlops braminus (Daudin)</i>	Serpentes	Typhlopidae	Jhabua
Diard's Blind snake	<i>Typhlops diardi (Schlegel)</i>	Serpentes	Typhlopidae	Jhabua
Russell's viper	<i>Vipera russelli (Shaw)</i>	Serpentes	Viperidae	Jhabua
Checkered keelback water snake	<i>Xenochrophis piscator (Schneider)</i>	Serpentes	Colubridae	Jhabua

Source: Data extracted from references 6,7 (ZSI)

Chapter 5: Biological Diversity in Narmada River Tributaries

This chapter presents an integrated overview of biological diversity in the tributaries of the Narmada River across the upper, middle, and lower reaches of the basin. The tributary systems contribute significantly to the ecological functioning of the river by supporting diverse aquatic communities, regulating nutrient dynamics, and providing critical habitats for breeding, feeding, and migration of biota. The biological composition of these tributaries varies longitudinally in response to changes in hydrology, geomorphology, flow regulation, land use, and salinity influence, resulting in distinct freshwater assemblages in the upper and middle reaches and more complex river–estuarine communities in the lower basin. The chapter focuses primarily on macrobenthos and fish communities, which serve as key bio-indicators of environmental quality and ecosystem health, and together provide a basin-scale understanding of biodiversity patterns, ecological connectivity, and conservation significance within the Narmada River system.

5.1 UPPER AND MIDDLE NARMADA REACHES

5.1.1 Macrobenthos

Macrobenthos data are available for the current period for both the Upper and Middle Basins. In the Upper Basin, four classes and 26 species were recorded at the Dudhi River, located on the left bank and one of the major tributaries of the Narmada River, which flows through the Chhindwara and Narsingpur districts. In the Middle Basin, there are four classes, and 50 species have been recorded at various locations, including Ganjal, Tawa, and the Bhagner River of the Middle Narmada Basin (Ref. 8-11). Checklist for macrobenthos tributaries data is in Annexure (5.1.1.1 and 5.1.1.2)

5.1.2 Fish

In the current zone, there are 10 families of fish and 30 species found in the tributaries of the Narmada River in the Upper Narmada Basin, shown in Figure 17. The fish are primarily found in the Dudhi, Bargi, Barna, and Tendwa rivers, which are tributaries of the Narmada River. Checklist of the Fishes is in the Annexure (5.1.2.1)

In the Middle Narmada Basin, the major locations where fish are found are the Chakaldi (Kolar River), Choral (Choral River), Harda (Anchal River), Sehore (Sip River), and Dewas (Jamner River) in the Sehore district. Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa, and Palakmati from Satpura and Kolar, Bhagner, Gadaria, and Juanri from the Vindhya ranges. In the intermediate zone, 13 families and 54 species were recorded. In the current scenario, there are 18 families, and a total of 87 fish species present in the tributaries of the Middle Narmada Basin, as illustrated in Figure 18 (Ref. 12-19). Checklist of the fish, location given in the Annexure (5.1.2.2, 5.1.2.3)

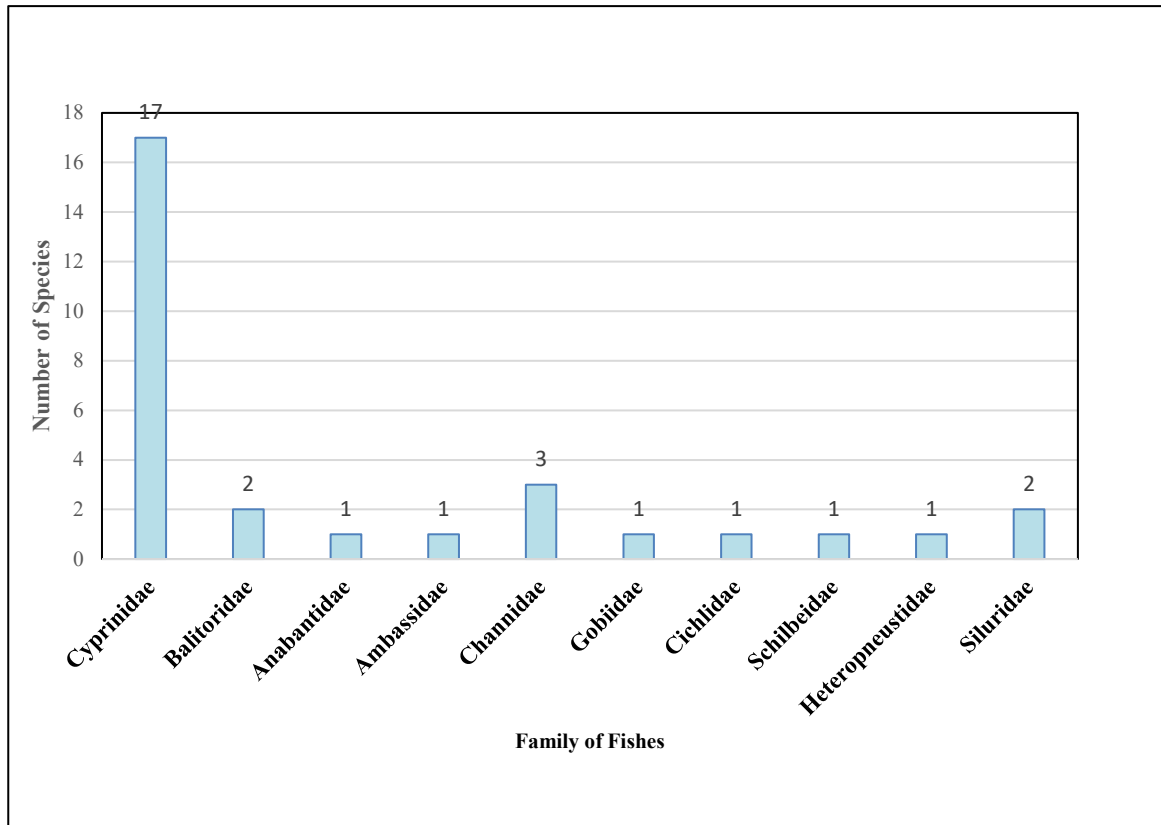


Figure 17: Different classes and species of fish in the different tributaries of the Upper Narmada Basin.

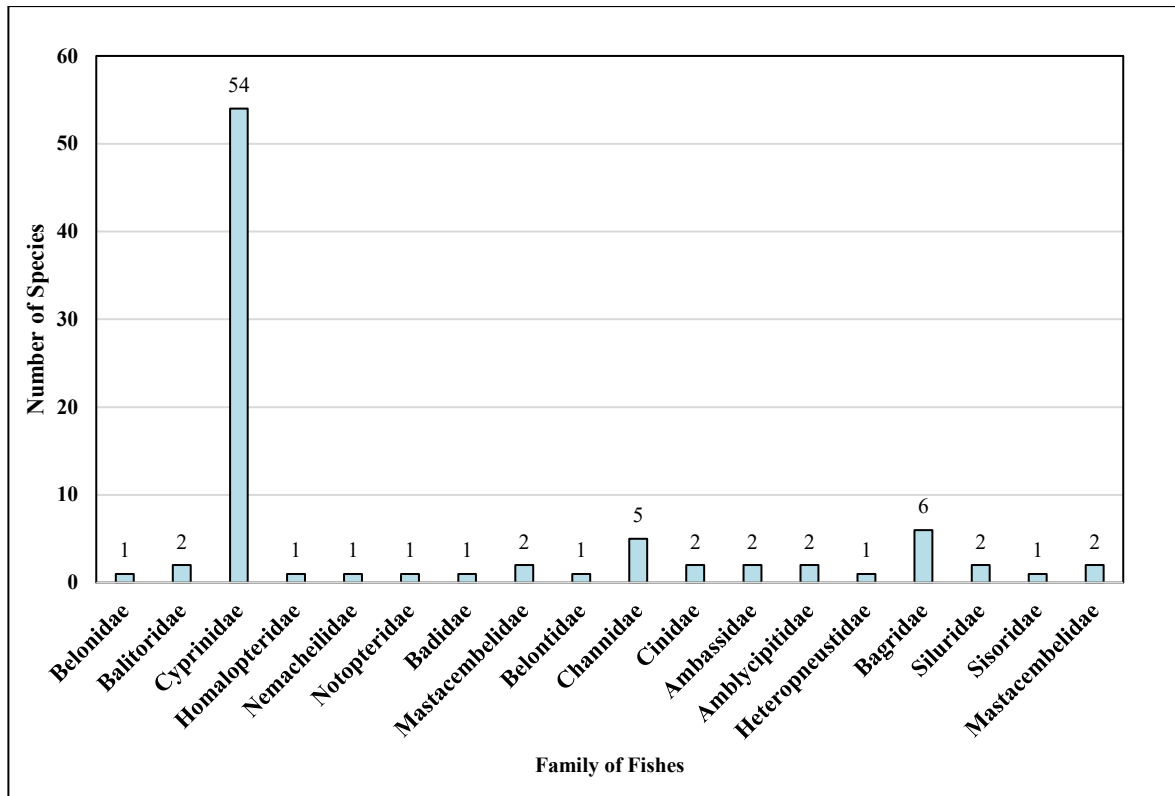


Figure 18: Different classes and species of fish in the different tributaries of the Middle Narmada Basin.

5.2 LOWER NARMADA REACH

The ecological architecture of the Lower Narmada River Basin constitutes a complex, highly dynamic hydro-biological system acting as a critical ecotone—a transition zone between the freshwater catchment of Central India and the marine expanse of the Arabian Sea. Unlike the rheophilic (fast-flowing) upper reaches, this lower basin (from Kevadia to the Gulf of Khambhat) is governed by the interplay of regulated freshwater discharge and semi-diurnal tidal ingress. This unique hydrological mixing creates a continuum of habitats, ranging from freshwater lotic systems in principal tributaries (Orsang, Karjan, Heran, Men, and Bhukhi) to oligohaline transition zones and polyhaline estuarine mudflats.

This chapter focuses on two primary biotic communities selected as key bio-indicators:

- **Macrobenthos:** Functioning as "historical archivists" of river health, their sessile nature allows them to integrate sediment quality and localized pollution events over time.
- **Fish:** Including migratory species like Hilsa and Mahseer, which reflect the longitudinal connectivity and hydrological integrity of the system.

5.2.1 Macrobenthos

Macrobenthic invertebrates, organisms that inhabit the sediment-water interface, serve as robust bio-indicators of long-term environmental quality in riverine systems. Their sessile or limited-mobility lifestyle makes them particularly sensitive to localized changes in sediment texture, organic enrichment, and salinity gradients. In the Lower Narmada Basin, the distribution and abundance of macrobenthos are governed by the flow regime released from the Sardar Sarovar Dam and the sediment loads contributed by tributaries like the Orsang and Karjan.

The assessment of macrobenthic diversity reveals significant spatial and temporal heterogeneity across the basin. The community structure shifts dramatically from freshwater-dominated assemblages in the upstream reaches (Kevadia to Poicha) to resilient, stress-tolerant communities in the saline-influenced downstream zones (Bharuch to Ambetha). The analysis below details these patterns across three distinct seasons.

Seasonal Abundance and Community Structure

The seasonal dynamics of macrobenthos in the Lower Narmada are characterized by cycles of colonization, disturbance (flush-out), and recovery. The total abundance of organisms ranges widely, from complete absence in highly stressed zones to prolific densities exceeding 5,000 individuals per liter in nutrient-rich habitats.

Pre-Monsoon (Summer) Season

The summer season, characterized by low but stable freshwater flows and stabilized riverbed sediments, supports the highest biomass and density of benthic organisms in the middle reaches of the Lower Basin. During this period, the riverbed is relatively undisturbed by high-velocity flows, allowing for the establishment of complex benthic communities.

Station-wise Analysis:

- **Kevadia:** Located immediately downstream of the Sardar Sarovar Dam, this station represents a modified lotic habitat influenced by reservoir tailwaters. The total macrobenthic abundance was recorded at 222 nos. per liter. The community exhibited a relatively balanced structure indicative of diverse microhabitats:
 - Mollusca (45.95%; 102 nos. per liter): The dominance of mollusks suggests stable, calcium-rich substrates.
 - Annelida (33.33%; 74 nos. per liter): A significant presence of oligochaetes indicates organic processing capabilities.
 - Diptera (16.67%; 37 nos. per liter): The presence of Dipteran larvae (likely Chironomidae) reflects the availability of soft sediments and slower water velocities typical of regulated flows.
 - Ostracoda (4.05%; 9 nos. per liter): These crustaceans contribute to the micro-benthic diversity.
- **Poicha (Orsang Confluence):** Situated near the confluence of the Orsang River, Poicha recorded a higher total abundance of 418 nos. per liter. The community was overwhelmingly dominated by Mollusca (77.76%; 325 nos. per liter), with Annelida constituting the remaining 22.24% (93 nos. per liter). The high density of mollusks at this station is likely driven by the sediment inputs from the Orsang tributary, which provides sandy-loam substrates ideal for gastropods and bivalves. The absence of insect larvae suggests a habitat preference here for hard-shelled invertebrates over soft-bodied insects.

- **Sisodara:** Moving downstream, the station at Sisodara displayed a further increase in benthic density, reaching 985 nos. per liter. The dominance of Mollusca intensified, accounting for 86.80% (855 nos. per liter) of the total population. Annelida comprised 13.20% (130 nos. per liter). This trend suggests a longitudinal continuum of stable, productive benthic habitats extending from the tributary confluences into the middle river segment.
- **Jhanor:** This station recorded the peak benthic abundance for the entire basin during summer, with a massive density of 5,158 nos. per liter.
 - Mollusca (80.91%; 4,173 nos. per liter): The hyper-abundance of mollusks at Jhanor indicates a zone of high primary productivity and organic deposition, possibly supported by nutrient inputs from surrounding agricultural lands.
 - Annelida (19.09%; 985 nos. per liter): While numerically significant, annelids were secondary to the prolific mollusk population. This site represents a "hotspot" of benthic productivity in the Lower Narmada.
- **Bharuch:** As the river enters the estuarine transition zone, a profound shift in community structure was observed. Total abundance was 1,311 nos. per liter, but the composition inverted completely compared to upstream sites.
 - Annelida (98.55%; 1,292 nos. per liter): The community was almost exclusively composed of annelids (likely Tubificidae). This dominance is a classic indicator of organic enrichment and hypoxic sediment conditions, often associated with urban discharge or industrial effluents in the estuarine mixing zone.
 - Mollusca (1.45%; 19 nos. per liter): The sharp decline in mollusks suggests that the substrate or salinity regime at Bharuch during summer is unsuitable for the freshwater gastropod assemblages found upstream.
- **Bhadbhut:** Located further into the estuary, this station showed signs of salinity stress. Total abundance dropped to 120 nos. per liter. The community was mixed, with Annelida (61.67%; 74 nos. per liter), Mollusca (30.83%; 37 nos. per liter), and the appearance of Nematoda (7.50%; 9 nos. per liter). The presence of nematodes is often associated with meiobenthic communities in brackish environments.
- **Ambetha:** The most downstream station, situated near the river mouth, not recorded organisms. This absence is attributed to the harsh environmental conditions during the

pre-monsoon period, where high salinity intrusion and strong tidal scouring create an unstable benthic environment that prevents colonization.

Monsoon Season

The onset of the monsoon brings high discharge volumes from the Sardar Sarovar Dam and the tributaries. This hydraulic force scours the riverbed, increases turbidity, and alters sediment composition, leading to a general "wash-out" of benthic populations in exposed areas, while favoring burrowing taxa in sheltered zones.

Station-wise Analysis:

- **Kevadia:** Contrary to the downstream trend, Kevadia exhibited a significant increase in abundance to 1,682 nos. per liter. The community became almost mono-specific, dominated by Mollusca (98.94%; 1,664 nos. per liter).
 - Ecological Insight: This resilience suggests that the regulated release from the dam might create a hydraulic refuge or a "shadow zone" immediately downstream, where mollusks are protected from the full force of the flood pulse. The low counts of Annelida and Diptera (0.53% each) indicate that soft sediments required by these groups may have been displaced.
- **Jhanor:** The community structure at Jhanor underwent a complete inversion compared to summer. While total abundance remained high at 3,550 nos. per liter, Annelida surged to dominance, comprising 80.62% (2,862 nos. per liter) of the population. Mollusca declined to 18.59% (660 nos. per liter).
 - Ecological Insight: This shift highlights the resilience of annelids to physical disturbance. Their burrowing capability allows them to withstand bed-load movement better than surface-dwelling mollusks during high flows.
- **Poicha & Sisodara:** These intermediate stations showed reduced abundances (83 nos. per liter and 232 nos. per liter respectively). Poicha maintained a 100% mollusk community, likely due to the continued influence of the Orsang River's rocky/sandy substrate. Sisodara showed a mixed recovery with Mollusca (71.98%), Diptera (15.95%), and Annelida (12.07%).
- **Estuarine Zone (Bharuch to Ambetha):** The monsoon flows effectively flushed out the estuarine benthos.

- Bharuch: Abundance dropped to 74 nos. per liter, dominated by Annelida (75.68%).
- Shakkarpura: Recorded 46 nos. per liter, with a mix of Annelida (60.87%) and Mollusca (39.13%).
- Bhadbhut & Ambetha: Both stations supported minimal populations of 9 nos. per liter, consisting entirely of Annelida (100%). This reflects the extreme instability of the estuarine bed during flood discharge.

Post-Monsoon (Winter) Season

The winter season marks a period of hydrological stabilization and biological recovery. As flows recede and turbidity decreases, benthic habitats are recolonized, leading to increased diversity and abundance across the basin.

Station-wise Analysis:

- **Sisodara:** This station emerged as the most diverse and abundant site during winter, with a total count of 1,245 nos. per liter. The community composition reflects a complex recovery:
 - Annelida (56.71%; 706 nos. per liter)
 - Mollusca (38.79%; 483 nos. per liter)
 - Diptera (2.25%; 28 nos. per liter)
 - Odonata (1.53%; 19 nos. per liter): The appearance of Odonata nymphs (dragonflies/damselflies) is a critical bio-indicator of improved water quality and the presence of riparian vegetation or macrophytes.
 - Ostracoda (0.72%; 9 nos. per liter).
- **Jhanor:** Continued to support high biomass (1,124 nos. per liter), with Annelida (76.87%) retaining dominance over Mollusca (23.13%). The persistence of annelids here suggests sustained organic inputs in this reach.
- **Estuarine Recovery (Bharuch, Bhadbhut, Mahegam, Ambetha):** The estuarine zone showed clear signs of recolonization, primarily by annelids.
 - Bharuch: Total abundance recovered to 186 nos. per liter, dominated by Annelida (84.95%). Notably, this season saw the appearance of Ephemeroptera (Mayfly larvae; 4.84%), which are sensitive to pollution, suggesting a seasonal improvement in water quality.

- Bhadbhut: Recorded 37 nos. per liter, with a significant presence of Malacostraca (75.68%; 28 nos. per liter). This indicates the ingress of estuarine crustaceans (prawns, shrimps, crabs) into the river channel, linking benthic productivity to the fisheries potential.
- Mahegam & Ambetha: Supported 539 and 186 nos. per liter respectively, exclusively composed of Annelida. This confirms that in the lower estuary, polychaetes/oligochaetes are the primary pioneer colonizers of the sediment.

Taxonomic Group Profiles and Functional Roles

The macrobenthic community of the Lower Narmada is structured around three primary phyla, each fulfilling distinct ecological roles within the riverine food web.

1. Phylum Mollusca

- **Composition:** The molluscan assemblage is the defining feature of the freshwater reaches (Kevadia to Jhanor). It includes Gastropoda (snails) and Bivalvia (clams). Dominant families likely include Thiaridae, Viviparidae, and Corbiculidae.
- **Ecological Role:** As filter feeders (bivalves) and grazers/scrapers (gastropods), they play a crucial role in benthic-pelagic coupling, transferring energy from phytoplankton and periphyton to higher trophic levels such as benthivorous fish (*Labeo* spp.) and birds. Their shell formation also contributes to the calcium cycle within the basin. The high abundance at Poicha suggests the Orsang River contributes calcium-rich waters or suitable granulometry for shell development.

2. Phylum Annelida

- **Composition:** Annelids are the most resilient and ubiquitous group, dominating the downstream and organically enriched zones. The community comprises freshwater Oligochaeta (*Tubifex*, *Limnodrilus*) in the Bharuch area and marine/estuarine Polychaeta (*Neanthes*, *Dendronereis*) in the Bhadbhut-Ambetha stretch.
- **Ecological Role:** Functioning primarily as deposit feeders, annelids process organic detritus within the sediment. Their hyper-dominance at Bharuch (near 98% in summer) serves as a bio-indicator of organic loading, likely from anthropogenic sources, and hypoxic sediment conditions where other taxa cannot survive.

3. Phylum Arthropoda

- **Composition:** Although numerically less dominant than mollusks or annelids, arthropods contribute significantly to functional diversity.
 - Insecta: Represented by Diptera (Chironomid larvae), Odonata (dragonfly nymphs), and Ephemeroptera (mayflies). These taxa are largely restricted to freshwater zones and are seasonally pulsed.
 - Crustacea: Includes Ostracoda and Malacostraca (prawns/crabs). The presence of Malacostraca at Bhadbhut is ecologically significant, marking the nursery grounds for commercially important prawn species like *Macrobrachium* and *Penaeus*.

5.2.2 Fish

The ichthyofaunal diversity of the Lower Narmada Basin serves as a barometer for the ecological integrity of the entire river system. As a transitional corridor connecting the inland catchment to the marine environment, the Lower Basin supports a unique assemblage of obligate freshwater species, estuarine residents, and marine migrants. Surveys conducted in the region, including comprehensive field identification studies by CIFRI (Bulletin No. 182), have documented an expanded diversity of 84 fish species and 3 commercially important prawn species.

Family-wise Diversity and Composition

The fish assemblage is structurally dominated by the Order Cypriniformes (approx. 60%), followed by Siluriformes (35%), and Perciformes. This updated assessment reveals a more complex ecological profile, confirming the presence of exotic carps (*Cyprinus carpio*, *Hypophthalmichthys molitrix*) in the upper stretches and a highly diverse marine-estuarine assemblage in the lower zone, including species such as *Lates calcarifer* and *Harpadon nehereus*. This composition reflects the typical ichthyofaunal hierarchy of large Indian river systems but is distinguished by a significant component of marine-vagrant species due to the estuarine influence.

Family Cyprinidae (Carps and Minnows)

This family exhibits the highest species richness and biomass contribution in the Lower Narmada.

- **Indian Major Carps (IMC):** The basin supports robust populations of Catla (Catla), Labeo rohita (Rohu), and Cirrhinus mrigala (Mrigal). These potamodromous species utilize the main river channel for feeding and the tributaries (Orsang, Karjan) for breeding migrations during the monsoon.

- **Labeo Complex:** The diversity of the genus *Labeo* is particularly high, including *L. calbasu*, *L. bata*, *L. gonius*, *L. fimbriatus*, and *L. boggut*. The presence of *Labeo pangusia*, a species often associated with swift, rocky waters, suggests that suitable hill-stream habitats persist in the upper reaches of the Lower Basin near the dam and tributary confluences.
- **Mahseer:** The endangered Mahseer, *Tor tor*, is recorded in the basin. Historically abundant, its populations are now fragmented. The deep rocky pools near the Orsang confluence and the upper Karjan reservoir likely serve as critical remnant habitats for this iconic sport fish.
- **Forage Species:** Smaller cyprinids such as *Puntius sophore*, *Puntius sarana*, *Osteobrama cotio*, *Salmostoma bacaila*, *Salmostoma phulo*, *Rasbora daniconius*, *Chela labuca*, *Garra gotyla*, and *Amblypharyngodon mola* form the base of the piscivorous food web. Their abundance supports the diverse populations of predatory catfish and piscivorous birds found in the basin.

Families Bagridae and Siluridae (Catfishes)

Catfishes are the second most dominant group, thriving in the benthic and demersal zones of the river.

- **Apex Predators:** The Siluridae family is represented by *Wallago attu* (Freshwater Shark) and *Ompok bimaculatus* (Butter Catfish). *Wallago attu* is a top predator in the riverine food chain, indicating a healthy abundance of prey species.
- **Bagrids:** The family Bagridae is diverse, including large predators like *Sperata seenghala* (Giant River-catfish) and *Sperata aor*, as well as smaller species like *Rita rita*, *Rita pavementata*, *Mystus cavasius*, and *Mystus bleekeri*. These species are highly adaptable to the variable flow conditions of the Lower Narmada.
- **Other Catfishes:** The Schilbeidae family (*Clupisoma garua*, *Eutropiichthys vacha*, *Ailia coila*) inhabits the open water column, while *Arius* (Family Ariidae) represents the estuarine catfish component found near the river mouth.

Family Clupeidae (Shads and Herrings)

- **Tenuulosa ilisha (Hilsa):** The Indian Shad (Hilsa) is the flagship anadromous species of the Narmada estuary. It migrates from the Arabian Sea upstream to Bharuch and Bhadbhut for spawning. The recruitment and abundance of Hilsa are critically linked to

the volume of freshwater discharge during the monsoon, which provides the necessary olfactory cues and hydraulic conditions for migration.

- **Other Clupeids:** The checklist also includes *Setipinna phasa*, a smaller clupeid inhabiting the brackish zones.

Estuarine and Marine Vagrants

As the river approaches the Gulf of Khambhat (Bhadbhut to Ambetha), the fish fauna becomes increasingly characterized by euryhaline and marine species that utilize the estuary as a nursery or feeding ground.

- **Gobiidae:** Includes the ubiquitous *Glossogobius giuris* (Tank Goby) and specialized mudskippers like *Periophthalmodon schlosseri* and *Periophthalmus variabilis*, which inhabit the intertidal mudflats.
- **Mugilidae (Mulletts):** *Mugil cephalus*, *Liza parsia*, and *Liza macrolepis* are common detritivores in the estuarine zone. The freshwater mullet *Rhinomugil corsula* is found in the upstream reaches.
- **Marine Incursions:** The influence of the sea is evident from the presence of marine species such as *Polynemus indicus* (Threadfin), *Eleutheronema tetradactylum* (Fourfinger Threadfin), *Harpodon nehereus* (Bombay Duck), *Cynoglossus elongatus* (Tongue Sole), *Scatophagus argus* (Spotted Scat), *Lutianus argentimaculatus* (Mangrove Snapper), *Chanos* (Milkfish), and *Scianoides biauritus*. These species rely on the nutrient-rich estuarine mixing zone for rapid growth.

Crustaceans (Prawns)

The Lower Narmada supports a commercially significant prawn fishery.

- **Freshwater Prawn:** *Macrobrachium rosenbergii* (Giant Freshwater Prawn) requires the river's salinity gradient to complete its lifecycle (larvae require brackish water, adults live in freshwater).
- **Marine Prawns:** *Penaeus indicus* and *Metapenaeus dobsoni* are marine penaeid prawns that enter the estuary for nursing, exploiting the rich benthic resources.

Fisheries Production Trends

The fisheries output of the Lower Narmada River downstream of the dam has exhibited significant fluctuations over the past 15 years, reflecting the dynamic environmental conditions and anthropogenic pressures.

- 2008-09: Production stood at 10,482.55 tonnes, indicating a productive baseline.
- 2014-15: A sharp decline was observed, with production dropping to 7,597.85 tonnes.
- 2020-21: The lowest point in the dataset was reached at 5,346.9 tonnes, likely correlating with periods of reduced environmental flows or poor monsoon rainfall.
- 2023-24: A remarkable recovery occurred, with production surging to 11,514.75 tonnes, the highest recorded in the recent dataset. This resurgence may be attributed to favorable hydrological years facilitating Hilsa migration, improved estuarine productivity, or successful fisheries management interventions.

Table 41: Total fish production of Lower Narmada River downstream of Sardar Sarovar Dam

Year	Production (Tonnes)	Year	Production (Tonnes)	Year	Production (Tonnes)
2008-09	10,482.55	2014-15	7,597.85	2020-21	5,346.90
2009-10	9,878.05	2015-16	6,886.75	2021-22	9,783.15
2010-11	10,004.80	2016-17	6,946.55	2022-23	9,276.80
2011-12	9,256.00	2017-18	7,696.65	2023-24	11,514.75
2012-13	9,236.50	2018-19	7,790.25		
2013-14	9,543.30	2019-20	7,834.45		

Tributary-Specific Ichthyofauna

The tributaries of the Lower Narmada act as critical ecological extensions, providing breeding grounds, refugia, and distinct habitats.

- **Orsang River:** As a major right-bank tributary joining near Chandod, the Orsang is characterized by a sandy bed and seasonal flow. Its confluence zone is a biodiversity hotspot, particularly for lithophilic breeders like *Labeo boggut*, *Cirrhinus reba*, and *Tor*

tor. During the monsoon, the sediment-laden waters of the Orsang create distinct turbidity plumes that may serve as cover for migrating fish.

- **Karjan River:** The Karjan, regulated by the Karjan Dam, offers a mix of lentic (reservoir) and lotic (riverine) habitats. The reservoir supports stocked populations of major carps, while the river downstream sustains hardy species like *Mastacembelus armatus* (Spiny Eel) and *Channa* spp. (Snakeheads). The avian diversity in the Karjan basin (62 species) suggests a high abundance of forage fish.
- **Bhukhi River:** Located near the river mouth, the Bhukhi functions as a tidal creek. It supports a community dominated by euryhaline species such as *Mugil cephalus*, *Arius*, and *Scatophagus argus*. It serves as a vital nursery for juvenile prawns, protecting them from the high-velocity currents of the main Narmada channel.
- **Heran and Men Rivers:** These tributaries drain the tribal districts of Chhota Udaipur and Narmada. They are characterized by rocky substrates and hill-stream dynamics, likely supporting rheophilic genera like *Garra*, *Salmostoma*, and *Rasbora*.

Conservation Status

A significant proportion of the fish fauna in the Lower Narmada Basin is categorized under various threat levels, necessitating urgent conservation attention. Conservation assessments based on IUCN criteria reveal a critical need for protection beyond just the Mahseer. While *Tor tor* remains a key species of concern (Near Threatened), several other species face ecological pressures. The Common Carp (*Cyprinus carpio*) is listed as Vulnerable (VU), though noted here as an introduced exotic. Crucially, the Near Threatened (NT) category now includes the butter catfish *Ompok bimaculatus*, the freshwater shark *Wallago attu*, the silver carp *Hypophthalmichthys molitrix*, and the estuarine catfish *Arius gogora*. The majority of the remaining assemblage, including *Labeo rohita* and *Catla catla*, falls under Least Concern (LC).

Table 42: Consolidated List of Fish Diversity, Habitat, and IUCN Conservation Status in the Lower Narmada Basin

S. No.	Family	Scientific Name	Common Name	IUCN Status	Habitat Note
1	Notopteridae	Notopterus notopterus	Grey Featherback	Least Concern	
2	Cyprinidae	Catla catla	Catla	Not Evaluated	Freshwater
3	Cyprinidae	Labeo rohita	Rohu	Least Concern	Freshwater
4	Cyprinidae	Tor tor	Tor Mahseer	Near Threatened	Freshwater
5	Cyprinidae	Cyprinus carpio	Common Carp	Vulnerable	Exotic/Introduced
6	Cyprinidae	Puntius ticto	Firefin Barb	Least Concern	Freshwater
7	Cyprinidae	Barilius bendelisis	Hamilton's Barila	Least Concern	Hill Stream
8	Bagridae	Rita rita	Rita	Least Concern	Freshwater/Tidal
9	Bagridae	Sperata aor	Long-whiskered Catfish	Least Concern	Freshwater
10	Siluridae	Wallago attu	Boal / FW Shark	Near Threatened	Freshwater Predator
11	Siluridae	Ompok bimaculatus	Butter Catfish	Near Threatened	Freshwater
12	Clupeidae	Tenulosa ilisha	Hilsa	Not Evaluated	Migratory

S. No.	Family	Scientific Name	Common Name	IUCN Status	Habitat Note
13	Ariidae	Arius gagora	Gagora Catfish	Near Threatened	Estuarine
14	Harpadontidae	Harpadon nehereus	Bombay Duck	Not Evaluated	Marine/Estuarine
15	Centropomidae	Lates calcarifer	Barramundi/Bhetki	Not Evaluated	Estuarine
16	Scatophagidae	Scatophagus argus	Spotted Scat	Least Concern	Estuarine

(Source: CIFRI Bulletin No. 182)

5.2.3 Planktonic Base of the Food Web

The productivity of the macrobenthos and fish communities is underpinned by the planktonic food web.

- **Phytoplankton:** The river is dominated by Myxophyceae (Blue-green algae), especially during the summer at downstream stations like Bharuch (up to 88.98%), indicating eutrophic conditions. Bacillariophyceae (Diatoms) become abundant in winter (up to 54.66% at Jhanor), providing high-quality food for grazers like rohu and mollusks.
- **Zooplankton:** The community includes Protozoa, Rotifera, and Copepoda. High zooplankton counts at Bhadbhut and Ambetha (up to 26% of total plankton) during summer correlate with the estuarine mixing zone, providing essential nutrition for larval fish and prawns.

5.2.4 Ecological Stressors and Conservation

The biological integrity of the basin is under siege from three primary vectors:

1. **Hydrological Alteration:** The loss of the natural flood pulse disrupts breeding cues for carps and migration signals for Hilsa. Reduced flushing has allowed the salt wedge to penetrate 30-40 km upstream, eliminating stenohaline freshwater species.

2. **Chemical Barriers:** The industrial discharge near Ankaleshwar acts as a "chemical dam," deterring migration and causing bioaccumulation in benthic predators like *Wallago attu*.
3. **Destructive Fishing:** The use of small-mesh nets, locally known as "Disco nets," destroys juvenile stocks of carps and Hilsa, preventing population recruitment.

Disclaimer: The data presented in this section is based on secondary sources and specific surveys conducted in 2023-24. Species lists and abundance values are subject to natural seasonal variability and methodological differences.

Chapter 6: Conclusion

In this report, data is compiled from various government agencies, books, and relevant research papers to gain a deeper understanding of the vast biodiversity of the Narmada Basin. Arranging the data by sub-basin and according to time zones helps to understand the status of different biological diversity in the river and its tributaries, such as the number of species in different sub-basins, common contributors in both basins, and how different animal and plant species are distributed across the area. As we concluded, the flora, specifically phytoplankton and periphyton, in the Upper Narmada Basin exhibits a high variety of plant life. It has also been found that several studies have been done on fish and reported a variety of fish in the Upper and Middle Basins. Many species are common to both sub-basins, as well as there are some fish species that are specific to each sub-basin. The common species in the upper and middle basins, whether flora or fauna, exhibit significant connectivity within the river eco-system. In addition to fish, the report also highlights the presence of waterbirds, water-associated birds, and water-dependent birds, as well as amphibians and reptiles, in both the Upper and Middle basins. The report also collected data for the tributaries and concluded that fish are more abundant in the tributaries of the Middle Narmada Basin compared to the Upper Narmada Basin.

The biological diversity of the Lower Narmada River Tributaries represents a resilient yet vulnerable ecological system. The Macrobenthos community serves as a sensitive recorder of environmental history, shifting from diverse freshwater assemblages in the upper reaches to stress-tolerant annelid communities in the industrialized estuarine zone. The Fish fauna, with its impressive diversity of 84 taxa, underscores the basin's role as a critical migration corridor and nursery. The recent recovery in fish production figures is a positive signal, but the reliance on specific flow regimes for species like Hilsa and Mahseer highlights the need for sustained environmental flows.

Overall, based on this report, it is easy to understand the ecological importance of the river system and the Narmada Basin. Understanding the habitat, location, and number of species is essential, as this information can inform ecological conservation strategies, restoration policies, and long-term protection of the natural environment.

Chapter 7: Recommendations

Based on the assessment of biological diversity across the tributaries of the Narmada River basin, the following recommendations are proposed to strengthen conservation, monitoring, and sustainable management of aquatic and riparian ecosystems:

1. Strengthening Biological Monitoring Program

A systematic, long-term monitoring framework should be established for flora and fauna across the Narmada basin. Regular surveys covering macrobenthos, fish, plankton, and riparian biodiversity will enable accurate assessment of species composition, population trends, and ecological health.

2. Seasonal and Long-Term Surveys

Seasonal monitoring (pre-monsoon, monsoon, and post-monsoon) should be institutionalized to capture temporal variations in species diversity and abundance. In addition, permanent biological monitoring stations should be set up at key tributaries and confluence zones to detect long-term ecological changes and early signs of stress.

3. Focused Research on Tributary Ecosystems

There is a need to enhance scientific research specifically on the tributaries of the Narmada River, as these systems function as critical breeding, nursery, and refuge habitats. Detailed ecological studies on lesser-studied tributaries will help improve basin-wide biodiversity understanding.

4. Environmental Flow Management

Adequate environmental flows should be maintained, particularly downstream of major hydraulic structures, to support fish migration, spawning of key species, sediment transport, and flushing of organic loads. Flow regulation should consider ecological requirements alongside hydropower and irrigation demands.

5. Protection of Tributary Confluence Zones

Confluence areas of major tributaries should be identified and protected as ecologically sensitive zones. Activities such as unregulated sand mining, encroachment, and waste disposal should be strictly controlled, as these areas often serve as critical spawning and feeding grounds for aquatic fauna.

6. Riparian Zone Restoration

Restoration and protection of riparian vegetation along tributaries and the main river channel should be prioritized. Healthy riparian zones provide shelter, breeding habitats, and food resources for birds, insects, and aquatic organisms, while also stabilizing riverbanks and improving water quality.

7. Pollution Reduction and Waste Management

Immediate measures are required to reduce pollution loads entering the river system. Untreated or inadequately treated sewage and industrial effluents should be strictly regulated through improved treatment infrastructure and enforcement of effluent discharge standards, as pollution directly affects river health and biodiversity.

8. Habitat Conservation and Legal Protection

Natural habitats such as reservoirs, wetlands, sanctuaries, and floodplain ecosystems should be protected and restored through strengthened legal frameworks and effective enforcement. Special attention should be given to habitats supporting near-threatened and endangered species.

9. Public Awareness and Community Participation

Awareness programs and outreach campaigns should be conducted in villages and towns along the Narmada basin to educate local communities about the ecological importance of river biodiversity. Community participation in conservation efforts can significantly enhance the long-term success of biodiversity protection initiatives.

Disclaimer

The data presented in this chapter are primarily based on secondary sources and specific survey records available at the time of assessment. Species names, population estimates, and distribution patterns may vary across sources due to differences in survey periods, methodologies, and reporting frameworks. Consequently, minor discrepancies in species numbers or locations may occur, and the information should be interpreted in the context of natural seasonal variability and data limitations.

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Annexure

Annexure 2.1.1: Checklist of Streams flowing through the comprising districts of the Upper Narmada River Basin

State	District	Area (sq. km)	Rivers
Chhattisgarh	Kabeerdham	622.12	Banjar River, Bagh Nala, Kukurbahra Nala, Manai Nadi, Bolda Nala, Dhawaipani, Phen River, Muradobra Nala, Jamuniya Nala, Halon Nadi, Lilari Nadi, Khudrai Nala B, Khudrai Nala A, Sunhadabra Nala, Jatla Nala, Harpan R, Katai Nadi, Kabripathra Nala
Chhattisgarh	Rajnandgaon	86.06	Gabdi Nala, Banjar River, Bansbhira, Jatla Nala
Chhattisgarh	Mungeli	0.41	-
Chhattisgarh	Bilaspur	0.32	-
Madhya Pradesh	Sagar	376.23	Sinhora Nadi, Madni Nala, Ghogra Nadi, Ghoghara Nala A, Dona Nala, Biranj Nadi, Lakha Nala, Ghora Nala
Madhya Pradesh	Damoh	419.82	Loraphalku N, Kair Nadi, Katai Nala, Kasalo N, Madhera Nala, Kohani Nala, Dhaprau Nala
Madhya Pradesh	Katni	1134.6	Silpuri N, Patpara N, Kair Nadi, Kobhi Nala, Datla Nala A, Damdama Nala, Heran Nadi, Labhera Nala, Mauri Nala, Katkahri Nala, Paterh N, Bah Nal, Suhar N, Hiran R
Madhya Pradesh	Raisen	4518.16	Narmada River, Palkamli N A, Chikhla N, Machherayi Nala, Pata Nala B, Kuhadoh Nala, Tendon Nala, Pathariya N B, Machhwai N C, Machhwai N B, Amdoh Nadi, Sinhora Nadi, Dudhi River, Jamner Nadi A, Dobi Nadi, Tendon Nadi, Kala Jhora, Magardha N, Gajanda Nadi, Surai N, Tirka Nala, Bandhan Nadi, Satdhara N, Renga Jhora, Puranapani Nala, Ikyawan Nala, Sukha Nala B, Pathariya N A, Narheri N, Gobi Nala, Patra Nala, Machhwai Nala A, Machhwai Nala B, Machhwai Nala C, Katkasiri N, Bagha N A, Gharbora N, Jhira Nala, Kukarsuta Nala, Mankundi N, Nakti Nadi, Surai Nala, Nandora Nala, Naharla Nadi, Rohiya Nala, Bagha Nala B, Ud N, Kelori Nadi, Rajadhar Nala, Gangaji Nala, Begam Nadi, Silatiya Nala, Ghorhra N, Kopikasa Nala, Piria Nala B, Chandni Nala, Garanju N, Ikyawan N, Gondiwala Jhora, Sukhchain Nala B, Bindar N, Supajhora N, Budka N, Bagha N B, Barna N, Barna Nadi, Machhwai Nala E, Lawajhir Nala, Gadaria N, Anjan River, Chamarsil N, Renga Nala, Khand N, Lamti N B, Jamandhora N, Sukar N, Narvadiya Nadi
Madhya Pradesh	Sehore	372.09	Narmada River, Bankur Nala, Tawa River, Kusumaili Nala, Bagli N, Dobi Nadi, Pilia N, Burhi Narmada, Katkasiri N, Potiya Nala, Surai Nala, Chandni Nala, Kosmi N

Biological Profile of Narmada Basin

Madhya Pradesh	Jabalpur	4796.75	Narmada River, Kalhar N, Chanta N, Madroti Nala, Ranwai Nadi, Khuja Nala, Matoria Nala, Narrai Nala, Hagni N, Matoria N, Parpara N, Matbaria N, Dev Nala C, Omti Nala, Newari Nala, Mugarha N, Moti Nala B, Kair Nadi, Kobhi Nala, Katai Nala, Kanchan Nala, Bargela N, Soner Nadi, Datla Nala A, Kasalo N, Dala Nala, Duraghat N, Beha Nala, Bidri N, Badheru Nala, Musra Nala, Heran Nadi, Labhera Nala, Hinauta N,
			Madhera Nala, Imarti Nala, Burungi Nala, Patparaha Nala, Bagher nala, Sayanashi Nala, Koha Nala, Khandari R, Teriya Nala, Karthai Nadi, Ghughara Nala, Kohani Nala, Keolari Nala, Pataha Nala, Phagua N, Imarati Nala B, Dhaprau Nala, Mangraha N, Barne N, Tanguna Nala, Banjar Nala, Uldana Nala A, Gajna N, Gaur Nadi, Moti N, Kanari N, Bah Nal, Suhar N, Hiran R, Pariat Nadi, Khurji N
Madhya Pradesh	Anuppur	535.74	Narmada River, Marjadi Nala, Radhan N, Kendu Nala, Sukhnal Nala, Rachhar N, Bijora Nala, Machhrar N, Karnai Nala, Barati, Lutiya Nala, Ruddi Nala, Pandripani Nala, Kaswani Nala, Dhobgarh Nala
Madhya Pradesh	Dindori	4773.07	Narmada River, Palothar Nala, Rigarh Nala, Marjadi Nala, Karwar Nala, Kirha N, Kukripota Nala, Boir Nala, Barnai Nala, Sontiragh Nadi, Gonrabal Nadi, Sukhnal Nala, Kukra N, Katori N, Kutri R, Bhari N, Kajla N, Machhrar N, Jalda N, Renga N, Dev Nala B, Dev Nala A, Udar N, Matiyari Nala, Karnai Nala, Devsara Nala, Manai Nadi, Batai Nala, Piparkhutta, Chichila N, Turar Nala, Kasaha Nala, Teriya N, Banari N, Ghutai Nala, Dandana Nala B, Chulhanani N, Roha N, Korrer N, Mochi Nala, Kanai N, Tethi Nala, Silgati N, Chachar Nala, Silgi N, Amadob Nala, Kharmer Nadi, Sanjari Nala, Sukha Nala A, Pandripani Nala, Dandana Nala A, Kanai Nala, Ghurdaur Nala, Dhamna Nala, Nakaldaiha Nala, Ghunsi N, Bijakhan Nala, Chakrar Nadi, Rora N, Harpan R, Harsinghi Nala, Tar Nala, Kuteli Nala, Chhipani Nala, Seoni R, Kachnari Nala, Ajar N, Kotar N, Karmandai Nadi, Sukhmer N, Kukra Nala, Dandala Nala, Burhner Nadi
Madhya Pradesh	Narsimhapur	4997.75	Narmada River, Gangdor N, Palkamli N A, Gahra Nala, Baghdolna N, Daku N, Machherayi Nala, Bhanga Nala, Pata Nala B, Chitarewa R, Sukha N, Sapan N, Bagha Nala A, Gheta N, Anvariya Nala, Bodhi N, Ihmana Nala, Kanchana N, Chandighoghra Nala, Machhwai N A, Lendia N, Sinhor Nadi, Datla Nala B, Dudhi River, Shankar N, Ulgana Nala B, Panaha Nala, Chhapra Nala, Madni Nala, Moti Nala B, Ghughari Nala, Gahedua N, Dandri Nala, Gahra N A, Gahra N B, Shrijot Nala, Umar Nala, Pathn N, Soner Nadi, Sanadhana Nala, Sanaghogra Nala, Tinsra N, Rohiya N, Sukha Nala B, Ol Nadi, Umar N, Singla N, Chuna N, Baghni Nala, Badheru Nala, Jhundapani N, Bagla N A, Bagla N B, Kivari Nala, Piria N, Pituwa Nala, Roriya Nala, Onra Nala, Bagher nala, Nakta Nala, Palkamli N B, Singri N, Ghoghra N, Gidhan Nala, Barurewa N A, Barurewa N B, Sher River, Kilchi N, Umar Nadi, Jugiya Nala, Ghagra Nala, Sukhchain Nala A, Ajan N, Dondhajhora Nala, Dhamani Nadi, Kethki N, Gehra N, Paraijhiriya N, Tarha N, Gokalkhatta Nala, Ghora Nala, Macharewa

Biological Profile of Narmada Basin

			R, Kanchna N, Siddh N, Shakkar Nadi, Hiran R, Kanera Nadi, Pandajhir Nadi, Ritha Nala, Dagasia Nala
Madhya Pradesh	Mandla	6645.74	Narmada River, Kalhar N, Patparahi Nala, Ranwai Nadi, Gonrabal Nadi, Narrai Nala, Banjar River, Gahir Nala, Garbari N, Mahodar Nala, Bhagorna Nala, Burburhi Nala, Parpara N, Sorma Jhori Nala, Sunha Nala, Gonda Nala, Bargi N, Kindri Nala, Kopedonda Nala, Dev Nala A, Datla N, Jhamjhum Nala, Newari Nala, Jejam Nala, Manai Nadi, Burburi Nala, Tirri Nala, Dulha Nadi, Chichila N, Datla Nala C, Kokchi Nala, Bijna Nala, Teriya N, Baghdhara Nala, Borada Nala, Dhawaipani, Mala Nala, Banari N, Amakol Nala, Ghoghra Nala, Shakraha N, Bidri N, Roha N, Balal Nadi, Muradobra Nala, Hati Nala, Matiar Nala, Kukura Nala,
			Patpara Nala A, Gurbini Nala, Kukri Nala, Anari N, Marjadi N, Musra Nala, Gurda Nala, Hingna Nala, Bhimori Nala, Bamiya Nala, Guptganga N, Chuhi Nala, Burungi Nala, Ghutku Nala, Sukha Nala A, Lilari Nadi, Bhai Nala, Murjhiriya N, Bakal N, Patpara Nala B, Matiar N, Baidar Nala, Sailwara Nadi, Kanha Nala, Gurbini N, Rajakachhar Nala, Bhaibahin Nala, Harpan R, Surpan N B, Surpan N A, Katai Nadi, Nigga Nala, Sukhram Nala, Khular N, Bagbura Nala, Mogaliya Nala, Kani Nala, Hargarhi Nala, Khalkhaii Nala, Ghanghar N, Tanguna Nala, Bitiyadhar Nala, Hiragarhi Nala, Mohanjhori Nala, Patpara Nala C, Gaur Nadi, Jhamil N, Surwahi Nadi, Nadiya N, Kop Nala, Kukra Nala, Burhner Nadi, Phadej Nala, Kabripathra Nala
Madhya Pradesh	Hoshangabad	4646.03	Narmada River, Ghorhri Nala, Bainganga N, Gahra Nala, Mithijhiri N, Langham Nadi, Parata N, Machherayi Nala, Gondiya N, Monpani Nadi, Bankur Nala, Manasgwari N, Bhimkind N, Bella Nala, Lehanra Nala, Kahuapani Nala, Kerapati N, Koti Nadi, Tawa River, Bori N, Dudhi River, Bagli N, Chikla Nala, Sainkuria Nala, Ulgana Nala B, Sonbhadra Nadi, Jhau Nala, Bhajipani Nala, Motighoghra N, Sula N, Madagin N, Lama Nala, Jhirpa N, Sukhi N B, Bhangra Nadi, Bija N, Nagan Nala, Kotipati N, Panari Nala, Burhi Narmada, Denwa River, Bhainsa Nala, Machla Nala, Bori Nadi, Fundna Nala, Mundi Nala, Ol Nadi,

			Palakmati Nadi A, Kummajhiri Nala, Brijta N, Sarra Nala, Bakod Nala, Dhamni Nadi, Machhwai Nala D, Hathidol Nala, Silpali Nala, Manwara N, Junkur N, Katkasiri N, Nagdwari Nadi, Korajhori Nala, Suk Tawa R, Potiya Nala, Periyar Nadi, Maru Nadi, Marraun N, Machldhar Nala, Ganjakunwar Nala, Malni Nadi, Giddh N, Suaria Nala, Sukha Nadi, Rain Nadi, Dabdaba Nala, Dagdaga Nala, Supara Nala, Baosiya Nala, Tilberi N, Ghoghara Nala B, Andharabawdi Nala, Marram Nala, Suki Nala, Shrijonth Nala, Bhardakkunda, Nala, Korn Nadi, Bordha N, Dhar N B, Khahra N, Lingi N, Anjan River, Patpani Nala, Khand N
Madhya Pradesh	Seoni	2251.37	Narmada River, Chanta N, Jamanjhira Nala, Bhagorna Nala, Datla Nala B, Patwara Nadi, Soner Nadi, Kurumunda Nala, Balal Nadi, Umar N, Paryat Nadi, Hinata N, Temur N, Teriya Nala, Sher River, Gadheri Nadi, Jhanda Nala, Sher Nala, Imarati Nala B, Dabki Nadi, Imarati Nala A, Bitiyadhar Nala, Baghora N, Hiragarhi Nala, Nagdhar Nala, Macharewa R, Gurda Nadi, Chakarpah N, Shakkar Nadi, Kanera N, Kanera Nadi
Madhya Pradesh	Chhindwara	3545.08	Bainganga N, Baghdolna N, Kudnl Nala, Bhadbhada N, Sukha N, Sapan N, Gheta N, Bodhi N, Chandighoghra Nala, Tharakna Nala, Kerapati N, Tawa River, Bodaljhora Nala, Dudhi River, Dudhi Nadi, Sonbhadra Nadi, Hard N, Jhau Nala, Bholki N, Bhajipani Nala, Sula N, Barchha Nala, Bija N, Bharadhar Nadi, Sanadhana Nala, Denwa River, Sanaghogra Nala, Lamti N A, Bori Nadi, Nishan Nala A, Lamti Nala, Tambiya Nadi, Dhar N A, Jhmru N, Kapur Nala, Chitarewa River, Silpati N, Piparpani Nala, Bairsa Nala, Parakhundra N, Dhubghat Nala, Dudher Nadi, Korajhori Nala, Katta N, Paraswara Nala, Mangrahi Nala, Jahrkot Nala, Karidoh Nala, Toparghogra Nala, Marraun N, Nikai Nala, Barurewa N B, Titoriya Nala, Goridhar Nala, Gui N, Rakhi Nadi, Macharewa R, Banki Nala, Shakkar Nadi, Anjan River, Patpani Nala
Madhya Pradesh	Balaghat	2291.76	Kharari Nala, Samrahi Nala, Gurur Nala, Banjar River, Bagh Nala, Gonda Nala, Halon N, Jhamjhum Nala, Burburi Nala, Bolda Nala, Banjari Nala, Kanhar Nala, Khudrahi Nala, Muradobra Nala, Hati Nala, Pati Nala, Pongar Nala, Halon Nadi, Bhai Nala, Khudrai Nala A, Tanno River, Kashmiri Nadi, Mimkur Nala, Bhongir Nala, Bhaibahin Nala, Kheni Nala, Minkor Nala, Aunrghaghra Nala, Mohanjhori Nala

Madhya Pradesh	Betul	2875.47	Chakora Nala, Machna R, Sawaniya N, Ghughra N, Dharakhoh N, Pata Nala A, Bhimkind N, Khirni Nala, Chip or Ghoghra N, Kerapati N, Koti Nadi, Mohta Nala, Salidhana N, Tawa River, Dudhi Nadi, Latiya Nala, Ghosiya Nala, Sonbhadra Nadi, Nimpani Nala, Raniyam Nala, Bharanga Nadi, Kubra Nala, Bharadhar Nadi, Lendi Nala, Kotipati N, Denwa Nadi, Nishan Nala A, Nishan Nala B, Sukhi N A, Bhaunra N, Silpati N, Golai Nala, Singanwari Nala, Amari N, Ampan Nala, Bhumkadhana N, Saranga Nadi, Mundawani N, Pindu Nala, Malni Nadi, Lobar Nadi, Kashi Nala, Bamandev N, Diydoh N, Jogimath Nala, Darwaza N, Dagdaga Nala, Juban Nala, Magardoh N, Phopas Nadi, Sampna, N, Mankar Nadi, Bordha N, Dhar N B, Karanji Nala, Chakdilya Nala
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Annexure 2.1.2: Checklist of Streams flowing through the districts comprising the Middle Narmada River Basin

STATE	DISTRICT	Area	River
Madhya Pradesh	Bhopal	57.96	Kolar Nadi
Madhya Pradesh	Raisen	309.99	Kolar Nadi, Bhagner N, Gadariya N, Kaliyadeh Nadi, Gunjari N, Utawali N
Madhya Pradesh	Sehore	4009.46	Narmada River, Tawa River, Ganjal Nadi, Kolar Nadi, Sip Nadi, Hather Nadi, Datuni N, Jamner Nadi B, Ajnal Nala, Bagdi Nadi
Gujarat	Panch Mahals	83.92	Hatni River
Madhya Pradesh	Dewas	4160.44	Narmada River, Ajnal Nadi, Machak Nadi, Datuni N, Kanar Nadi, Khari N B, Jamner Nadi B, Bagdi Nadi, Lohar N, Chankeshar Nadi
Madhya Pradesh	Jhabua	2982.96	Narmada River, Hatni River, Udai Nadi, Uri Wagh River, Wagh or Bagh River, Sukar nadi, Jharkal River, Ankhar Nadi, Dohi Nadi, Panchhi N
Madhya Pradesh	Dhar	5208.52	Narmada River, Goi Nadi, Deb Nadi, Man Nadi, Uri Wagh River, Wagh or Bagh River, Karam Nadi, Borar Nadi, Khuj Nadi, Chiri Nadi
Madhya Pradesh	Indore	1548.81	Karam Nadi, Kanar Nadi, Choral R, Kholar N, Lohar N, Malan N, Maheshwari Nala, Jamner N, Dabri N, Odhali N A
Madhya Pradesh	Hoshangabad	4462.05	Morand Nadi, Ganjal Nadi, Ajnal Nadi, Machak Nadi, Syani Nadi, Hather Nadi, Ghorapachhar Nadi, Bhaji Nadi, Kalimachak Nadi, Midkul N
Gujarat	Vadodara	413.14	Narmada River, Khat Nadi, Katri Nadi, Indarvar Nala
Madhya Pradesh	West Nimar	10869.71	Narmada River, Goi Nadi, Beda Nadi, Sukta Nadi B, Kunda River, Deb Nadi, Kanar Nadi, Choral R, Borar Nadi, Kholar N
Madhya Pradesh	East Nimar	5905.70	Narmada River, Sukta Nadi B, Ghorapachhar Nadi, Kalimachak Nadi, Chhota Tawa River, Bham N, Ajnal N, Ruparel Nadi, Kaveri Nadi, Sivji Nala
Madhya Pradesh	Betul	554.51	Morand Nadi, Bhaji Nadi, Duni Nala, Bhaura N, Ghogra N, Sukha Nala, Kotmi Nala, Dakidoh Nala, Junapani N C
Gujarat	Bharuch	358.73	Narmada River, Devganga
Maharashtra	Nandurbar	1230.61	Narmada River, Udai Nadi, Devganga, Jharkal River, Devnand Nadi, Bokrata Nadi, Khat Nadi, Katri Nadi, Devganga spring L, Devganga spring R
Maharashtra	Dhule	9.01	N/A

Annexure 2.1.3: Checklist of Streams flowing through the districts comprising the Lower Narmada River Basin

State	District	Area (sq. km)	Rivers
Madhya Pradesh	Alirajpur	967.71	Dhudhwal Nala, Pogvi Nala, Nakda Nadi, Dhamni Nala, Ani Nadi, Heran Nadi A, Sukta Nadi A, Mathwad nala, Sukti Nadi, Kara Nadi, Orsang River, Bharoj Nala, Undo Nala
Gujarat	Panch Mahals	133.98	Suki Nadi, Meriya
Gujarat	Dohad	65.88	Bharaj Nadi, Vaswa Nadi
Gujarat	Vadodara	594.65	Heran Nadi A, Aina Nadi, Narmada River, Orsang River, Asan Nala, Veru Nala, Bhukhi River
Gujarat	Bharuch	2705.55	Amala Khadi, Ratanpurni, Amravati Nadi, Mohan Nadi, Karedi, Karjan, Narmada River, Hakran, Kansa ki khadi, Asan Nala, Lindia Nala, Chandrai Khadi, Bhukhi River, Kaveri, Madhumati Nadi
Gujarat	Narmada	2205.44	Men, Tarav, Nandi Khadi, Heran Nadi A, Mohan Nadi, Aswan Nadi, Sukli, Karjan, Dhamao Khadi, Narmada River, Bogla Nadi, Shamarlyo Khadi, Kaludi, Dhamani Khadi, Chandrai Khadi, Makodi Nadi, Damni
Gujarat	Surat	206.71	Mohan Nadi, Karjan
Gujarat	Chota Udaipur	3158.39	Suki Nadi, Dhudhwal Nala, Meriya, Pogvi Nala, Men, Nakda Nadi, Dhamni Nala, Ani Nadi, Bharda Nadi, Surya Nadi, Heran Nadi A, Sukta Nadi A, Aina Nadi, Aswan Nadi, Mathwad nala, Maniyan Nadi, Uchh, Kara Nadi, Huktu, Waswa Nala, Orsang River, Jarawa Nala, Rami, Jharkhadi Nadi, Heran Nadi B, Bharaj Nadi, Vaswa Nadi, Suket Nadi, Bharoj Nala, Undo Nala
Maharashtra	Nandurbar	33.2	Bogla Nadi

Annexure 3.3.1: Checklist of Riparian and Floodplain Vegetation

Aquatic Herbs in Narmada Basin, India						
Family	Scientific name	Habitat	Annual/Perennial	Native/Exotic	IUCN status	Zone
Menyanthaceae	<i>Nymphoides indica</i> (L.) Kuntze	Aquatic	Perennial	Native	LC	UZ
Asteraceae	<i>Cyathocline purpurea</i> (Buch.-Ham. ex D.Don) Kuntze	Semi aquatic	Annual	Native	LC	UZ
Araceae	<i>Colocasia esculenta</i> (L.) Schott	Semi aquatic	Perennial	Native	LC	UZ
Nelumbonaceae	<i>Nelumbo nucifera</i> Gaertn.	Aquatic	Perennial	Native	NE	UZ
Nymphaeaceae	<i>Nymphaea nouchali</i> Burm. f.	Aquatic	Perennial	Native	LC	UZ
Lythraceae	<i>Rotala serpyllifolia</i> (Roth) Bremek.	Aquatic	Perennial	Native	LC	UZ
Trapaceae	<i>Trapa natans</i> L.	Aquatic	Perennial	Native	LC	UZ
Menyanthaceae	<i>Nymphoides hydrophylla</i> (Loureiro) Kuntze	Aquatic	Perennial	Native	LC	UZ
Lentibulariaceae	<i>Utricularia australis</i> R. Br.	Aquatic	Perennial	Native	LC	UZ
Ceratophyllaceae	<i>Ceratophyllum demersum</i> L.	Aquatic	Perennial	Native	LC	UZ
Hydrocharitaceae	<i>Hydrilla verticillata</i> (L.f.) Royle	Aquatic	Perennial	Native	LC	UZ
Hydrocharitaceae	<i>Nechamandra alternifolia</i> (Roxb. ex Wight) Thwaites	Aquatic	Perennial	Native	LC	UZ
Hydrocharitaceae	<i>Najas indica</i> (Willd.) Cham.	Aquatic	Perennial	Native	LC	UZ
Hydrocharitaceae	<i>Najas minor</i> All.	Aquatic	Perennial	Native	LC	UZ
Hydrocharitaceae	<i>Ottelia alismoides</i> (L.) Pers.	Aquatic	Perennial	Native	LC	UZ
Hydrocharitaceae	<i>Vallisneria spiralis</i> L.	Aquatic	Perennial	Native	LC	UZ
Pontederiaceae	<i>Eichhornia crassipes</i> (Mart.) Solms	Aquatic	Perennial	Exotic	NE	UZ
Araceae	<i>Lemna perpusilla</i> Torr.	Aquatic	Annual	Native	LC	UZ
Araceae	<i>Pistia stratiotes</i> L.	Aquatic	Perennial	Native	LC	UZ
Araceae	<i>Spirodela polyrhiza</i> (L.) Schleid.	Aquatic	Perennial	Native	LC	UZ
Alismataceae	<i>Caldesia oligococca</i> (F.Muell.) Buchenau	Aquatic	Annual	Native	LC	UZ
Aponogetonaceae	<i>Aponogeton natans</i> (L.) Engl. & K.Krause	Aquatic	Perennial	Native	LC	UZ

Potamogetonaceae	<i>Potamogeton crispus</i> L.	Aquatic	Perennial	Native	LC	UZ
Potamogetonaceae	<i>Potamogeton pectinatus</i> L.	Aquatic	Perennial	Native	LC	UZ
Salviniaceae	<i>Azolla pinnata</i>	Aquatic	Annual	Native	LC	UZ
Marsileaceae	<i>Marsilea quadrifolia</i> L.	Aquatic	Annual	Exotic	LC	UZ
Portulacaceae	<i>Portulaca oleracea</i> L.	Semi aquatic	Annual	Exotic	LC	UZ
Fabaceae	<i>Neptunia oleracea</i> Lour.	Semi aquatic	Annual	Native	LC	UZ
Apiaceae	<i>Centella asiatica</i> (L.) Urb.	Semi aquatic	Annual	Native	LC	UZ
Convolvulaceae	<i>Ipomoea aquatica</i> Forsskal	Semi aquatic	Perennial	Native	LC	UZ
Verbenaceae	<i>Phyla nodiflora</i> (L.) Greene	Semi aquatic	Perennial	Native	LC	UZ
Plantaginaceae	<i>Bacopa monnieri</i> (L.) Pennell	Semi aquatic	Perennial	Native	LC	UZ
Plantaginaceae	<i>Limnophila indica</i> (L.) Druce	Semi aquatic	Perennial	Native	LC	UZ
Plantaginaceae	<i>Limnophila sessiliflora</i> (Vahl) Bl.	Semi aquatic	Perennial	Native	LC	UZ
Polygonaceae	<i>Persicaria barbata</i> (L.) H.Hara	Semi aquatic	Annual	Native	LC	UZ
Polygonaceae	<i>Persicaria glabra</i> (Willd.) Gomez de la Maza	Semi aquatic	Annual	Native	LC	UZ
Commelinaceae	<i>Commelina hasskarlii</i> C.B.Clarke	Semi aquatic	Annual	Native	LC	UZ
Commelinaceae	<i>Cyanotis axillaris</i> (L.) D.Don ex Sweet	Semi aquatic	Annual	Native	LC	UZ
Commelinaceae	<i>Murdannia nudiflora</i> (L.) Brenan	Semi aquatic	Annual	Native	NE	UZ
Commelinaceae	<i>Murdannia spirata</i> (L.) G.Brückn.	Semi aquatic	Annual	Native	LC	UZ
Alismataceae	<i>Limnophyton obtusifolium</i> (L.) Miq.	Semi aquatic	Annual	Native	LC	UZ

Aquatic and Semi- Aquatic Grasses in the Narmada Basin, India						
Family	Scientific name	Habitat	Annual/Perennial	Native/Exotic	IUCN status	Reference
Typhaceae	<i>Typha angustifolia</i> L.	Aquatic	Perennial	Native	LC	UZ
Eriocaulaceae	<i>Eriocaulon brownianum</i> Mart.	Semi aquatic	Perennial	Native	LC	UZ
Cyperaceae	<i>Cyperus alulatus</i> J.Kern	Semi aquatic	Annual	Native	LC	UZ
Cyperaceae	<i>Cyperus corymbosus</i> Rottb.	Semi aquatic	Perennial	Native	LC	UZ
Cyperaceae	<i>Cyperus difformis</i> L.	Semi aquatic	Annual	Exotic	LC	UZ
Cyperaceae	<i>Cyperus digitatus</i> Roxb.	Semi aquatic	Perennial	Native	LC	UZ
Cyperaceae	<i>Cyperus exaltatus</i> Retz.	Semi aquatic	Perennial	Native	LC	UZ
Cyperaceae	<i>Cyperus iria</i> L.	Semi aquatic	Annual	Native	LC	UZ
Cyperaceae	<i>Cyperus nutans</i> Vahl	Semi aquatic	Perennial	Native	LC	UZ
Cyperaceae	<i>Cyperus pangorei</i> Rottb.	Semi aquatic	Perennial	Native	LC	UZ
Cyperaceae	<i>Pycnus pumilus</i> (L.) Nees	Semi aquatic	Annual	Native	LC	UZ
Cyperaceae	<i>Cyperus rotundus</i> L.	Semi aquatic	Perennial	Native	LC	UZ
Cyperaceae	<i>Eleocharis geniculata</i> (L.) Roem. & Schult.	Semi aquatic	Annual	Native	LC	UZ
Cyperaceae	<i>Eleocharis atropurpurea</i> (Retz.) J.Presl & C.Presl	Semi aquatic	Annual	Native	LC	UZ
Cyperaceae	<i>Eleocharis dulcis</i> (Burm.f.) Trin. ex Hensch.	Semi aquatic	Perennial	Native	LC	UZ
Cyperaceae	<i>Fimbristylis dichotoma</i> (L.) Vahl	Semi aquatic	Perennial	Native	LC	UZ
Cyperaceae	<i>Fimbristylis falcata</i> (Vahl) Kunth	Semi aquatic	Perennial	Native	NE	UZ
Cyperaceae	<i>Fimbristylis ferruginea</i> (L.) Vahl	Semi aquatic	Perennial	Native	LC	UZ
Cyperaceae	<i>Fimbristylis ovata</i> (Burm.f.) J.Kern	Semi aquatic	Perennial	Native	LC	UZ
Cyperaceae	<i>Fuirena ciliaris</i> (L.) Roxb.	Semi aquatic	Annual	Native	LC	UZ
Cyperaceae	<i>Kyllinga bulbosa</i> P.Beauv.	Semi aquatic	Perennial	Native	LC	LZ
Poaceae	<i>Coix lacryma-jobi</i> L.	Semi aquatic	Annual	Native	NE	UZ

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Poaceae	<i>Coix aquatica</i> Roxb.	Semi aquatic	Perennial	Native	NE	UZ
Poaceae	<i>Oryza rufipogon</i> Griff.	Semi aquatic	Perennial	Native	LC	UZ
Poaceae	<i>Panicum psilopodium</i> Trin	Semi aquatic	Perennial	Native	LC	UZ
Poaceae	<i>Paspalidium punctatum</i> (Burm.f.) A.Camus	Semi aquatic	Perennial	Native	LC	UZ
Poaceae	<i>Vetiveria zizanioides</i> (L.) Nash	Semi aquatic	Perennial	Native	NE	UZ

Annexure 4.4.1: Checklist of fish (CIFRI – Biological Investigations on Fisheries of Narmada River (1958–1966))

Order	Family	Fish Taxa
Clupeiformes	Clupeidae	<i>Hilsailisha</i> (Hamilton)
	Clupeidae	<i>Gonialosamanminna</i> (Hamilton)
	Notopteridae	<i>Notopterusnotopterus</i> (Pallas)
Cypriniformes	Cyprinidae	<i>Chelalaubuca</i> (Hamilton)
	Cyprinidae	<i>Oxygasterclupeoides</i> (Bloch)SubfamilyRASBORINAE
	Cyprinidae	<i>Bariliusbarila</i> (Hamilton)
	Cyprinidae	<i>Bariliusbendelisisvarchedra</i> (Hamilton)
	Cyprinidae	<i>Bariliusevezardi</i> Day
	Cyprinidae	<i>Bariliusradiolatus</i> Gunther
	Cyprinidae	<i>Danio</i> (<i>Danio</i>) <i>aequipinnatus</i> (McClelland)
	Cyprinidae	<i>Danio</i> (<i>Danio</i>) <i>devario</i> (Hamilton)
	Cyprinidae	<i>Danio</i> (<i>Brachydanio</i>) <i>rerio</i> (Hamilton)
	Cyprinidae	<i>Esomusdanrica</i> (Hamilton)
	Cyprinidae	<i>Rasboradaniconius</i> (Hamilton)
	Cyprinidae	<i>Amblypharyngodonmola</i> (Hamilton)
	Cyprinidae	<i>Torkhudree</i> Sykes
	Cyprinidae	<i>Torputitora</i> (Hamilton)
	Cyprinidae	<i>Tortor</i> (Hamilton)
	Cyprinidae	<i>Puntiusambassis</i> (Day)
	Cyprinidae	<i>Puntiusamphibius</i> (CuvierandValenciennes)
	Cyprinidae	<i>Puntiuschrysopoma</i> Cuvier
	Cyprinidae	<i>Puntiusconchoni</i> (Hamilton)
	Cyprinidae	<i>Puntiusdorsalis</i> (Jerdon)
	Cyprinidae	<i>Puntiusguganio</i> (Hamilton)
	Cyprinidae	<i>Puntiuspinnauratus</i> (Day)
	Cyprinidae	<i>Puntiussarana</i> (Hamilton)

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Cyprinidae	<i>Puntiussophore</i> (Hamilton)
Cyprinidae	<i>Puntiustictoticto</i> (Hamilton)
Cyprinidae	<i>Puntiustitius</i> (Hamilton)
Cyprinidae	<i>Oreichthyscosuatis</i> (Hamilton)
Cyprinidae	<i>Catlacatla</i> (Hamilton)
Cyprinidae	<i>Cirrhinusmrigala</i> (Hamilton)
Cyprinidae	<i>Cirrhinusreba</i> (Hamilton)
Cyprinidae	<i>Garragotyla</i> (Gray)
Cyprinidae	<i>Garralamta</i> (Hamilton)
Cyprinidae	<i>Garramullya</i> (Sykes)
Cyprinidae	<i>Labeobata</i> (Hamilton)
Cyprinidae	<i>Labeobogut</i> Sykes
Cyprinidae	<i>Labeocalbasu</i> (Hamilton)
Cyprinidae	<i>Labeodyocheilus</i> (McClelland)
Cyprinidae	<i>Labeofimbriatus</i> (Bloch)
Cyprinidae	<i>Labeogonius</i> (Hamilton)
Cyprinidae	<i>Labeorohita</i> (Hamilton)
Cyprinidae	<i>Osteobramacotiocotio</i> (Hamilton)
Cyprinidae	<i>Crossocheiluslatiuslatius</i> (Hamilton)
Psilorhynchidae	<i>Parapsilorhynchustentaculatus</i> (Ann)
Cobitidae	<i>Noemacheilusbotia</i> (Hamilton)
Cobitidae	<i>Noemacheilusdayi</i> Hora
Cobitidae	<i>Noemacheilusevezardi</i> Day
Cobitidae	<i>Lepidocephalichthysguntea</i> (Hamilton)
Siluridae	<i>Ompokbimaculatus</i> (Bloch)
Siluridae	<i>Wallagoattu</i> (Bloch&Schneider)
Bagridae	<i>Mystus(Mystus)bleekeri</i> (Day)
Bagridae	<i>Mystus(Mystus)cavasius</i> (Hamilton)

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	Bagridae	<i>Mystus(Mystus)vittatus(Bloch)</i>
	Bagridae	<i>Mystus(Osteobagrus)aor(Hamilton)</i>
	Bagridae	<i>Mystus(Osteobagrus)seenghala(sykes)</i>
	Bagridae	<i>RitapavimentataGunther</i>
	Amblycipitidae	<i>Amblycepsmangois(Hamilton)</i>
	Sisoridae	<i>Gagataitchkeea(Sykes)</i>
	Sisoridae	<i>Glyptothoraxlonah(Sykes)</i>
	Sisoridae	<i>LaguviaribeiroiHora</i>
	Schilbeidae	<i>Clupisomagarua(Hamilton)</i>
	Heteropneustidae	<i>Heteropneustesfossilis(Bloch)</i>
	Claridae	<i>Clariasbatrachus(Linnaeus)</i>
Anguilliformes	Anguillidae	<i>Anguillabengalensis(Gray&Hardw)</i>
Beloniformes	Belonidae	<i>Xenentodoncancila(Hamilton)</i>
Ophiocephaliformes	Channidae	<i>Channagachua(Hamilton)</i>
	Channidae	<i>Channamarulius(Hamilton)</i>
	Channidae	<i>Channapunctatus(Bloch)</i>
Perciformes	Centropomidae	<i>Chandanama(Hamilton)</i>
	Centropomidae	<i>Chandaranga(Hamilton)</i>
	Nandidae	<i>Badisbadis(Hamilton)</i>
	Nandidae	<i>Nandusnandus(Hamilton)</i>
	Gobiidae	<i>Glossogobiusgiurisgiuris(Hamilton)</i>
Mastocembeliformes	Mastocembelidae	<i>Mastocembelusarmatusarmatus(Lacepede)</i>
	Mastocembelidae	<i>Mastocembeluspancalus(Hamilton)</i>

Annexure 4.4.2: Checklist of Fishes from (2. CIFRI – River Narmada: Its Environment and Fisheries (Bulletin No. 157) 2009

Family	Fish Taxa	Location
Cyprinidae	<i>Labeo bata</i> (Hamilton-Buchanan)	Amarkantak to Gadarwada
Cyprinidae	<i>Labeo gonius</i> (Hamilton-Buchanan)	Amarkantak to Gadarwada
Cyprinidae	<i>Labeo rohita</i> (Hamilton-Buchanan)	Amarkantak to Gadarwada
Bagridae	<i>Aorichthys aor</i> (Hamilton-Buchanan)	Amarkantak to Gadarwada
Bagridae	<i>Aorichthys seenghala</i> (Sykes)	Amarkantak to Gadarwada
Cyprinidae	<i>Aspidoparia morar</i> (Hamilton-Buchanan)	Amarkantak to Gadarwada
Cyprinidae	<i>Catla catla</i> (Hamilton-Buchanan)	Amarkantak to Gadarwada
Channidae	<i>Channa punctatus</i> (Hamilton-Buchanan)	Amarkantak to Gadarwada
Cyprinidae	<i>Cirrhinus mrigala</i> (Hamilton-Buchanan)	Amarkantak to Gadarwada
Cyprinidae	<i>Cirrhinus reba</i> (Hamilton-Buchanan)	Amarkantak to Gadarwada
Clariidae	<i>Clarias batrachus</i> (Scopoli)	Amarkantak to Gadarwada
Schilbeidae	<i>Clupisoma garua</i> (Hamilton-Buchanan)	Amarkantak to Gadarwada
Heteropneustidae	<i>Heteropneustes fossilis</i> (Bloch)	Amarkantak to Gadarwada
Cyprinidae	<i>Labeo calbasu</i> (Hamilton-Buchanan)	Amarkantak to Gadarwada
Cyprinidae	<i>Labeo fimbriatus</i> (Bloch)	Amarkantak to Gadarwada
Mastacembelidae	<i>Mastacembelus armatus</i> (Lacepède)	Amarkantak to Gadarwada
Notopteridae	<i>Notopterus notopterus</i> (Pallas)	Amarkantak to Gadarwada
Cyprinidae	<i>Puntius sophore</i> (Hamilton-Buchanan)	Amarkantak to Gadarwada
Bagridae	<i>Rita pavimentata</i> (Valenciennes)	Amarkantak to Gadarwada
Cyprinidae	<i>Salmostoma bacaila</i> (Hamilton-Buchanan)	Amarkantak to Gadarwada
Cyprinidae	<i>Tor tor</i> (Hamilton-Buchanan)	Amarkantak to Gadarwada
Siluridae	<i>Wallago attu</i> (Schneider)	Amarkantak to Gadarwada
Bagridae	<i>Mystus bleekeri</i> (Day)	Gadarwada to Harsud
Bagridae	<i>Aorichthys aor</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Bagridae	<i>Aorichthys seenghala</i> (Hamilton-Buchanan)	Gadarwada to Harsud

Sisoridae	<i>Bagarius bagarius</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Barilius barila</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Barilius bendelisis</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Catla catla</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Ambassidae	<i>Chanda nama</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Channidae	<i>Channa marulius</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Channidae	<i>Channa punctatus</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Channidae	<i>Channa striatus</i> (Bloch)	Gadarwada to Harsud
Cyprinidae	<i>Chela laubuca</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Cirrhinus mrigala</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Cirrhinus reba</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Schilbeidae	<i>Clupisoma garua</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Esomus danricus</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Schilbeidae	<i>Eutropiichthys vacha</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Garra gotyla gotyla</i> (Gray)	Gadarwada to Harsud
Gobiidae	<i>Glossogobius giuris</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Labeo bata</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Labeo boggut</i> (Sykes)	Gadarwada to Harsud
Cyprinidae	<i>Labeo calbasu</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Labeo dyocheilus</i> (McClelland)	Gadarwada to Harsud
Cyprinidae	<i>Labeo fimbriatus</i> (Bloch)	Gadarwada to Harsud
Cyprinidae	<i>Labeo gonius</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Labeo pangusia</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Labeo rohita</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cobitidae	<i>Lepidocephalichthys guntea</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Prawns	<i>Macrobrachium lamarrei</i>	Gadarwada to Harsud
Prawns	<i>Macrobrachium rosenbergii</i>	Gadarwada to Harsud

Mastacembelidae	<i>Macrognathus pancalus</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Mastacembelidae	<i>Mastacembelus armatus</i> (Lacepède)	Gadarwada to Harsud
Bagridae	<i>Mystus cavasius</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Bagridae	<i>Mystus vittatus</i> (Bloch)	Gadarwada to Harsud
Nandidae	<i>Nandus nandus</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Nemacheilus beavani</i> (Günther)	Gadarwada to Harsud
Cyprinidae	<i>Nemacheilus botia</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Notopteridae	<i>Notopterus notopterus</i> (Pallas)	Gadarwada to Harsud
Siluridae	<i>Ompok bimaculatus</i> (Bloch)	Gadarwada to Harsud
Siluridae	<i>Ompok pabda</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Channidae	<i>Ophiocephalus gachua</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Osteobrama cotio cotio</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Osteobrama vigorsii</i> (Sykes)	Gadarwada to Harsud
Cyprinidae	<i>Parluciosoma daniconius</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Ambassidae	<i>Pseudambassis ranga</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Puntius sarana sarana</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Puntius sophore</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Puntius ticto</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Bagridae	<i>Rita pavimentata</i> (Valenciennes)	Gadarwada to Harsud
Cyprinidae	<i>Salmostoma bacaila</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Schilbeidae	<i>Silonia silondia</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Tor putitora</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Tor tor</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Siluridae	<i>Wallago attu</i> (Schneider)	Gadarwada to Harsud
Belonidae	<i>Xenentodon cancila</i> (Hamilton-Buchanan)	Gadarwada to Harsud
Cyprinidae	<i>Cyprinus carpio</i> (Linnaeus)	Harsud to Surpan
Cyprinidae	<i>Labeo bata</i> (Hamilton-Buchanan)	Harsud to Surpan

Cyprinidae	<i>Labeo gonius</i> (Hamilton-Buchanan)	Harsud to Surpan
Cyprinidae	<i>Labeo rohita</i> (Hamilton-Buchanan)	Harsud to Surpan
Cyprinidae	<i>Puntius sarana</i> (Hamilton-Buchanan)	Harsud to Surpan
Cyprinidae	<i>Puntius sophore</i> (Hamilton-Buchanan)	Harsud to Surpan
Cyprinidae	<i>Salmostoma bacaila</i> (Hamilton-Buchanan)	Harsud to Surpan
Schilbeidae	<i>Ailia coila</i> (Hamilton-Buchanan)	Harsud to Surpan
Bagridae	<i>Aorichthys aor</i> (Hamilton-Buchanan)	Harsud to Surpan
Bagridae	<i>Aorichthys seenghala</i> (Hamilton-Buchanan)	Harsud to Surpan
Cyprinidae	<i>Aspidoparia morar</i> (Hamilton-Buchanan)	Harsud to Surpan
Cyprinidae	<i>Catla catla</i> (Hamilton-Buchanan)	Harsud to Surpan
Ambassidae	<i>Chanda nama</i> (Hamilton-Buchanan)	Harsud to Surpan
Channidae	<i>Channa punctatus</i> (Hamilton-Buchanan)	Harsud to Surpan
Channidae	<i>Channa striatus</i> (Bloch)	Harsud to Surpan
Cyprinidae	<i>Cirrhinus mrigala</i> (Hamilton-Buchanan)	Harsud to Surpan
Cyprinidae	<i>Cirrhinus reba</i> (Hamilton-Buchanan)	Harsud to Surpan
Schilbeidae	<i>Clupisoma garua</i> (Hamilton-Buchanan)	Harsud to Surpan
Schilbeidae	<i>Eutropiichthys vacha</i> (Hamilton-Buchanan)	Harsud to Surpan
Gobiidae	<i>Glossogobius giuris</i> (Hamilton-Buchanan)	Harsud to Surpan
Cyprinidae	<i>Labeo calbasu</i> (Hamilton-Buchanan)	Harsud to Surpan
Cyprinidae	<i>Labeo fimbriatus</i> (Bloch)	Harsud to Surpan
Prawns	<i>Macrobrachium rosenbergii</i>	Harsud to Surpan
Mastacembelidae	<i>Macrognathus pancalus</i> (Hamilton-Buchanan)	Harsud to Surpan
Mastacembelidae	<i>Mastacembelus armatus</i> (Lacepède)	Harsud to Surpan
Bagridae	<i>Mystus cavasius</i> (Hamilton-Buchanan)	Harsud to Surpan
Notopteridae	<i>Notopterus chitala</i> (Hamilton-Buchanan)	Harsud to Surpan
Notopteridae	<i>Notopterus notopterus</i> (Pallas)	Harsud to Surpan
Siluridae	<i>Ompok bimaculatus</i> (Bloch)	Harsud to Surpan

Pangasiidae	<i>Pangasius pangasius</i> (Hamilton-Buchanan)	Harsud to Surpan
Ambassidae	<i>Pseudambassis ranga</i> (Hamilton-Buchanan)	Harsud to Surpan
Bagridae	<i>Rita pavimentata</i> (Valenciennes)	Harsud to Surpan
Bagridae	<i>Rita rita</i> (Hamilton-Buchanan)	Harsud to Surpan
Schilbeidae	<i>Silonia silondia</i> (Hamilton-Buchanan)	Harsud to Surpan
Cyprinidae	<i>Tor tor</i> (Hamilton-Buchanan)	Harsud to Surpan
Siluridae	<i>Wallago attu</i> (Schneider)	Harsud to Surpan
Belonidae	<i>Xenentodon cancila</i> (Hamilton-Buchanan)	Harsud to Surpan
Channidae	<i>Channa striatus</i> (Bloch)	Surpan to Gulf of Cambay
Schilbeidae	<i>Ailia coila</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Amblypharyngodon mola</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Bagridae	<i>Aorichthys aor</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Bagridae	<i>Aorichthys bleekeri</i> (Day)	Surpan to Gulf of Cambay
Bagridae	<i>Aorichthys seenghala</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Bagridae	<i>Aorichthys vittatus</i> (Bloch)	Surpan to Gulf of Cambay
Ariidae	<i>Arius arius</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Catla catla</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Centropomidae	<i>Chanda nama</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Centropomidae	<i>Chanda ranga</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Channidae	<i>Channa marulius</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Channidae	<i>Channa punctatus</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Chanidae	<i>Chanos chanos</i> (Forsskal)	Surpan to Gulf of Cambay
Cyprinidae	<i>Chela laubuca</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Cirrhinus mrigala</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Cirrhinus reba</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Schilbeidae	<i>Clupisoma garua</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cynoglossidae	<i>Cynoglossus elongatus</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay

Cichlidae	<i>Etroplus maculatus</i> (Bloch)	Surpan to Gulf of Cambay
Cichlidae	<i>Etroplus suratensis</i> (Bloch)	Surpan to Gulf of Cambay
Schilbeidae	<i>Eutropiichthys vacha</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Garra gotyla gotyla</i> (Gray)	Surpan to Gulf of Cambay
Gobiidae	<i>Glossogobius giuris</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Harpodontidae	<i>Harpadon nehereus</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Labeo bata</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Labeo boggut</i> (Sykes)	Surpan to Gulf of Cambay
Cyprinidae	<i>Labeo calbasu</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Labeo fimbriatus</i> (Bloch)	Surpan to Gulf of Cambay
Cyprinidae	<i>Labeo gonius</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Labeo pangusia</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Labeo rohita</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Centropomidae	<i>Lates calcarifer</i> (Bloch)	Surpan to Gulf of Cambay
Mugilidae	<i>Liza macrolepis</i> (Smith)	Surpan to Gulf of Cambay
Mugilidae	<i>Liza parsia</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Lutianidae	<i>Lutjanus argentimaculatus</i> (Forsskal)	Surpan to Gulf of Cambay
Prawns	<i>Macrobrachium rosenbergii</i>	Surpan to Gulf of Cambay
Mastacembelidae	<i>Mastacembelus armatus</i> (Lacepède)	Surpan to Gulf of Cambay
Mugilidae	<i>Mugil cephalus</i> (Linnaeus)	Surpan to Gulf of Cambay
Bagridae	<i>Mystus cavasius</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Notopteridae	<i>Notopterus notopterus</i> (Pallas)	Surpan to Gulf of Cambay
Siluludae	<i>Ompok bimaculatus</i> (Bloch)	Surpan to Gulf of Cambay
Cyprinidae	<i>Osteobrama cotio</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Osteobrama vigorsii</i> (Sykes)	Surpan to Gulf of Cambay
Prawns	<i>Penaeus indicus</i>	Surpan to Gulf of Cambay
Gobiidae	<i>Periophthalmodon schlosseri</i> (Pallas)	Surpan to Gulf of Cambay

Gobiidae	<i>Periophthalmus variabilis</i> (Eggert)	Surpan to Gulf of Cambay
Polynemoidae	<i>Polynemus indicus</i> (Shaw)	Surpan to Gulf of Cambay
Polynemoidae	<i>Polynemus tetradactylus</i> (Shaw)	Surpan to Gulf of Cambay
Cyprinidae	<i>Puntius sarana</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Puntius sophore</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Rasbora daniconius</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Mugilidae	<i>Rhinomugil corsula</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Bagridae	<i>Rita pavimentata</i> (Valenciennes)	Surpan to Gulf of Cambay
Bagridae	<i>Rita rita</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Salmostoma bacaila</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Cyprinidae	<i>Salmostoma phulo</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Sciaenidae	<i>Sciaenoides biauritus</i> (Blyth)	Surpan to Gulf of Cambay
Clupeidae	<i>Setipinna phasa</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Clupeidae	<i>Tenualosa ilisha</i> (Whitehead)	Surpan to Gulf of Cambay
Cyprinidae	<i>Tor tor</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay
Siluludae	<i>Wallago attu</i> (Schneider)	Surpan to Gulf of Cambay
Belonidae	<i>Xenentodon cancila</i> (Hamilton-Buchanan)	Surpan to Gulf of Cambay

Annexure: 4.4.3 Checklist of the fishes of the Narmada Basin (WII, 2022)

Order	Family	Fish Taxa
Siluriformes	Ailiidae	<i>Ailia coila</i> (Hamilton, 1822)
Siluriformes	Amblycepidae	<i>Amblyceps mangois</i> (Hamilton, 1822)
Cypriniformes	Danionidae	<i>Amblypharyngodon microlepis</i> (Bleeker, 1853)
Cypriniformes	Danionidae	<i>Amblypharyngodon mola</i> (Hamilton, 1822)
Anabantiformes	Anabantidae	<i>Anabas testudineus</i> (Bloch, 1792)
Anguilliformes	Anguillidae	<i>Anguilla bengalensis</i> (McClelland, 1844)
Cyprinodontiformes	Aplocheilidae	<i>Aplocheilus panchax</i> (Hamilton, 1822)
Siluriformes	Ariidae	<i>Arius arius</i> (Hamilton, 1822)
Siluriformes	Ariidae	<i>Arius gagora</i> (Hamilton, 1822)
Anabantiformes	Badidae	<i>Badis badis</i> (Bleeker, 1855)
Siluriformes	Sisoridae	<i>Bagarius bagarius</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Bangana dero</i> (Hamilton, 1822)
Cypriniformes	Danionidae	<i>Barilius barila</i> (Hamilton, 1822)
Gobiiformes	Gobiidae	<i>Boleophthalmus dussumieri</i> (Valenciennes, 1837)
Cypriniformes	Danionidae	<i>Cabdio jaya</i> (Hamilton 1822)
Cypriniformes	Danionidae	<i>Cabdio morar</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Chagunius chagunio</i> (Hamilton, 1822)
Perciformes	Ambassidae	<i>Chanda nama</i> (Hamilton, 1822)
Anabantiformes	Channidae	<i>Channa gachua</i> (Hamilton, 1822)
Anabantiformes	Channidae	<i>Channa marulius</i> (Hamilton, 1822)
Anabantiformes	Channidae	<i>Channa punctatus</i> (Bloch, 1793)
Anabantiformes	Channidae	<i>Channa striatus</i> (Bloch, 1793)
Gonorynchiformes	Chanidae	<i>Chanos chanos</i> (Forsskal, 1775)
Cypriniformes	Cyprinidae	<i>Cirrhinus cirrhosus</i> (Bloch, 1795)
Cypriniformes	Cyprinidae	<i>Cirrhinus mirgala</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Cirrhinus reba</i> (Hamilton, 1822)
Siluriformes	Clariidae	<i>Clarias gariepinus</i> (Bruchell, 1822)
Siluriformes	Clariidae	<i>Clarias magur</i> (Hamilton, 1822)
Siluriformes	Ailiidae	<i>Clupisoma garua</i> (Hamilton, 1822)
Cypriniformes	Xenocyprididae	<i>Ctenopharyngodon idella</i> (Valenciennes, 1844)
Cypriniformes	Cyprinidae	<i>Cyprinus carpio</i> (Linnaeus, 1752)
Cypriniformes	Danionidae	<i>Danio rerio</i> (Hamilton, 1822)
Cypriniformes	Danionidae	<i>Devario aequipinnatus</i> (McClelland, 1842)
Cypriniformes	Danionidae	<i>Devario devario</i> (Hamilton 1822)
Gobiiformes	Eleotridae	<i>Eleotris fusca</i> (Forster, 1801)
Perciformes	Polynemidae	<i>Eleutheronema tetradactylum</i> (Shaw, 1804)
Clupiformes	Clupeidae	<i>Escualosa thoracata</i> (Valenciennes, 1847)
Cypriniformes	Danionidae	<i>Esomos danricus</i> (Hamilton, 1822)
Cichliformes	Cichlidae`	<i>Etroplus suratensis</i> (Bloch, 1790)
Siluriformes	Ailiidae	<i>Eutropiichthys murius</i> (Hamilton, 1822)
Siluriformes	Ailiidae	<i>Eutropiichthys vacha</i> (Hamilton, 1822)

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Siluriformes	Sisoridae	<i>Gagata cenia</i> (Hamilton, 1822)
Siluriformes	Sisoridae	<i>Gagata itchkeea</i> (Skyles, 1839)
Cypriniformes	Cyprinidae	<i>Garra gotyla</i> (Grey, 1830)
Cypriniformes	Cyprinidae	<i>Garra lamta</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Garra mullya</i> (Skyles, 1839)
Perciformes	Gerreidae	<i>Gerres filamentosus</i> (Cuvier, 1829)
Gobiiformes	Gobiidae	<i>Glossogobius giuris</i> (Hamilton, 1822)
Siluriformes	Sisoridae	<i>Glyptothorax lonah</i> (Skyles, 1839)
Siluriformes	Sisoridae	<i>Glyptothorax telchitta</i> (Hamilton, 1822)
Clupiformes	Clupeidae	<i>Gonialosa manmina</i> (Hamilton, 1822)
Clupeiformes	Clupeidae	<i>Gudusia chapara</i> (Hamilton, 1822)
Siluriformes	Heteropneustidae	<i>Heteropneustes fossilis</i> (Bloch, 1794)
Cypriniformes	Cyprinidae	<i>Hypophthalmichthys molitrix</i> (Valenciennes, 1844)
Cypriniformes	Cyprinidae	<i>Hypophthalmichthys nobilis</i> (Richardson, 1845)
Anguilliformes	Hemiramphidae	<i>Hyporhamphus limbatus</i> (Valenciennes, 1847)
Anguilliformes	Hemiramphidae	<i>Hyporhamphus quoyi</i> (Valenciennes, 1847)
Acanthuriformes	Sciaenidae	<i>Johnius belangerii</i> (Cuvier, 1830)
Cypriniformes	Cyprinidae	<i>Labeo angra</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Labeo bata</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Labeo boga</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Labeo boggut</i> (Skyles, 1839)
Cypriniformes	Cyprinidae	<i>Labeo calbasu</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Labeo catla</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Labeo dyocheilus</i> (McClelland, 1839)
Cypriniformes	Cyprinidae	<i>Labeo fimbriatus</i> (Bloch, 1795)
Cypriniformes	Cyprinidae	<i>Labeo gonius</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Labeo pangusia</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Labeo rohita</i> (Hamilton, 1822)
Perciformes	Latidae	<i>Lates calcarifer</i> (Bloch, 1790)
Cypriniformes	Danionidae	<i>Laubuka laubuca</i> (Hamilton, 1822)
Cypriniformes	Cobitidae	<i>Lepidocephalichthys guntea</i> (Hamilton, 1822)
Perciformes	Polynemidae	<i>Leptomelanosoma indicum</i> (Shaw, 1804)
Synbranchiformes	Mastacembelidae	<i>Macrognathus pancalus</i> (Hamilton, 1822)
Synbranchiformes	Mastacembelidae	<i>Mastacembelus armatus</i> (Lacepede, 1800)
Elopiformes	Megalopidae	<i>Megalops cyprinoides</i> (Broussonet, 1782)
Mugiliformes	Mugilidae	<i>Mugil cephalus</i> (Linnaeus, 1758)
Siluriformes	Bagridae	<i>Mystus bleekeri</i> (Day, 1877)
Siluriformes	Bagridae	<i>Mystus cavasius</i> (Hamilton, 1822)
Siluriformes	Bagridae	<i>Mystus gulio</i> (Bloch, 1794)
Siluriformes	Bagridae	<i>Mystus tengara</i> (Hamilton, 1822)
Siluriformes	Bagridae	<i>Mystus vittatus</i> (Bloch, 1794)
Perciformes	Nandidae	<i>Nandus nandus</i> (Hamilton, 1822)
Osteoglossiformes	Notopteridae	<i>Notopterus chitala</i> (Hamilton, 1822)
Osteoglossiformes	Notopteridae	<i>Notopterus synurus</i> (Bloch & Schneider 1801)

Siluriformes	Siluridae	<i>Ompok bimaculatus</i> (Bloch, 1794)
Siluriformes	Siluridae	<i>Ompok pabda</i> (Hamilton, 1822)
Siluriformes	Siluridae	<i>Ompok pabo</i> (Hamilton, 1822)
Cypriniformes	Danionidae	<i>Opsarius bendelisis</i> (Hamilton, 1807)
Cypriniformes	Danionidae	<i>Opsarius radiolatus</i> (Günther 1868)
Cypriniformes	Cyprinidae	<i>Oreichthys cosuatis</i> (Hamilton, 1822)
Cichliformes	Cichilidae	<i>Oreochromis mossambica</i> (Peters, 1852)
Cypriniformes	Cyprinidae	<i>Osteobrama cotio</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Osteobrama vigorsii</i> (Skyles, 1839)
Siluriformes	Ariidae	<i>Osteogeneiosus militaris</i> (Linnaeus, 1758)
Acanthuriformes	Sciaenidae	<i>Otolithoides pama</i> (Hamilton, 1822)
Siluriformes	Pangasiidae	<i>Pangasius pangasius</i> (Hamilton, 1822)
Cypriniformes	Nemacheilidae	<i>Paracanthocobitis botia</i> (Hamilton, 1822)
Perciformes	Ambassidae	<i>Parambassis baculis</i> (Hamilton, 1822)
Perciformes	Ambassidae	<i>Parambassis ranga</i> (Hamilton, 1822)
Clupiformes	Pristigasteridae	<i>Pellona ditchela</i> (Valenciennes, 1847)
Cypriniformes	Cyprinidae	<i>Pethia conchonius</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Pethia gelius</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Pethia guganio</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Pethia ticto</i> (Hamilton 1822)
Mugiliformes	Mugilidae	<i>Planiliza macrolepis</i> (Smith 1846)
Mugiliformes	Mugilidae	<i>Planiliza macrolepis</i> (Smith, 1846)
Mugiliformes	Mugilidae	<i>Planiliza planiceps</i> (Valenciennes, 1836)
Cyprinodontiformes	Poeciliidae	<i>Poecilia reticulata</i> (Peters, 1859)
Cichliformes	Cichlidae	<i>Pseudetroplus maculatus</i> (Bloch, 1795)
Siluriformes	Sisoridae	<i>Pseudolaguvia ribeiroi</i> (Hora, 1921)
Cypriniformes	Cyprinidae	<i>Puntius ambassis</i> (Day, 1869)
Cypriniformes	Cyprinidae	<i>Puntius amphibius</i> (Valenciennes, 1842)
Cypriniformes	Cyprinidae	<i>Puntius chola</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Puntius dorsalis</i> (Jerdon, 18492)
Cypriniformes	Cyprinidae	<i>Puntius sophore</i> (Hamilton, 1823)
Cypriniformes	Danionidae	<i>Rasbora daniconius</i> (Hamilton, 1822)
Cypriniformes	Danionidae	<i>Rasbora rasbora</i> (Hamilton, 1822)
Mugiliformes	Mugilidae	<i>Rhinomugil corsula</i> (Hamilton, 1822)
Siluriformes	Bagridae	<i>Rita gogra</i> (Skyles, 1839)
Siluriformes	Bagridae	<i>Rita rita</i> (Hamilton, 1822)
Cypriniformes	Danionidae	<i>Salmophasia bacalia</i> (Hamilton, 1822)
Cypriniformes	Danionidae	<i>Salmophasia boopis</i> (Day, 1874)
Cypriniformes	Danionidae	<i>Salmostoma balookee</i> (Skyles, 1839)
Cypriniformes	Danionidae	<i>Salmostoma orissaensis</i> (Banarescu, 1968)
Cypriniformes	Danionidae	<i>Salmostoma phulo</i> (Hamilton, 1822)
Perciformes	Scatophagidae	<i>Scatophagus argus</i> (Linnaeus, 1766)
Clupiformes	Engraulidae	<i>Setipinna phasa</i> (Hamilton, 1822)
Perciformes	Sillaginidae	<i>Sillago sihama</i> (Forsskal, 1775)
Siluriformes	Ailiidae	<i>Silonia silondia</i> (Hamilton, 1822)
Siluriformes	Bagridae	<i>Sperata aor</i> (Hamilton, 1822)

Siluriformes	Bagridae	<i>Sperata seenghala</i> (Skyles, 1839)
Cypriniformes	Cyprinidae	<i>Systomus sarana</i> (Hamilton, 1822)
Cypriniformes	Cyprinidae	<i>Tariqilabeo latius</i> (Hamilton 1822)
Clupiformes	Clupeidae	<i>Tenuالosa ilisha</i> (Hamilton, 1822)
Centrarchiformes	Teraponitidae	<i>Therapon jarbua</i> (Forsskal, 1775)
Cypriniformes	Cyprinidae	<i>Tor khudree</i> (Skyles, 1839)
Cypriniformes	Cyprinidae	<i>Tor tor</i> (Hamilton, 1822)
Anabantiformes	Osphronemidae	<i>Trichogaster chuna</i> (Hamilton, 1822)
Anabantiformes	Osphronemidae	<i>Trichogaster fasciata</i> (Bloch, 1801)
Anabantiformes	Osphronemidae	<i>Trichogaster lalius</i> (Hamilton, 1822)
Gobiiformes	Oxudercidae	<i>Trypauchen vagina</i> (Bloch, 1801)
Mulliformes	Mullidae	<i>Upeneus vittatus</i> (Forsskal, 1775)
Siluriformes	Siluridae	<i>Wallago attu</i> (Schneider, 1801)
Beloniformes	Belonidae	<i>Xenentodon cancila</i> (Hamilton, 1822)

Annexure 4.4.4: Checklist of the Upper Narmada Basin fishes(CIFRI – **Fishes of River Narmada: A Field Identification Manual**)

Order	Family	Zoological Name	Location
Cypriniformes	Balitoridae	<i>Acanthocobitis botia</i>	Bargi (Bargi River), Dindori (Narmada River), Rusa (Tudar River), Jabalpur, Mandla, Sagar, Shahdol, Betul, Dindori, Narsinghpur
Siluriformes	Schilbeidae	<i>Ailia coila</i>	Shahdol
Siluriformes	Amblycipitidae	<i>Amblyceps mangois</i>	Mandla, Dindori, Jabalpur, Narsinghpur, Shahdol
Cypriniformes	Cyprinidae	<i>Amblypharyngodon melettinus</i>	Jabalpur, Mandla, Narsinghpur
Cypriniformes	Cyprinidae	<i>Amblypharyngodon mola</i>	Mandla, Tilwaraghat (Jabalpur) and Kareli (Narsinghpur), Dudhi Basin, Jabalpur, Shahdol, Dindori, Narsinghpur, Mandla, Betul
Perciformes	Anabantidae	<i>Anabas testudineus</i>	Bergi (Jabalpur), Mandla, Kareli (Narsinghpur), Chandan Ghat (Dindori), Mandla (Banjar River), Jabalpur
Anguilliformes	Anguillidae	<i>Anguilla bengalensis bengalensis</i>	Shahdol, Dindori, Mandla, Narsinghpur, Jabalpur
Siluriformes	Bagridae	<i>Aorichthys aor</i>	Amarkantak, Mandla, Jabalpur, Narsinghpur, Kareli and Gadawada, Jabalpur
Siluriformes	Bagridae	<i>Aorichthys seenghala</i>	Amarkantak, Mandla, Jabalpur, Narsinghpur, Kareli and Gadawada
Cyprinodontiformes	Aplocheilidae	<i>Aplocheilus lineatus</i>	Dindori, Mandla, Narsinghpur, Jabalpur
Cyprinodontiformes	Aplocheilidae	<i>Aplocheilus panchax</i>	Tilwaraghat (Jabalpur)
Cypriniformes	Cyprinidae	<i>Aristichthys nobilis</i>	Jabalpur
Cypriniformes	Cyprinidae	<i>Aspidoparia jaya</i>	Narsinghpur
Cypriniformes	Cyprinidae	<i>Aspidoparia morar</i>	Bergi (Jabalpur), Kareli (Narsinghpur), Mandla, Amarkantak, Kareli and Gadawada, Jabalpur, Shahdol
Perciformes	Badidae	<i>Badis badis</i>	Mandla, Kareli (Narsinghpur), Jabalpur, Mandla, Sagar, Dindori, Narsinghpur
Siluriformes	Sisoridae	<i>Bagarius bagarius</i>	Chandanghat (Dindori), Jabalpur, Sagar, Shahdol
Cypriniformes	Cyprinidae	<i>Barilius barila</i>	Amarkantak and Mandla Dudhi Basin, Chhindwara, Shahdol, Dindori, Mandla, Jabalpur, Narsinghpur, Betul
Cypriniformes	Cyprinidae	<i>Barilius barna</i>	Chhindwara, Narsinghpur, Sagar, Shahdol
Cypriniformes	Cyprinidae	<i>Barilius bendelisis</i>	Mandla, Pushkar Dam (Amarkantak), Bhimkhundi (Dindori), Dindori, Mandla (Banjar River), Rusa (Tudar

			River), Tawa (Tendwa River), Betul, Chhindwara, Jabalpur, Mandla, Narsinghpur, Sagar, Shahdol
Cypriniformes	Cyprinidae	<i>Barilius evezardi</i>	Jabalpur, Dindori, Narsinghpur, Betul
Cypriniformes	Cyprinidae	<i>Barilius radiolatus</i>	Mandla, Jabalpur
Cypriniformes	Cyprinidae	<i>Barilius shacra</i>	Jabalpur, Shahdol
Cypriniformes	Cyprinidae	<i>Barilius vagra</i>	Jabalpur, Shahdol, Dindori, Mandla
Cypriniformes	Cyprinidae	<i>Catla catla</i>	Mandla to Gadawara, Kareli (Narsinghpur), Jabalpur, Dudhi Basin, Kapildhara (Amarkantak), Chandanghat (Dindori), Bargi (Bargi River), Dindori (Narmada River), Mandla (Banjar River), Tawa (Tendwa River), Indore, Jabalpur, Sagar, Shahdol, Dindori, Mandla, Narsinghpur, Betul
Cypriniformes	Cyprinidae	<i>Chagunius chagunio</i>	Mandla, Shahdol
Perciformes	Ambassidae	<i>Chanda baculis</i>	Jabalpur
Perciformes	Ambassidae	<i>Chanda nama</i>	Jabalpur, Chandan Ghat (Dindori), Pushkar Dam (Amarkantak), Bhimkhundi (Dindori), Mandla, Bargi (Bargi River), Mandla (Banjar River), Tawa (Tendwa River), Dindori, Mandla, Chhindwara, Jabalpur, Narsinghpur, Raisen
Perciformes	Ambassidae	<i>Chanda ranga</i>	Chandan Ghat (Dindori), Jabalpur, Raisen
Perciformes	Channidae	<i>Channa marulius</i>	Bergi (Jabalpur), Kareli (Narsinghpur), Chandan Ghat (Dindori), Bhimkhundi (Dindori), Jabalpur, Dindori (Narmada River), Sultanpur (Barna River), Jabalpur, Sagar, Shahdol, Dindori, Mandla, Narsinghpur, Betul
Perciformes	Channidae	<i>Channa orientalis</i>	Mandla, Jabalpur, Betul, Damoh, Shahdol, Sagar, Dindori, Narsinghpur, Bergi (Jabalpur), Mandla, Kareli (Narsinghpur), Tilwaraghat (Jabalpur), Kapildhara (Amarkantak), Chandanghat (Dindori), Bhamni (Banjar River), Dindori (Narmada River), Mandla (Banjar River)
Perciformes	Channidae	<i>Channa punctatus</i>	Bergi (Jabalpur), Mandla, Kareli (Narsinghpur), Tilwaraghat (Jabalpur), Gadawara, Amarkantak, Dudhi Basin, Jabalpur, Mandla, Chandan Ghat (Dindori), Bhimkhundi (Dindori), Kosamghat (Jabalpur), Bhamni (Banjar River), Mandla (Banjar River), Sultanpur (Barna River), Tawa (Tendwa River), Jabalpur, Raisen, Mandla, Seoni, Narsinghpur, Sagar, Shahdol, Dindori, Betul

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Perciformes	Channidae	<i>Channa striatus</i>	Jabalpur, Chhindwara, Sagar, Shahdol, Chandan Ghat (Dindori), Jabalpur, Pushkar Dam (Amarkantak), Bergi (Jabalpur), Kareli (Narsinghpur)
Cypriniformes	Cyprinidae	<i>Chela cachius</i>	Jabalpur, Narsinghpur, Raisen, Shahdol
Cypriniformes	Cyprinidae	<i>Chela laubuca</i>	Kapildhara (Amarkantak), Mandla, Dindori, Dudhi Basin Tilwaraghat (Jabalpur), Betul, Raisen, Shahdol, Narsinghpur, Jabalpur
Cypriniformes	Cyprinidae	<i>Chela dadyburjori</i>	Narsinghpur, Raisen
Osteoglossiformes	Notopteridae	<i>Chitala chitala</i>	Jabalpur, Shahdol
Cypriniformes	Cyprinidae	<i>Cirrhinus cirrhosus</i>	Jabalpur, Jabalpur, Shahdol, Dindori, Mandla, Narsinghpur, Betul
Cypriniformes	Cyprinidae	<i>Cirrhinus mrigala</i>	Amarkantak, Mandla, Jabalpur, Narsinghpur, Kareli and Gadawada, Chandan Ghat (Dindori), Jabalpur, Bargi (Bargi River), Dindori (Narmada River), Tawa (Tendwa River)
Cypriniformes	Cyprinidae	<i>Cirrhinus reba</i>	Amarkantak, Gadawada, Kareli (Narsinghpur), Tilwaraghat (Jabalpur), Mandla, Jabalpur, Dudhi Basin, Chandan Ghat (Dindori), Bargi (Bargi River), Dindori (Narmada River), Mandla (Banjar River)
Siluriformes	Clariidae	<i>Clarias batrachus</i>	Mandla, Gadawada, Kareli (Narsinghpur), Chandan Ghat (Dindori), Tilwaraghat (Jabalpur), Bergi (Jabalpur), Kapildhara (Amarkantak), Jabalpur, Narsinghpur, Sagar, Seoni, Dindori, Barwani, Mandla
Siluriformes	Clariidae	<i>Clarias gariepinus</i>	Shahdol
Siluriformes	Schilbeidae	<i>Clupisoma garua</i>	Amarkantak, Mandla, Jabalpur, Narsinghpur, Kareli and Gadawada, Bargi (Bargi River), Mandla (Banjar River), Jabalpur, Indore, Shahdol, Sagar, Dindori, Mandla, Narsinghpur, Betul
Siluriformes	Schilbeidae	<i>Clupisoma montana</i>	Shahdol
Perciformes	Belontiidae	<i>Colisa fasciatus</i>	Shahdol
Perciformes	Belontiidae	<i>Colisa lalia</i>	Mandla, Tilwaraghat (Jabalpur)
Clupeiformes	Clupeidae	<i>Corica soborna</i>	Mandla, Tilwaraghat (Jabalpur)
Cypriniformes	Cyprinidae	<i>Crossocheilus latius latius</i>	Mandla, Tilwaraghat (Jabalpur), Jabalpur, Shahdol, Betul, Dindori, Mandla, Narsinghpur, Indore
Cypriniformes	Cyprinidae	<i>Cyprinus carpio</i>	Mandla, Jabalpur, Chandan Ghat (Dindori), Dindori (Narmada River), Mandla (Banjar River), Jabalpur, Narsinghpur
Cypriniformes	Cyprinidae	<i>Danio aequipinnatus</i>	Raisen, Sagar, Shahdol, Dindori, Mandla, Narsinghpur, Jabalpur, Betul

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Cypriniformes	Cyprinidae	<i>Danio rerio</i>	Narmada Kund (Amarkantak), Kapildhara (Amarkantak), Dindori, Chhindwara, Jabalpur, Mandla, Raisen, Shahdol, Dindori, Narsinghpur, Betul
Cypriniformes	Cyprinidae	<i>Devario devario</i>	Narsinghpur, Shahdol, Betul, Chhindwara, Jabalpur, Mandla, Narsinghpur, Raisen, Kareli (Narsinghpur) and Tilwaraghat (Jabalpur), Narmada Kund (Amarkantak), Pushkar Dam (Amarkantak), Kapildhara (Amarkantak), Bhimkhundi (Dindori), Dindori, Kosamghat (Jabalpur)
Cypriniformes	Cyprinidae	<i>Devario malabaricus</i>	Bargi (Bargi River), Mandla (Banjar River), Tawa (Tendwa River), Sagar
Siluriformes	Sisoridae	<i>Erethistes pusillus</i>	Shahdol
Siluriformes	Sisoridae	<i>Erethistoides montana</i>	Shahdol
Cypriniformes	Cyprinidae	<i>Esomus danricus</i>	Kosamghat (Jabalpur), Mandla (Banjar River), Chhindwara, Raisen, Shahdol, Dindori, Mandla, Narsinghpur, Jabalpur, Betul
Siluriformes	Schilbeidae	<i>Eutropiichthys murius</i>	Tilwaraghat (Jabalpur), Dindori (Narmada River), Shahdol
Siluriformes	Schilbeidae	<i>Eutropiichthys vacha</i>	Mandla, Jabalpur, Sagar, Shahdol, Dindori, Mandla, Narsinghpur
Siluriformes	Sisoridae	<i>Gagata cenia</i>	Narsinghpur, Shahdol, Dindori, Mandla, Jabalpur, Betul
Siluriformes	Sisoridae	<i>Gagata itchkeea</i>	Jabalpur
Cypriniformes	Cyprinidae	<i>Garra gotyla</i>	Dudhi Basin, Jabalpur, Raisen, Shahdol, Dindori, Mandla, Narsinghpur
Cypriniformes	Cyprinidae	<i>Garra lamta</i>	Chandan Ghat (Dindori), Pushkar Dam (Amarkantak), Kapildhara (Amarkantak), Kosamghat (Jabalpur), Chhindwara, Jabalpur, Indore, Sagar, Shahdol, Dindori, Mandla, Narsinghpur
Cypriniformes	Cyprinidae	<i>Garra mullya</i>	Jabalpur, Mandla, Narsinghpur, Dindori, Shahdol, Betul, Kapildhara (Amarkantak), Chandanghat (Dindori), Dindori (Narmada River), Sultanpur (Barna River)
Perciformes	Gobiidae	<i>Glossogobius giuris</i>	Bergi (Jabalpur), Tilwaraghat (Jabalpur), Kareli (Narsinghpur) Chandan Ghat (Dindori), Dudhi Basin, Bargi (Bargi River), Mandla (Banjar River), Sultanpur (Barna River), Tawa (Tendwa River), Bhimkhundi (Dindori), Chandanghat (Dindori), Mandla
Perciformes	Gobiidae	<i>Glossogobius giuris giuris</i>	Bhimkhundi (Dindori), Chandanghat (Dindori), Mandla, Jabalpur, Sagar, Shahdol, Betul, Dindori, Mandla, Narsinghpur
Siluriformes	Sisoridae	<i>Glyptothorax annandalei</i>	Shahdol

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Siluriformes	Sisoridae	<i>Glyptothorax indicus</i>	Shahdol
Siluriformes	Sisoridae	<i>Glyptothorax lonah</i>	Dindori (Narmada River), Mandla, Narsinghpur, Jabalpur
Siluriformes	Sisoridae	<i>Glyptothorax telchitta</i>	Jabalpur, Sagar, Shahdol
Siluriformes	Sisoridae	<i>Gogangra viridescens</i>	Shahdol
Clupeiformes	Clupeidae	<i>Goniolosa manmina</i>	Shahdol, Dindori, Mandla, Narsinghpur, Jabalpur
Siluriformes	Heteropneustidae	<i>Heteropneustes fossilis</i>	Jabalpur, Seoni, Sagar, Shahdol, Dindori, Mandla, Narsinghpur, Mandla, Kareli (Narsinghpur), Tilwaraghat (Jabalpur), Amarkantak, Gadawada, Chandanghat (Dindori), Sultanpur (Barna River), Tawa (Tendwa River)
Cypriniformes	Cyprinidae	<i>Hypophthalmichthys molitrix</i>	Mandla, Dindori, Mandla, Narsinghpur, Jabalpur
Cypriniformes	Cyprinidae	<i>Hypophthalmichthys nobilis</i>	Mandla
Cypriniformes	Cyprinidae	<i>Hypselobarbus curmuca</i>	Jabalpur
Cypriniformes	Homalopteridae	<i>Indoreonectes evezardi</i>	Dindori, Mandla, Betul, Narsinghpur, Jabalpur
Perciformes	Sciaenidae	<i>Johnius coitor</i>	Shahdol
Cypriniformes	Cyprinidae	<i>Labeo angra</i>	Dudhi Basin, Sagar
Cypriniformes	Cyprinidae	<i>Labeo ariza</i>	Jabalpur, Sagar, Shahdol, Dindori, Mandla, Narsinghpur, Betul
Cypriniformes	Cyprinidae	<i>Labeo bata</i>	Amarkantak, Gadawada, Kareli (Narsinghpur), Tilwaraghat (Jabalpur), Mandla, Jabalpur, Chandan Ghat (Dindori), Sultanpur (Barna River), Jabalpur, Shahdol, Mandla, Dindori, Narsinghpur, Betul
Cypriniformes	Cyprinidae	<i>Labeo boga</i>	Kareli (Narsinghpur), Tilwaraghat (Jabalpur), Mandla, Jabalpur, Shahdol
Cypriniformes	Cyprinidae	<i>Labeo boggut</i>	Bargi (Bargi River), Chakaldi (Kolar River), Dindori (Narmada River), Dudhi Basin, Jabalpur, Sagar, Shahdol, Dindori, Mandla, Narsinghpur, Betul
Cypriniformes	Cyprinidae	<i>Labeo calbasu</i>	Amarkantak, Kareli (Narsinghpur), Tilwaraghat (Jabalpur), Mandla to Gadawara, Jabalpur, Chandan Ghat (Dindori), Bhimkhundi (Dindori), Bargi (Bargi River), Dindori (Narmada River), Mandla (Banjar River), Sultanpur (Barna River), Tawa (Tendwa River), Betul, Jabalpur, Sagar, Shahdol, Dindori, Mandla, Narsinghpur
Cypriniformes	Cyprinidae	<i>Labeo dyocheilus</i>	Bhamni (Banjar River), Sultanpur (Barna River), Shahdol, Dindori, Mandla, Betul, Narsinghpur, Jabalpur

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Cypriniformes	Cyprinidae	<i>Labeo fimbriatus</i>	Amarkantak, Mandla, Jabalpur, Narsinghpur, Kareli and Gadawada, Bargi (Bargi River), Jabalpur, Shahdol, Dindori, Barwani, Mandla, Narsinghpur, Betul
Cypriniformes	Cyprinidae	<i>Labeo gonius</i>	Amarkantak, Gadawada, Kareli (Narsinghpur), Tilwaraghat (Jabalpur), Mandla, Kapildhara (Amarkantak), Jabalpur, Shahdol, Narsinghpur, Dindori, Mandla
Cypriniformes	Cyprinidae	<i>Labeo pangusia</i>	Dindori, Mandla, Narsinghpur, Jabalpur, Shahdol
Cypriniformes	Cyprinidae	<i>Labeo rohita</i>	Amarkantak, Mandla, Gadawada, Kareli (Narsinghpur), Tilwaraghat (Jabalpur), Bergi (Jabalpur), Chandan Ghat (Dindori), Mandla (Banjar River), Tawa (Tendwa River), Jabalpur, Sagar, Shahdol, Dindori, Mandla, Narsinghpur, Betul
Siluriformes	Sisoridae	<i>Laguvia ribeiroi ribeiroi</i>	Jabalpur
Cypriniformes	Cobitidae	<i>Lepidocephalichthys berdmoei</i>	Jabalpur
Cypriniformes	Cobitidae	<i>Lepidocephalichthys guntea</i>	Narmada Kund (Amarkantak), Chandanghat (Dindori), Mandla, Tilwaraghat (Jabalpur), Kareli (Narsinghpur), Betul, Mandla, Jabalpur, Chhindwara, Narsinghpur, Raisen, Sagar, Shahdol, Dindori, Indore
Cypriniformes	Cobitidae	<i>Lepidocephalus thermalis</i>	Jabalpur
Cypriniformes	Homalopteridae	<i>Longischistura striata</i>	Jabalpur, Narsinghpur
Synbranchiformes	Mastacembelidae	<i>Macrognathus aculeatus</i>	Bergi (Jabalpur), Kareli (Narsinghpur), Tilwaraghat (Jabalpur), Gadawada, Jabalpur, Shahdol
Synbranchiformes	Mastacembelidae	<i>Macrognathus pancalus</i>	Betul, Jabalpur, Chhindwara, Mandla, Narsinghpur, Tilwaraghat (Jabalpur), Raisen, Sagar, Shahdol, Dindori
Synbranchiformes	Mastacembelidae	<i>Mastacembelus armatus</i>	Chhindwara, Jabalpur, Mandla, Narsinghpur, Raisen, Betul, Sagar, Shahdol, Dindori, Kareli (Narsinghpur), Tilwaraghat (Jabalpur)
Anguilliformes	Syngnathidae	<i>Microphis cuncalus</i>	Betul
Siluriformes	Bagridae	<i>Mystus bleekeri</i>	Amarkantak, Gadawada, Kareli (Narsinghpur), Tilwaraghat (Jabalpur), Mandla, Betul, Jabalpur, Sagar, Dindori, Mandla, Narsinghpur
Siluriformes	Bagridae	<i>Mystus cavasius</i>	Jabalpur, Betul, Sagar, Shahdol, Dindori, Mandla, Narsinghpur, Amarkantak, Mandla, Gadawada, Kareli (Narsinghpur), Tilwaraghat (Jabalpur)

Siluriformes	Bagridae	<i>Mystus vittatus</i>	Raisen, Jabalpur, Sagar, Seoni, Shahdol, Dindori, Mandla, Narsinghpur
Perciformes	Nandidae	<i>Nandus nandus</i>	Jabalpur, Narsinghpur, Sagar, Dindori, Mandla
Siluriformes	Sisoridae	<i>Nangra itchkea</i>	Jabalpur
Siluriformes	Sisoridae	<i>Nangra nangra</i>	Sagar
Cypriniformes	Cyprinidae	<i>Nemacheilus sikmaiensis</i>	Mandla
Cypriniformes	Cyprinidae	<i>Nemacheilus anguilla</i>	Chhindwara
Cypriniformes	Cyprinidae	<i>Nemacheilus beavani</i>	Chhindwara
Cypriniformes	Cyprinidae	<i>Nemacheilus botia</i>	Jabalpur, Mandla
Cypriniformes	Cyprinidae	<i>Nemacheilus denisoni dayi</i>	Mandla
Cypriniformes	Cyprinidae	<i>Nemacheilus denisoni denisoni</i>	Jabalpur
Cypriniformes	Cyprinidae	<i>Nemacheilus multifasciata</i>	Jabalpur
Cypriniformes	Cyprinidae	<i>Nemacheilus striatus</i>	Jabalpur, Narsinghpur
Osteoglossiformes	Notopteridae	<i>Notopterus notopterus</i>	Raisen, Sagar, Dindori, Mandla, Narsinghpur, Jabalpur, Shahdol, Betul, Kareli and Gadawada
Siluriformes	Siluridae	<i>Ompok bimaculatus</i>	Amarkantak, Mandla, Jabalpur, Narsinghpur, Gadawara, Bergi (Jabalpur), Kareli (Narsinghpur), Mandla, Dudhi Basin, Tilwaraghat (Jabalpur), Betul, Seoni, Sagar, Shahdol, Dindori, Mandla, Narsinghpur
Siluriformes	Siluridae	<i>Ompok pabda</i>	Raisen, Jabalpur, Dindori, Mandla, Narsinghpur
Cypriniformes	Cyprinidae	<i>Oreochthys cosuatis</i>	Dindori, Mandla, Narsinghpur, Jabalpur
Perciformes	Cichlidae	<i>Oreochromis mossambicus</i>	Betul, Sultanpur (Barna River)
Cypriniformes	Cyprinidae	<i>Osteobrama cotio cotio</i>	Jabalpur, Shahdol, Narsinghpur, Sagar, Dindori, Mandla, Betul
Cypriniformes	Cyprinidae	<i>Osteobrama vigorsii</i>	Jabalpur
Siluriformes	Pangasiidae	<i>Pangasius pangasius</i>	Shahdol
Cypriniformes	Homalopteridae	<i>Pangio pangia</i>	Shahdol
Perciformes	Chandidae	<i>Parambassis ranga</i>	Betul, Jabalpur, Chhindwara, Narsinghpur, Raisen, Shahdol
Cypriniformes	Cyprinidae	<i>Parapsilorhynchus tentaculatus</i>	Dindori, Mandla, Narsinghpur, Jabalpur
Siluriformes	Sisoridae	<i>Pseudolaguvia ribeiroi</i>	Jabalpur, Shahdol

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Cypriniformes	Cyprinidae	<i>Puntius amphibius</i>	Chhindwara, Raisen, Sagar, Shahdol, Dindori, Mandla, Narsinghpur, Jabalpur
Cypriniformes	Cyprinidae	<i>Puntius chola</i>	Jabalpur, Raisen, Sagar, Shahdol
Cypriniformes	Cyprinidae	<i>Puntius conchoni</i>	Betul, Chhindwara, Jabalpur, Narsinghpur, Raisen, Sagar, Shahdol, Dindori, Mandla, Indore
Cypriniformes	Cyprinidae	<i>Puntius curmuca</i>	Jabalpur
Cypriniformes	Cyprinidae	<i>Puntius dorsalis</i>	Jabalpur, Narsinghpur, Dindori, Mandla
Cypriniformes	Cyprinidae	<i>Puntius gelius</i>	Jabalpur, Sagar
Cypriniformes	Cyprinidae	<i>Puntius guganio</i>	Raisen, Dindori, Mandla, Narsinghpur, Jabalpur
Cypriniformes	Cyprinidae	<i>Puntius mahecola</i>	Jabalpur
Cypriniformes	Cyprinidae	<i>Puntius melanostigma</i>	Jabalpur
Cypriniformes	Cyprinidae	<i>Puntius punjabensis</i>	Chhindwara, Jabalpur
Cypriniformes	Cyprinidae	<i>Puntius sarana</i>	Betul, Jabalpur, Raisen, Sagar, Shahdol, Dindori, Mandla, Narsinghpur
Cypriniformes	Cyprinidae	<i>Puntius sarana sarana</i>	Raisen, Jabalpur, Mandla
Cypriniformes	Cyprinidae	<i>Puntius sophore</i>	Mandla, Jabalpur, Narsinghpur, Shahdol, Sagar, Dindori, Betul
Cypriniformes	Cyprinidae	<i>Puntius ticto</i>	Chhindwara, Jabalpur, Narsinghpur, Betul, Hoshangabad, Shahdol, Sagar, Dindori, Mandla
Cypriniformes	Cyprinidae	<i>Puntius waagenii</i>	Jabalpur, Mandla
Cypriniformes	Cyprinidae	<i>Raiamas bola</i>	Dindori, Mandla, Narsinghpur
Cypriniformes	Cyprinidae	<i>Rasbora daniconius daniconius</i>	Betul, Chhindwara, Jabalpur, Mandla, Narsinghpur, Raisen, Damoh, Sagar, Shahdol, Dindori
Cypriniformes	Cyprinidae	<i>Rasbora rasbora</i>	Jabalpur
Perciformes	Mugilidae	<i>Rhinomugil corsula</i>	Jabalpur, Shahdol
Siluriformes	Bagridae	<i>Rita gogra</i>	Jabalpur, Sagar, Dindori, Mandla, Narsinghpur
Siluriformes	Bagridae	<i>Rita rita</i>	Jabalpur, Shahdol
Cypriniformes	Cyprinidae	<i>Salmophasia balooke</i>	Betul, Jabalpur, Sagar, Shahdol, Dindori, Mandla, Narsinghpur
Cypriniformes	Cyprinidae	<i>Salmostoma acinaces</i>	Damoh
Cypriniformes	Cyprinidae	<i>Salmostoma bacaila</i>	Betul, Jabalpur, Sagar, Dindori, Mandla, Narsinghpur, Shahdol
Cypriniformes	Cyprinidae	<i>Salmostoma boops</i>	Shahdol
Cypriniformes	Cyprinidae	<i>Salmostoma phulo</i>	Betul
Cypriniformes	Cyprinidae	<i>Schismatorhynchus nukta</i>	Jabalpur, Sagar
Cypriniformes	Balitoridae	<i>Schistura beavani</i>	Chhindwara, Dindori, Mandla, Narsinghpur, Jabalpur, Shahdol

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Cypriniformes	Balitoridae	<i>Schistura dayi</i>	Mandla, Seoni, Shahdol, Dindori, Narsinghpur, Jabalpur, Betul
Cypriniformes	Balitoridae	<i>Schistura denisonii</i>	Jabalpur, Sagar, Dindori, Mandla, Narsinghpur
Cypriniformes	Balitoridae	<i>Schistura savona</i>	Jabalpur
Cypriniformes	Balitoridae	<i>Schistura scaturigina</i>	Jabalpur, Shahdol
Cypriniformes	Cyprinidae	<i>Securicula gora</i>	Shahdol, Dindori, Mandla, Narsinghpur, Jabalpur
Perciformes	Mugilidae	<i>Sicamugil cascasia</i>	Shahdol
Siluriformes	Schilbeidae	<i>Silonia silondia</i>	Sagar, Shahdol, Dindori, Mandla, Narsinghpur, Jabalpur
Cypriniformes	Cyprinidae	<i>Sinilabeo dero</i>	Jabalpur, Shahdol
Siluriformes	Sisoridae	<i>Sisor rabdophorus</i>	Shahdol
Siluriformes	Bagridae	<i>Sperata aor</i>	Jabalpur, Shahdol, Dindori, Mandla, Narsinghpur, Betul
Siluriformes	Bagridae	<i>Sperata seenghala</i>	Jabalpur, Shahdol, Sagar, Dindori, Mandla, Narsinghpur, Betul
Clupeiformes	Clupeidae	<i>Tenualosa ilisha</i>	Shahdol, Dindori, Mandla, Narsinghpur, Jabalpur, Betul
Cypriniformes	Cyprinidae	<i>Tor khudree</i>	Shahdol, Dindori, Mandla, Narsinghpur, Jabalpur
Cypriniformes	Cyprinidae	<i>Tor putitora</i>	Jabalpur, Mandla, Dindori, Narsinghpur
Cypriniformes	Cyprinidae	<i>Tor tor</i>	Mandla, Betul, Jabalpur, Shahdol, Dindori, Narsinghpur
Siluriformes	Siluridae	<i>Wallago attu</i>	Betul, Jabalpur, Shahdol, Narsinghpur, Sagar, Dindori, Mandla, Bargi (Bargi River), Dindori (Narmada River), Mandla (Banjar River), Sultanpur (Barna River), Tawa (Tendwa River), Amarkantak, Mandla, Gadawada, Kareli (Narsinghpur), Tilwaraghat (Jabalpur)
Beloniformes	Belonidae	<i>Xenentodon cancila</i>	Betul, Jabalpur, Raisen, Sagar, Shahdol, Dindori, Mandla, Narsinghpur

Annexure 4.4.5: Checklist of the Middle Narmada Basin fishes (**CIFRI – Fishes of River Narmada: A Field Identification Manual**)

Order	Family	Zoological Name	Location
Cypriniformes	Balitoridae	<i>Acanthocobitis botia</i>	Choral (Choral River), Harda (Anchinal River), Pati (Goi River), Sehore (Sip River) & Dewas (Jamner River) (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Hoshangabad, Indore, Dewas, Khandwa, Barwani, Khargone
Siluriformes	Amblycipitidae	<i>Amblyceps mangois</i>	(Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada-Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Hoshangabad, Nimar, Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Dewas, Khandwa, Khargone
Cypriniformes	Cyprinidae	<i>Amblypharyngodon mola</i>	Sehore (Sip River) & Dewas (Jamner River), Hoshangabad (Sethani ghat and Dongarwada), Barwani, Dhar, Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Indore
Perciformes	Anabantidae	<i>Anabas testudineus</i>	Hoshangabad, Khandwa, Barwani, Khargone, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Nimar, Mortakka (Khandwa), Dhar, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Anguilliformes	Anguillidae	<i>Anguilla bengalensis</i>	Barwani, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Indirasagar Dam
Anguilliformes	Anguillidae	<i>Anguilla bengalensis bengalensis</i>	Hoshangabad, Dewas, Barwani
Siluriformes	Bagridae	<i>Aorichthys seenghala</i>	Barwani, Dhar, Khandwa, Khargone, Hoshangabad
Cyprinodontiformes	Aplocheilidae	<i>Aplocheilus lineatus</i>	Dewas, Barwani, Hoshangabad
Cyprinodontiformes	Aplocheilidae	<i>Aplocheilus panchax</i>	Hoshangabad
Cypriniformes	Cyprinidae	<i>Aspidoparia morar</i>	Indore

Perciformes	Badidae	<i>Badis badis</i>	Barwani, Dhar, Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Sehore (Sip River), (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada-Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Bhopal, Hoshangabad, Dewas, Indore
Siluriformes	Sisoridae	<i>Bagarius bagarius</i>	Hoshangabad (Sethani ghat and Dongarwada), Nimar, Indore
Cypriniformes	Cyprinidae	<i>Barilius barila</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur) Omkareshwar, Hoshangabad (Sethani ghat and Dongarwada), Mortakka (Khandwa), Khargone, Dewas, Indore, Khandwa
Cypriniformes	Cyprinidae	<i>Barilius bendelisis</i>	(Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Sehore (Sip River) & Dewas (Jamner River), Omkareshwar Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad (Sethani ghat and Dongarwada), Mortakka (Khandwa)
Cypriniformes	Cyprinidae	<i>Barilius bendelisis bendelisis</i>	Hoshangabad, Khandwa, Barwani, Khargone, Dewas, Indore
Cypriniformes	Cyprinidae	<i>Barilius evezardi</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Khandwa, Khargone, Dewas
Cypriniformes	Cyprinidae	<i>Barilius radiolatus</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Khandwa, Khargone, Dewas
Cypriniformes	Cyprinidae	<i>Barilius shacra</i>	Hoshangabad
Cypriniformes	Cyprinidae	<i>Barilius vagra</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Sehore (Sip River) & Dewas (Jamner River), Indore
Cypriniformes	Cyprinidae	<i>Catla catla</i>	Barwani, Barwani (Bhilkhedra, Pendra), Dhar, Hoshangabad (Narmada River), Indira Sagar (Narmada River), Indirasagar Dam, Khandwa, Khargone, Kolar dam (Kolar River), Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Sehore (Sip River) & Dewas (Jamner River), Maheshwar dam, Hoshangabad (Sethani

Biological Profile of Narmada Basin

			ghat and Dongarwada), Nimar, Khargone/Barwani, Dhar, Omkareshwar
Cypriniformes	Cyprinidae	<i>Chagunius chagunio</i>	Hoshangabad
Perciformes	Ambassidae	<i>Chanda baculis</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Perciformes	Ambassidae	<i>Chanda nama</i>	Dewas, Hoshangabad, Indirasagar Dam, Barwani, Barwani (Bhilkheda, Pendra), Dhar, Khandwa, Khargone, Kolar dam (Kolar River), Maheshwar (Narmada River), Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Dewas (Jamner River) Maheshwar dam Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad (Sethani ghat and Dongarwada) Nimar, Khargone/Barwani (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Sehore (Sip River)
Perciformes	Ambassidae	<i>Chanda ranga</i>	Indirasagar Dam, Barwani, Barwani (Bhilkheda, Pendra), Dhar, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad (Sethani ghat and Dongarwada), Sehore (Sip River), Hoshangabad, Bhopal
Perciformes	Channidae	<i>Channa gachua</i>	Barwani, Dhar, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Sehore (Sip River) & Dewas (Jamner River), Maheshwar dam, Hoshangabad (Sethani ghat and Dongarwada)
Perciformes	Channidae	<i>Channa marulius</i>	Barwani, Dhar, Khandwa, Khargone, Kolar dam (Kolar River), Maheshwar, Maheshwar (Narmada River), Mandleshwar, Morttak (Sanawat)(Narmada River), Omkareshwar, Punasa, Sardar Sarovar Reservoir, Maheshwar dam, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad (Sethani ghat and Dongarwada), Nimar Khargone/Barwani, Hoshangabad and Sehore, Narmadapuram, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati

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			from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Indore, Bhopal
Perciformes	Channidae	<i>Channa orientalis</i>	Hoshangabad, Indore, Khandwa, Barwani, Khargone, Dewas, Bhopal, Bhamnia, Choral (Choral River), Maheshwar (Narmada River), (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada-Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
Perciformes	Channidae	<i>Channa punctatus</i>	Barwani, Barwani (Bhilkheda, Pendra), Dhar, Harda (Anchinal River), Indirasagar Dam, Khandwa, Khargone, Maheshwar, Maheshwar (Narmada River), Mandleshwar, Morttak (Sanawat)(Narmada River), Omkareshwar, Punasa, Sardar Sarovar Reservoir, Sehore (Sip River) & Dewas (Jamner River), Maheshwar dam, Hoshangabad (Sethani ghat and Dongarwada) Nimar, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Indore, Bhopal
Perciformes	Channidae	<i>Channa striatus</i>	Hoshangabad, Indore, Bhopal, Barwani (Bhilkheda, Pendra), Dhar, Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Dewas (Jamner River) Maheshwar dam, Hoshangabad (Sethani ghat & Dongarwada) Nimar, Sehore (Sip River) Khargone
Cypriniformes	Cyprinidae	<i>Chela laubuca</i>	Barwani, Dhar, Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Dewas (Jamner River), Hoshangabad (Sethani ghat and Dongarwada) (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada-

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			Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Indore
Osteoglossiformes	Notopteridae	<i>Chitala chitala</i>	Narmadapuram, Nimar, Khandwa, Barwani, Khargone, Indore
Cypriniformes	Cyprinidae	<i>Cirrhinus cirrhosus</i>	Hoshangabad, Indore, Khandwa, Barwani, Khargone, Dewas, Hoshangabad (Sethani ghat & Dongarwada), Omkareshwar, Nimar, Khargone/ Barwani
Cypriniformes	Cyprinidae	<i>Cirrhinus latia</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Cypriniformes	Cyprinidae	<i>Cirrhinus mrigala</i>	Barwani, Dhar, Hoshangabad (Narmada River), Indirasagar Dam, Khandwa, Khargone, Kolar dam (Kolar River), Maheshwar, Maheshwar (Narmada River), Mandleshwar, Omkareshwar, Pati (Goi River), Punasa, Sardar Sarovar Reservoir, Sehore (Sip River), Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur) Khargone/Barwani, Omkareshwar, Hoshangabad, Indirasagar Dam (Khandwa)
Cypriniformes	Cyprinidae	<i>Cirrhinus reba</i>	Barwani, Dhar, Hoshangabad (Narmada River), Indirasagar Dam, Khandwa, Khargone, Kolar dam (Kolar River), Maheshwar, Maheshwar (Narmada River), Mandleshwar, Morttak (Badwani)(Narmada River), Morttak (Sanawat)(Narmada River), Omkareshwar, Punasa, Sardar Sarovar Reservoir, Sehore (Sip River) & Dewas (Jamner River), Maheshwar dam, Indirasagar Dam (Khandwa) (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges Nimar
Siluriformes	Clariidae	<i>Clarias batrachus</i>	Hoshangabad, Indore, Khandwa, Khargone, Dewas, Hoshangabad (Sethani ghat and Dongarwada) Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur) Maheshwar dam, Nimar, Barwani, Dhar, Hoshangabad (Powerkheda) (Narmada River), Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Siluriformes	Schilbeidae	<i>Clupisoma bastari</i>	Khandwa, Barwani, Khargone

Siluriformes	Schilbeidae	<i>Clupisoma garua</i>	Khandwa, Barwani, Khargone, Barwani (Dhar, Hoshangabad (Narmada River), Indira Sagar (Narmada River), Maheshwar, Maheshwar (Narmada River), Mandleshwar, Morttak (Badwani)(Narmada River), Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad (Sethani ghat and Dongarwada), Dewas
Perciformes	Belontiidae	<i>Colisa fasciatus</i>	Hoshangabad, Indore, Khandwa, Barwani, Khargone, Barwani (Bhilkheda, Pendra), Dhar, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad (Sethani ghat and Dongarwada)
Cypriniformes	Cyprinidae	<i>Crossocheilus latius latius</i>	Barwani, Dhar, Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Pati (Goi River), Punasa, Sardar Sarovar Reservoir Hoshangabad, Dewas
Cypriniformes	Cyprinidae	<i>Ctenopharyngodon idella</i>	Barwani, Barwani (Bhilkheda, Pendra), Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Nimar Hoshangabad, Khargone/Barwani, Dhar
Cypriniformes	Cyprinidae	<i>Cyprinus carpio</i>	Barwani, Barwani (Bhilkheda, Pendra), Dhar, Hoshangabad (Narmada River), Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Maheshwar dam, Hoshangabad (Sethani ghat and Dongarwada), Indirasagar Dam (Khandwa), Nimar, Khargone/Barwani/ Dhar, Mortakka (Khandwa)
Cypriniformes	Cyprinidae	<i>Cyprinus carpio carpio</i>	Hoshangabad
Cypriniformes	Cyprinidae	<i>Danio rerio</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad, Khargone, Dewas, Indore, Khandwa
Cypriniformes	Cyprinidae	<i>Devario aequipinnatus</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad, Khandwa, Khargone, Dewas, Bhopal, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhyan ranges Sehore (Sip River)

Cypriniformes	Cyprinidae	<i>Devario devario</i>	Barwani, Dhar, Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Sehore (Sip River) & Dewas (Jamner River), Hoshangabad, Omkareshwar, Indore
Cypriniformes	Cyprinidae	<i>Esomus danricus</i>	Khandwa, Hoshangabad, Barwani, Khargone, Dewas, Indore, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada-Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Sehore (Sip River)
Siluriformes	Schilbeidae	<i>Eutropiichthys vacha</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad, Khandwa, Khargone, Dewas, Indore
Siluriformes	Sisoridae	<i>Gagata cenia</i>	Maheshwar (Narmada River), Dewas, Barwani, Hoshangabad, Khandwa
Siluriformes	Sisoridae	<i>Gagata itchkeea</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Cypriniformes	Cyprinidae	<i>Garra gotyla</i>	Hoshangabad (Sethani ghat and Dongarwada), Sehore (Sip River) & Dewas (Jamner River), Omkareshwar, Mortakka (Khandwa), Barwani, Dhar, Khandwa, Khargone, Indore, Dewas
Cypriniformes	Cyprinidae	<i>Garra lamta</i>	Barwani, Dhar, Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad, Dewas
Cypriniformes	Cyprinidae	<i>Garra mullya</i>	Hoshangabad, Indore, Khandwa, Barwani, Khargone, Dewas, Chakaldi (Kolar River), Choral (Choral River), Harda (Anchinal River), Indirasagar Dam, Maheshwar, Mandleshwar, Morttak (Badwani)(Narmada River), Morttak (Sanawat)(Narmada River), Omkareshwar, Punasa, Sardar Sarovar Reservoir, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada-Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges Sehore (Sip River)

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Perciformes	Nandidae	<i>Glossogobius giuris</i>	Barwani, Barwani (Dhar, Harda (Anchinal River), Hoshangabad (Narmada River), Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Pati (Goi River), Punasa, Sardar Sarovar Reservoir, Sehore (Sip River) & Dewas (Jamner River), Hoshangabad (Sethani ghat and Dongarwada) (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Bhopal, Indore
Perciformes	Nandidae	<i>Glossogobius giuris giuris</i>	Bhopal, Hoshangabad, Indore, Dewas, Khandwa, Khargone, Barwani
Siluriformes	Sisoridae	<i>Glyptothorax lonah</i>	Barwani, Indirasagar Dam, Mandleshwar, Mortakka (Narmada River), Omkareshwar, Punasa, Sardar Sarovar Reservoir, Maheshwar (Narmada River), Hoshangabad, Khandwa, Dewas, Khargone
Siluriformes	Sisoridae	<i>Glyptothorax ribeiroi</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Siluriformes	Sisoridae	<i>Glyptothorax telchitta</i>	Barwani, Dhar, Khandwa, Khargone, Indore
Siluriformes	Sisoridae	<i>Gogangra viridescens</i>	Indore
Clupeiformes	Clupeidae	<i>Goniolosa manmina</i>	Hoshangabad, Khandwa, Barwani, Khargone, Dewas, Bhopal, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Clupeiformes	Clupeidae	<i>Gudusia chapra</i>	Khargone/Barwani, Indirasagar Dam (Khandwa), Barwani, Indore
Siluriformes	Heteropneustidae	<i>Heteropneustes fossilis</i>	Hoshangabad, Indore, Khandwa, Barwani, Dewas, Khargone, Bhopal, Barwani (Bhilkheda, Pendra), Indirasagar Dam, Maheshwar, Maheshwar (Narmada River), Mandleshwar, Morttak (Sanawat)(Narmada River), Omkareshwar, Punasa, Sardar Sarovar Reservoir, Sehore (Sip River) & Dewas (Jamner River), (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Hoshangabad (Sethani ghat and Dongarwada), Indirasagar Dam (Khandwa)
Clupeiformes	Clupeidae	<i>Hilsa ilisha</i>	Indirasagar Dam

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Cypriniformes	Cyprinidae	<i>Hypophthalmichthys molitrix</i>	Hoshangabad, Dewas, Barwani, Hoshangabad (Sethani ghat and Dongarwada) Khargone/ Barwani/ Dhar, Omkareshwar
Cypriniformes	Cyprinidae	<i>Hypselobarbus curmuca</i>	Hoshangabad
Cypriniformes	Cyprinidae	<i>Hypselobarbus jerdoni</i>	Indore
Cypriniformes	Homalopteridae	<i>Indoreonectes evezardi</i>	Hoshangabad, Indore, Khandwa, Barwani, Khargone, Dewas
Cypriniformes	Cyprinidae	<i>Labeo ariza</i>	Hoshangabad, Khandwa, Barwani, Khargone, Dewas
Cypriniformes	Cyprinidae	<i>Labeo bata</i>	Barwani, Barwani (Bhilkheda), Dhar, Indirasagar Dam, Khandwa, Khargone, Kolar dam (Kolar River), Maheshwar, Maheshwar (Narmada River), Mandleshwar, Nimar Omkareshwar, Punasa, Sardar Sarovar Reservoir, Sehore (Sip River) & Dewas (Jamner River), Maheshwar dam, Mandleshwar, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad (Sethani ghat and Dongarwada), Hoshangabad, Khandwa, Barwani, Khargone, Dewas
Cypriniformes	Cyprinidae	<i>Labeo boga</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Sehore (Sip River) & Dewas (Jamner River), Indirasagar Dam (Khandwa), Indore
Cypriniformes	Cyprinidae	<i>Labeo boggut</i>	Hoshangabad, Khandwa, Barwani, Khargone, Dewas, Bhopal, Barwani, Chakaldi (Kolar River), Choral (Choral River), Harda (Anchinal River), Hoshangabad (Narmada River), Indirasagar Dam, Maheshwar, Maheshwar (Narmada River), Mandleshwar, Morttak (Badwani), Omkareshwar, Punasa, Sardar Sarovar Reservoir, Sehore (Sip River) & Dewas (Jamner River), (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
Cypriniformes	Cyprinidae	<i>Labeo calbasu</i>	Barwani, Barwani (Bhilkheda, Pendra), Dhar, Hoshangabad (Narmada River), Indira Sagar (Narmada River), Indirasagar Dam, Khandwa, Kolar dam (Kolar River), Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Sehore (Sip River) & Dewas (Jamner River), Maheshwar dam, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala,

			Holipur), Hoshangabad (Sethani ghat and Dongarwada), Nimar, Khargone/Barwani, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada-Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Hoshangabad, Indore, Khandwa, Barwani, Khargone, Dewas
Cypriniformes	Cyprinidae	<i>Labeo dussumieri</i>	Hoshangabad (Narmada River), Indore
Cypriniformes	Cyprinidae	<i>Labeo dyocheilus</i>	Hoshangabad, Khandwa, Khargone, Dewas, Barwani, Indore, Hoshangabad (Narmada River), Indirasagar Dam, Kolar dam (Kolar River), Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Cypriniformes	Cyprinidae	<i>Labeo fimbriatus</i>	Hoshangabad, Khandwa, Khargone, Dewas, Maheshwar dam, Khargone, Barwani, Dhar, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad (Sethani ghat and Dongarwada), Indirasagar Dam (Khandwa), Kolar dam (Kolar River), Punasa, Sardar Sarovar Reservoir, Nimar, Khargone/Barwani, Omkareshwar, Mandleshwar
Cypriniformes	Cyprinidae	<i>Labeo gonius</i>	Barwani, Dhar, Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada-Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Hoshangabad, Indore, Khandwa, Barwani, Khargone, Dewas
Cypriniformes	Cyprinidae	<i>Labeo pangusia</i>	Sehore (Sip River) & Dewas (Jamner River), Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada-Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad, Dewas, Barwani
Cypriniformes	Cyprinidae	<i>Labeo rohita</i>	Sehore (Sip River) & Dewas (Jamner River), Maheshwar dam Hoshangabad (Sethani ghat and Dongarwada), Nimar, Barwani (Bhilkhedra, Pendra), Dhar, Hoshangabad (Narmada River), Indira Sagar (Narmada River), Indirasagar Dam, Khandwa, Khargone, Maheshwar, Maheshwar (Narmada River), Mandleshwar, Morttak

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			(Badwani)(Narmada River), Omkareshwar, Punasa, Sardar Sarovar Reservoir, Bhopal, Hoshangabad, Indore, Barwani, Khargone, Dewas, Khandwa
Siluriformes	Sisoridae	<i>Laguvia ribeiroi ribeiroi</i>	Hoshangabad
	Poeciliidae	<i>Lebistes reticulata</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Cypriniformes	Homalopteridae	<i>Lepidocephalichthys guntea</i>	Sehore (Sip River) & Dewas (Jamner River), Omkareshwar, Mandleshwar, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Barwani, Dhar, Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad, Khandwa, Barwani, Khargone, Dewas, Bhopal
Cypriniformes	Homalopteridae	<i>Lepidocephalus thermalis</i>	Indore
Synbranchiformes	Mastacembelidae	<i>Macrognathus aculeatus</i>	Hoshangabad
Synbranchiformes	Mastacembelidae	<i>Macrognathus pancalus</i>	Sehore (Sip River) & Dewas (Jamner River), Maheshwar dam, (Hoshangabad and Sehore district), Sampling sites across seven tributaries of the river Narmada-Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Nimar, Dhar, Hoshangabad (Sethani ghat & Dongarwada), Khargone/Barwani, Hoshangabad, Indore, Khandwa, Barwani, Dewas, Bhopal, Barwani (Bhilkheda, Pendra), Indirasagar Dam, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Synbranchiformes	Mastacembelidae	<i>Mastacembelus armatus</i>	Barwani, Barwani (Bhilkheda, Pendra), Choral (Choral River), Dhar, Harda (Anchinal River), Hoshangabad (Narmada River), Indira Sagar (Narmada River), Indirasagar Dam, Khandwa, Khargone, Kolar dam (Kolar River), Maheshwar, Maheshwar (Narmada river), Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Bhopal, Hoshangabad, Indore, Dewas
Anguilliformes	Syngnathidae	<i>Microphis cuncalus</i>	Khandwa, Barwani, Khargone
Siluriformes	Bagridae	<i>Mystus aor</i>	Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Nimar, Khargone, Barwani, Indirasagar Dam,

			Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Siluriformes	Bagridae	<i>Mystus bleekeri</i>	Sehore (Sip River) & Dewas (Jamner River), Maheshwar dam, Omkareshwar, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad (Sethani ghat and Dongarwada), Nimar, Khargone/ Barwani, Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Indirasagar Dam (Khandwa), Barwani (Bhilkheda), Choral (Choral River), Dhar, Harda (Anchinal River), Indirasagar Dam, Khandwa, Khargone, Kolar dam (Kolar River), Mandleshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad, Indore, Bhopal
Siluriformes	Bagridae	<i>Mystus cavasius</i>	Sehore (Sip River) & Dewas (Jamner River), (Hoshangabad and Sehore district), Sampling sites across seven tributaries of the river Narmada-Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Hoshangabad (Sethani ghat and Dongarwada), Barwani, Barwani (Bhilkheda, Pendra), Dhar, Hoshangabad (Narmada River), Indirasagar Dam, Khandwa, Khargone, Maheshwar, Maheshwar (Narmada River), Mandleshwar, Morttak (Badwani)(Narmada River), Morttak (Sanawat)(Narmada River), Omkareshwar, Punasa, Sardar Sarovar Reservoir, Bhopal, Indore, Dewas
Siluriformes	Bagridae	<i>Mystus montanus</i>	Hoshangabad
Siluriformes	Bagridae	<i>Mystus seenghala</i>	Maheshwar dam, Nimar, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad (Sethani ghat and Dongarwada), Khargone/ Barwani, Omkareshwar, Indirasagar Dam (Khandwa), Mandleshwar, Barwani, Barwani (Bhilkheda, Pendra), Indirasagar Dam, Maheshwar, Sardar Sarovar Reservoir
Siluriformes	Bagridae	<i>Mystus tengara</i>	Nimar, Sehore (Sip River) & Dewas (Jamner River), Indirasagar Dam (Khandwa), Barwani (Bhilkheda, Pendra), Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad, Khandwa, Khargone

Siluriformes	Bagridae	<i>Mystus vittatus</i>	Hoshangabad, Khandwa, Barwani, Khargone, Dewas, Indore, Nimar, Sehore (Sip River), Maheshwar dam, Hoshangabad (Sethani ghat and Dongarwada), Barwani (Bhilkheda, Pendra), Dhar, Indirasagar Dam, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Perciformes	Nandidae	<i>Nandus nandus</i>	Sehore (Sip River), Nimar, Maheshwar dam, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad (Sethani ghat and Dongarwada), (Hoshangabad and Sehore district), Sampling sites across seven tributaries of the river Narmada-Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Barwani (Bhilkheda, Pendra), Chakaldi (Kolar River), Choral (Choral River), Dhar, Indirasagar Dam, Khandwa, Khargone, Mandleshwar, Omkareshwar, Pati (Goi River), Punasa, Sardar Sarovar Reservoir, Bhopal, Indore, Dewas, Barwani
Cypriniformes	Cyprinidae	<i>Nemacheilus beavani</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Cypriniformes	Cyprinidae	<i>Nemacheilus botia</i>	Maheshwar dam, Nimar, Mandleshwar, Sehore (Sip River), Dewas (Jamner River), Omkareshwar, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Barwani, Dhar, Indirasagar Dam, Khandwa, Khargone, Punasa, Sardar Sarovar Reservoir, Bhopal, Hoshangabad
Cypriniformes	Cyprinidae	<i>Nemacheilus dayi</i>	Sehore (Sip River) & Dewas (Jamner River), Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Cypriniformes	Cyprinidae	<i>Nemacheilus denisoni denisoni</i>	Hoshangabad
Cypriniformes	Cyprinidae	<i>Nemacheilus evezardi</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad
Cypriniformes	Cyprinidae	<i>Nemacheilus sikmaiensis</i>	Hoshangabad
Osteoglossiformes	Notopteridae	<i>Notopterus chitala</i>	Barwani, Barwani (Bhilkheda, Pendra), Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir

Osteoglossiformes	Notopteridae	<i>Notopterus notopterus</i>	Khargone/Barwani/ Dhar, Nimar, Sehore (Sip River) & Dewas (Jamner River), Maheshwar dam, Hoshangabad (Sethani ghat and Dongarwada), Narmadapuram, (Hoshangabad and Sehore district), Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Indirasagar Dam (Khandwa), Barwani (Bhilkheda, Pendra), Choral (Choral River), Dhar, Hoshangabad (Narmada River), Indirasagar Dam, Khandwa, Kolar dam (Kolar River), Maheshwar, Maheshwar (Narmada River), Mandleshwar, Morttak (Sanawat)(Narmada River), Omkareshwar, Pati (Goi River), Punasa, Sardar Sarovar Reservoir, Indore
Siluriformes	Siluridae	<i>Ompok bimaculatus</i>	Hoshangabad (Sethani ghat and Dongarwada), Nimar, Maheshwar dam, Sehore (Sip River) & Dewas (Jamner River), (Hoshangabad and Sehore district), Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Indirasagar Dam (Khandwa), Barwani, Barwani (Bhilkheda, Pendra), Choral (Choral River), Dhar, Harda (Anchinal River), Hoshangabad (Narmada River), Indirasagar Dam, Khandwa, Khargone, Kolar dam (Kolar River), Mandleshwar, Morttak (Badwani) (Narmada River), Omkareshwar, Punasa, Sardar Sarovar Reservoir, Indore, Dewas
Siluriformes	Siluridae	<i>Ompok pabda</i>	Barwani, Dhar, Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad, Dewas
Cypriniformes	Cyprinidae	<i>Oreochthys cosuatis</i>	Barwani, Dhar, Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad, Dewas
Cypriniformes	Cyprinidae	<i>Osteobrama cotio</i>	Sehore (Sip River) & Dewas (Jamner River), Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Barwani, Barwani (Bhilkheda), Dhar, Harda (Anchinal River), Hoshangabad (Narmada River),

			Indirasagar Dam, Khandwa, Khargone, Kolar dam (Kolar River), Maheshwar, Maheshwar (Narmada River), Mandleshwar, Morttak (Badwani)(Narmada River), Omkareshwar, Pati (Goi River), Punasa, Sardar Sarovar Reservoir, Indore
Cypriniformes	Cyprinidae	<i>Osteobrama vigorsii</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Cypriniformes	Cyprinidae	<i>Oxygaster bacaila</i>	Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Sehore (Sip River), Khargone, Dhar, Omkareshwar, Barwani, Barwani (Bhilkheda, Pendra), Indirasagar Dam, Maheshwar, Mandleshwar, Punasa, Sardar Sarovar Reservoir
Cypriniformes	Cyprinidae	<i>Oxygaster clupeoides</i>	Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Cypriniformes	Cyprinidae	<i>Oxygaster gora</i>	Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Sehore (Sip River), Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Siluriformes	Pangasiidae	<i>Pangasius pangasius</i>	Khargone, Barwani (Bhilkheda, Pendra), Dhar, Nimar, Indore
Perciformes	Ambassidae	<i>Parambassis ranga</i>	(Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Sehore (Sip River), Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad, Indore, Khandwa, Barwani, Khargone
Cypriniformes	Cyprinidae	<i>Parapsilorhynchus tentaculatus</i>	Hoshangabad, Khandwa, Barwani, Khargone, Dewas, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Siluriformes	Sisoridae	<i>Pseudolaguvia ribeiroi</i>	(Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Hoshangabad, Khandwa, Barwani, Khargone
Cypriniformes	Cyprinidae	<i>Puntius ambassis</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir

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Cypriniformes	Cyprinidae	<i>Puntius amphibious</i>	Hoshangabad, Khandwa, Khargone, Dewas, Barwani, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Sehore (Sip River) & Dewas (Jamner River)
Cypriniformes	Cyprinidae	<i>Puntius arenatus</i>	Khandwa
Cypriniformes	Cyprinidae	<i>Puntius chola</i>	Sehore (Sip River) & Dewas (Jamner River), Nimar, Hoshangabad (Sethani ghat and Dongarwada), Barwani (Bhilkhedha, Pendra), Dhar, Khandwa, Khargone, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Bhopal, Hoshangabad, Barwani
Cypriniformes	Cyprinidae	<i>Puntius chrysopoma</i>	Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Cypriniformes	Cyprinidae	<i>Puntius conchoniis</i>	Sehore (Sip River) and Dewas (Jamner River), (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Bhopal, Hoshangabad, Khandwa, Dewas, Khargone
Cypriniformes	Cyprinidae	<i>Puntius curmuca</i>	Hoshangabad
Cypriniformes	Cyprinidae	<i>Puntius dorsalis</i>	Hoshangabad, Khandwa, Khargone, Dewas, Barwani, Sehore (Sip River), Nimar, Omkareshwar, Indirasagar Dam, Maheshwar, Mandleshwar, Punasa, Sardar Sarovar Reservoir
Cypriniformes	Cyprinidae	<i>Puntius guganio</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir Maheshwar dam, Hoshangabad, Raisen, Khandwa, Khargone, Dewas
Cypriniformes	Cyprinidae	<i>Puntius phutunio</i>	Indore
Cypriniformes	Cyprinidae	<i>Puntius pinnauratus</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad

Cypriniformes	Cyprinidae	<i>Puntius sarana</i>	Barwani (Bhilkheda, Pendra), Bhamni (Banjar River), Chakaldi (Kolar River), Choral (Choral River), Dhar, Harda (Anchinal River), Indira Sagar (Narmada River), Indirasagar Dam, Khandwa, Kolar dam (Kolar River), Maheshwar, Maheshwar (Narmada River), Mandleshwar, Morttak (Sanawat), Morttak (Sanawat)(Narmada River), Punasa, Sardar Sarovar Reservoir, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad (Sethani ghat and Dongarwada), Maheshwar dam, Nimar, Sehore (Sip River), Omkareshwar, Khargone/ Dhar, Indirasagar Dam (Khandwa), Dewas (Jamner River), Indore
Cypriniformes	Cyprinidae	<i>Puntius sarana sarana</i>	Hoshangabad, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhyan ranges
Cypriniformes	Cyprinidae	<i>Puntius sophore</i>	Barwani, Barwani (Bhilkheda, Pendra), Choral (Choral River), Dhar, Harda (Anchinal River), Hoshangabad (Narmada River), Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Pati (Goi River), Punasa, Sardar Sarovar Reservoir, Sehore (Sip River) & Dewas (Jamner River), Khargone/ Barwani, Hoshangabad (Sethani ghat and Dongarwada), (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Mortakka (Khandwa), Indore
Cypriniformes	Cyprinidae	<i>Puntius stigma</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Cypriniformes	Cyprinidae	<i>Puntius ticto</i>	Hoshangabad, Indore, Khandwa, Barwani, Khargone, Dewas, Barwani (Bhilkheda, Pendra), Chakaldi (Kolar River), Choral (Choral River), Dhar, Harda (Anchinal River), Hoshangabad (Narmada River), Pati (Goi River), Tawa (Tendwa River Maheshwar dam, Hoshangabad (Sethani ghat and Dongarwada), Nimar, Sehore (Sip River), Mortakka (Khandwa), (Hoshangabad and Sehore district)

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			Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
Cypriniformes	Cyprinidae	<i>Puntius titius</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur)
Cypriniformes	Cyprinidae	<i>Raiamas bola</i>	Hoshangabad, Indore, Dewas, Barwani
Cypriniformes	Cyprinidae	<i>Rasbora daniconius</i>	Sehore (Sip River) & Dewas (Jamner River), Maheshwar dam, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad (Sethani ghat and Dongarwada), Nimar, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Indirasagar Dam (Khandwa) Barwani, Barwani (Bhilkheda, Pendra), Dhar, Indirasagar Dam, Khandwa, Khargone, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Cypriniformes	Cyprinidae	<i>Rasbora daniconius daniconius</i>	Hoshangabad, Bhopal, Indore, Khandwa, Khargone, Dewas, Barwani
Perciformes	Mugilidae	<i>Rhinomugil corsula</i>	Barwani, Barwani (Bhilkheda, Pendra), Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Khandwa, Barwani, Khargone, Indore
Siluriformes	Bagridae	<i>Rita gogra</i>	Hoshangabad, Khandwa, Khargone, Dewas, Bhopal, Hoshangabad (Sethani ghat and Dongarwada), Narmadapuram, Dhar
Siluriformes	Bagridae	<i>Rita kuturnee</i>	Hoshangabad
Siluriformes	Bagridae	<i>Rita pavimentata</i>	Indirasagar Dam (Khandwa), Nimar, Khargone, Barwani, Dhar, Barwani, Barwani (Bhilkheda, Pendra), Dhar, Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Siluriformes	Bagridae	<i>Rita rita</i>	Barwani, Barwani (Bhilkheda, Pendra), Dhar, Indirasagar Dam, Khandwa, Khargone, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Maheshwar dam, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad (Sethani ghat and Dongarwada) Nimar,

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			Hoshangabad, Khandwa, Khargone, Indore
Cypriniformes	Cyprinidae	<i>Salmophasia balookee</i>	Hoshangabad, Khandwa, Barwani, Khargone, Dewas, Indore
Cypriniformes	Cyprinidae	<i>Salmostoma bacaila</i>	Choral (Choral River), Harda (Anchinal River), Hoshangabad (Narmada River), Maheshwar (Narmada River), Morttak (Badwani)(Narmada River), Pati (Goi River), Sultanpur (Barna River), Hoshangabad, Khandwa, Barwani, Khargone, Dewas, Indore, Sehore (Sip River), Nimar, Dhar
Cypriniformes	Cyprinidae	<i>Salmostoma clupeoides</i>	Bhopal, Hoshangabad
Cypriniformes	Cyprinidae	<i>Salmostoma phulophulo</i>	Khandwa
Cypriniformes	Balitoridae	<i>Schistura beavani</i>	Hoshangabad, Dewas, Barwani, Indore
Cypriniformes	Balitoridae	<i>Schistura dayi</i>	Hoshangabad, Barwani, Khargone, Dewas, Khandwa
Cypriniformes	Balitoridae	<i>Schistura denisonii</i>	Hoshangabad, Khandwa, Indore, Barwani, Khargone, Dewas
Cypriniformes	Cyprinidae	<i>Securicula gora</i>	Sehore (Sip River), Nimar, Indore, Dewas, Barwani, Hoshangabad
Siluriformes	Schilbeidae	<i>Silonia silondia</i>	Barwani, Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad, Khandwa, Barwani, Khargone, Dewas, Indore
Cypriniformes	Cyprinidae	<i>Sinilabeo dero</i>	Indore
Siluriformes	Bagridae	<i>Sperata aor</i>	(Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Hoshangabad (Narmada River), Maheshwar (Narmada River), Hoshangabad, Barwani, Khargone, Dewas, Indore, Khandwa
Siluriformes	Bagridae	<i>Sperata seenghala</i>	Sehore (Sip River) & Dewas (Jamner River) Hoshangabad (Narmada River), Kolar dam (Kolar River), Maheshwar (Narmada River), Hoshangabad, Indore, Khandwa, Barwani, Khargone, Dewas
Clupeiformes	Clupeidae	<i>Tenuulosa ilisha</i>	Barwani, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir, Hoshangabad, Khandwa, Barwani, Khargone, Dewas, Indore
Cypriniformes	Cyprinidae	<i>Tor khudree</i>	Hoshangabad, Khandwa, Barwani, Khargone, Dewas, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges,

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			Indirasagar Dam, Maheshwar, Mandleshwar, Omkareshwar, Punasa, Sardar Sarovar Reservoir
Cypriniformes	Cyprinidae	<i>Tor putitora</i>	Hoshangabad (Sethani ghat and Dongarwada), Hoshangabad, Khandwa, Barwani, Khargone, Dewas
Cypriniformes	Cyprinidae	<i>Tor tor</i>	Bhopal, Hoshangabad, Indore, Khandwa, Barwani, Khargone, Dewas, Sehore (Sip River), Maheshwar dam, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad (Sethani ghat and Dongarwada), Nimar, Narmadapuram, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges Barwani, Barwani (Bhilkheda, Pendra), Dhar, Harda (Anchinal River), Indira Sagar (Narmada River), Indirasagar Dam, Kolar dam (Kolar River), Maheshwar, Mandleshwar, Mortakka (Narmada River), Morttak (Badwani)(Narmada River), Omkareshwar, Punasa, Sardar Sarovar Reservoir
Siluriformes	Siluridae	<i>Wallago attu</i>	Sehore (Sip River) & Dewas (Jamner River), Maheshwar dam, Dhar, Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Hoshangabad (Sethani ghat and Dongarwada) Nimar, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada-Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Indirasagar Dam (Khandwa), Barwani (Bhilkheda, Pendra), Dhar, Khandwa, Khargone, Kolar dam (Kolar River), Maheshwar, Mandleshwar, Morttak (Sanawat)(Narmada River), Omkareshwar, Punasa, Sardar Sarovar Reservoir, Barwani, Dewas, Indore, Bhopal
Beloniformes	Belonidae	<i>Xenentodon cancila</i>	Hoshangabad (Sethani ghat and Dongarwada), Dewas (Jamner River), Maheshwar dam, Indirasagar Dam (Khandwa), Hoshangabad and Sehore district (5 sites- Shahganj, Bandrabhan, Budhni ghat, Moukala, Holipur), Nimar, (Hoshangabad and Sehore district) Sampling sites across seven tributaries of the river Narmada- Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria,

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		Juanri from Vindhya ranges, Barwani (Bhilkheda, Pendra), Dhar, Harda (Anchinal River), Hoshangabad (Narmada River), Khandwa, Khargone, Maheshwar, Maheshwar (Narmada River), Mandleshwar, Mortakka (Narmada River), Omkareshwar, Pati (Goi River), Punasa, Sardar Sarovar Reservoir, Dewas, Bhopal, Indore
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Annexure 5.1.1.1: Checklist of Macro Benthos of the Narmada tributaries of the Upper Basin.

Phylum	Class	Species	Location
Arthropoda	Arachnida	<i>Dolomedes sp.</i>	Dudhi River
		<i>Tetragnatha sp.</i>	Dudhi River
	Insecta	<i>Anax sp.</i>	Dudhi River
		<i>Archilestes sp.</i>	Dudhi River
		<i>Bellamya sp.</i>	Dudhi River
		<i>Berosus sp.</i>	Dudhi River
		<i>Caenis sp.</i>	Dudhi River
		<i>Cordulegaster sp.</i>	Dudhi River
		<i>Cybister sp.</i>	Dudhi River
		<i>Dineutus sp.</i>	Dudhi River
		<i>Dytiscus sp.</i>	Dudhi River
		<i>Enallagma sp.</i>	Dudhi River
		<i>Ephemerella sp.</i>	Dudhi River
		<i>Gomphus sp.</i>	Dudhi River
		<i>Lestes sp.</i>	Dudhi River
		<i>Nepa sp.</i>	Dudhi River
		<i>Notonecta sp.</i>	Dudhi River
		<i>Pelocoris sp.</i>	Dudhi River
		<i>Ranatra sp.</i>	Dudhi River
		<i>Siagra sp.</i>	Dudhi River
Mollusca	Bivalvia	<i>Parreysia occata sp.</i>	Dudhi River
		<i>Lamellidens marginalis sp.</i>	Dudhi River, Amarkantak to Gadarwada
	Gastropoda	<i>Tarebia lineata sp.</i>	Dudhi River, Jabalpur
		<i>Thiara scabra sp.</i>	Dudhi River, Jabalpur, Amarkantak to Gadarwada
		<i>Thiara tuberculata sp.</i>	Dudhi River, Jabalpur, Amarkantak to Gadarwada
		<i>Bellamya bengalensis sp.</i>	Dudhi River, Jabalpur, Amarkantak to Gadarwada

Annexure 5.1.1.2: Checklist of Macro Benthos of the Narmada Basin tributaries of the Middle Basin.

Phylum	Class	Species	Location
Annelida	Oligochaeta	<i>Lumbriculus sp.</i>	Ganjal
		<i>Tubifex sp.</i>	Ganjal
		<i>Nasiaeschna sp.</i>	Ganjal
Arthropoda	Insecta	<i>Berosus sps</i>	Ganjal
		<i>Caenis sp.</i>	Ganjal
		<i>Chauliodes sp.</i>	Ganjal
		<i>Chironomous sp.</i>	Ganjal
		<i>Cordulegaster sp.</i>	Ganjal
		<i>Cybister sp.</i>	Ganjal
		<i>Dineutus sp.</i>	Ganjal
		<i>Dolomedes sp.</i>	Ganjal
		<i>Dytiscus sp.</i>	Ganjal
		<i>Enallagma sp.</i>	Ganjal
		<i>Ephemerella sp.</i>	Ganjal
		<i>Eristalis sp.</i>	Ganjal
		<i>Gammarus sp.</i>	Ganjal
		<i>Gomphus sp.</i>	Ganjal
		<i>Hagenius sp.</i>	Ganjal
		<i>Hydrometra sp.</i>	Ganjal
		<i>Hydroporus sp.</i>	Ganjal
		<i>Lestes sp.</i>	Ganjal
		<i>Nepa sp.</i>	Ganjal
		<i>Notonecta sp.</i>	Ganjal
		<i>Octogomphus sp.</i>	Ganjal
		<i>Pelocoris sp.</i>	Ganjal
		<i>Peltodytes sp.</i>	Ganjal
		<i>Ranatra sp.</i>	Ganjal
		<i>Sigara sp.</i>	Ganjal
		<i>Tabanus sp.</i>	Ganjal
		<i>Tachopteryx sp.</i>	Ganjal
	Bivalvia	<i>Lamellidens corrianus sp.</i>	Ganjal
		<i>Lamellidens marginalis(Lamarck) sp.</i>	Ganjal and Chandni Nala, Tawa River, Kaliya deh River, Bhagner River
		<i>Parreysia occata sp.</i>	Chandni Nala, Tawa River, Gujnari Nalla, Gadaria Nalla, Kaliya deh River, Bhagner River, Ganjal
		<i>Parreysia shurtleffiana sp.</i>	Ganjal and, Gadaria Nalla, Kaliya deh River
		<i>Parreysia corrugata sp.</i>	Ganjal
		<i>Pisidium nevillianum sp.</i>	Bhagner River

Mollusca	Gastropoda	<i>Bellamya dissimilis</i> sp.	Ganjal Chandni Nala, Tawa River, Gujnari Nalla, Gadaria Nalla, Kaliya deh River, Bhagner River,
		<i>Bithynia pulchella</i> sp.	Gujnari Nalla, Gadaria Nalla, Kaliya deh River
		<i>Gyraulus convexiusculus</i> sp.	Ganjal, Tawa River, Gujnari Nalla, Gadaria Nalla, Bhagner River
		<i>Gyraulus labiatus</i> sp.	Chandni Nala, Tawa River, Gujnari Nalla, Gadaria Nalla, Kaliya deh River, Bhagner River, Narmadapuram -Bandrabhan, Sethani Ghat, Dongarwada, Anwaliya Ghat),
		<i>Indoplanorbis exustus</i> sp.	Tawa River, Gujnari Nalla, Gadaria Nalla, Kaliya deh River Ganjal
		<i>Lymnaea acuminata</i> sp.	Ganjal Tawa River, Gujnari Nalla, Gadaria Nalla,
		<i>Pila globosa</i> sp.	Gujnari Nalla, Gadaria Nalla
		<i>Tarebia granifera</i> sp.	Ganjal and Chandni Nala, Tawa River, Gujnari Nalla, Gadaria Nalla, Kaliya deh River, Bhagner River, -(Sakatpur, Sethani Ghat, Anwaliya Ghat)
		<i>Tarebia lineata</i> sp.	Ganjal
		<i>Thiara rudis</i> (Lea) sp.	Chandni Nala
		<i>Thiara scabra</i> sp.	Ganjal, Chandni Nala, Tawa River, Gujnari Nalla, Gadaria Nalla, Kaliya deh River, Bhagner River
		<i>Thiara tuberculata</i> sp.	Ganjal , Chandni Nala, Tawa River, Gujnari Nalla, Gadaria Nalla, Kaliya deh River, Bhagner River
		<i>Bellamya bengalensis</i> sp.	Chandni Nala, Tawa River, Gujnari Nalla, Gadaria Nalla, Kaliya deh River, Bhagner River, Ganjal
	Pelecypoda	<i>Corbicula striatella</i> sp.	Bhagner River, Ganjal Chandni Nala, Tawa River, Gujnari Nalla, Gadaria Nalla,

Annexure 5.1.2.1 Checklist of the fisheries tributaries data of the Current zone of the Upper Narmada Basin.

Order	Family	Fish Taxa	Location
Cypriniformes	Cyprinidae	<i>Barilius barila</i> (Hamilton-Buchanan)	Dudhi Basin
		<i>Labeo bata</i> (Hamilton-Buchanan)	Barna River
		<i>Catla catla</i> (Hamilton-Buchanan)	Dudhi Basin, Bargi (Bargi River), Banjar River, Tawa(Tendwa River
		<i>Cyprinus carpio</i>	Banjar River
		<i>Labeo dyocheilus</i> (McClelland)	Bhamni, Banjar River, Barna River
		<i>Esomus danricus</i> (Hamilton, 1822)	Banjar River
		<i>Labeo boggut</i> (Sykes)	Bargi River, Chakaldi, Kolar River, Dudhi Basin
	Balitoridae	<i>Acanthocobitis botia</i> (Hamilton-Buchanan)	Bargi River, Tudar River
		<i>Barilius bendelisis</i> (Hamilton, 1822)	Banjar River, Tudar River, Tawa (Tendwa River)
	Cyprinidae	<i>Chela (Chela) laubuca</i> (Hamilton-Buchanan)	Dudhi Basin
		<i>Labeo calbasu</i> (Hamilton-Buchanan)	Bargi River, Banjar River, Barna River, Tawa (Tendwa River)
		<i>Amblypharyngodon mola</i> (Hamilton-Buchanan)	Dudhi Basin
		<i>Cirrhinus mrigala</i> (Hamilton, 1822)	Bargi River, Tawa (Tendwa River)

		<i>Labeo fimbriatus</i> (Bloch)	Bargi River
		<i>Cirrhinus reba</i> (Hamilton, 1822)	Dudhi Basin, Bargi River, Banjar River
		<i>Labeo rohita</i> (Hamilton-Buchanan)	Banjar River, Tawa (Tendwa River)
		<i>Garra mullya</i> (Sykes)	Barna River
		<i>Garra gotyla</i>	Dudhi Basin
		<i>Labeo angra</i> (Hamilton-Buchanan)	Dudhi Basin
Perciformes	Anabantidae	<i>Anabas testudineus</i> (Bloch)	Banjar River
	Ambassidae	<i>Chanda nama</i> (Hamilton-Buchanan)	Bargi (Bargi River), Banjar River, Tawa (Tendwa River)
	Channidae	<i>Channa marulius</i> (Hamilton, 1822)	Barna River
		<i>Channa orientalis</i> (Bloch & Schneider)	Bhamni, Banjar River, Barna River, Tawa (Tendwa River)
		<i>Channa punctatus</i>	Dudhi Basin, Bhamni, Banjar River, Barna River, Tawa (Tendwa River)
	Gobiidae	<i>Glossogobius giuris</i>	Dudhi Basin, Bargi (Bargi River), Banjar River, Barna River, Tawa (Tendwa River)
	Cichlidae	<i>Oreochromis mossambicus</i> (Peters)	Barna River
Siluriformes	Schilbeidae	<i>Clupisoma garua</i> (Hamilton-Buchanan)	Bargi River, Banjar River

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	Heteropneustidae	<i>Heteropneustes fossilis</i> (Bloch, 1794)	Barna River, Tawa (Tendwa River)
	Siluridae	<i>Ompok bimaculatus</i> (Bloch)	Dudhi Basin
		<i>Wallago attu</i> (Bloch & Schneider)	Bargi (Bargi River), Banjar River, Barna River, TawaTendwa River

Annexure 5.1.2.2. Checklist of the fisheries tributaries data of the Intermediate zone of the Middle Narmada Basin.

Order	Family	FishTaxa	Location
Beloniformes	Belonidae	<i>Xenentodon cancila</i> (Hamilton-Buchanan)	Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Harda (Anchinal River), Pati (Goi River)
Cypriniformes	Cyprinidae	<i>Amblypharyngodon mola</i> (Hamilton-Buchanan)	Sehore (Sip River) & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Barilius bendelisis</i> (Hamilton-Buchanan)	Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Sehore (Sip River) & Dewas (Jamner River)
		<i>Barilius vagra</i> (Hamilton-Buchanan)	Sehore (Sip River) & Dewas (Jamner River)
		<i>Catla catla</i> (Hamilton-Buchanan)	Kolar (Kolar River), Sehore (Sip River) & Dewas (Jamner River)
		<i>Chela laubuca</i> (Hamilton-Buchanan)	Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Cirrhinus mrigala</i> (Hamilton, 1822)	Kolar (Kolar River), Pati (Goi River), Sehore (Sip River)
		<i>Cirrhinus reba</i> (Hamilton, 1822)	Kolar (Kolar River), Sehore (Sip River) & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Crossocheilus latius</i> (Hamilton-Buchanan)	Pati (Goi River)

		<i>Devario aequipinnatus</i> (McClelland)	Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Sehore (Sip River)
		<i>Devario devario</i> (Hamilton-Buchanan)	Sehore (Sip River) & Dewas (Jamner River)
		<i>Esomus danricus</i> (Hamilton-Buchanan)	Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Sehore (Sip River)
		<i>Garra mullya</i> (Sykes)	Chakaldi (Kolar River), Choral (Choral River), Harda (Anchinal River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Sehore (Sip River)
		<i>Labeo bata</i> (Hamilton-Buchanan)	Kolar (Kolar River), Sehore (Sip River) & Dewas (Jamner River)
		<i>Labeo boga</i> (Hamilton-Buchanan)	Sehore (Sip River) & Dewas (Jamner River)
		<i>Labeo boggut</i> (Sykes, 1839)	Chakaldi (Kolar River), Choral (Choral River), Harda (Anchinal River), Sehore (Sip River) & Dewas (Jamner River), Sehore district, Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Labeo calbasu</i> (Hamilton-Buchanan)	Kolar (Kolar River), Sehore (Sip River) & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Labeo dyocheilus</i> (McClelland, 1839)	Kolar (Kolar River)

		<i>Labeo fimbriatus</i> (Bloch, 1795)	Kolar (Kolar River)
		<i>Labeo gonius</i> (Hamilton-Buchanan)	Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Labeo pangusia</i> (Hamilton-Buchanan)	Sehore (Sip River) & Dewas (Jamner River), Sehore district, Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Labeo rohita</i> (Hamilton-Buchanan)	Sehore (Sip River) & Dewas (Jamner River)
	Homalopteridae	<i>Lepidocephalichthys guntea</i> (Hamilton-Buchanan)	Sehore (Sip River) & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Nemacheilus botia</i>	Sehore (Sip River), Dewas (Jamner River)
		<i>Nemacheilus dayi</i>	Sehore (Sip River) & Dewas (Jamner River)
		<i>Osteobrama cotio</i> (Hamilton, 1822)	Sehore (Sip River) & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Harda (Anchinal River), Kolar (Kolar River), Pati (Goi River)
		<i>Oxygaster bacaila</i> (Hamilton, 1822)	Sehore (Sip River)
		<i>Oxygaster gora</i> (Hamilton, 1822)	Sehore (Sip River)
		<i>Puntius conchoni</i> (Hamilton-Buchanan)	Sehore (Sip River) & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Puntius dorsalis</i> (Jerdon)	Sehore (Sip River)

		<i>Puntius sarana</i> (Hamilton-Buchanan)	Bhamni (Banjar River), Chakaldi (Kolar River), Choral (Choral River), Harda (Anchinal River), Kolar (Kolar River), Sehore (Sip River), Dewas (Jamner River)
		<i>Puntius sophore</i> (Hamilton-Buchanan)	Choral (Choral River), Harda (Anchinal River), Pati (Goi River), Sehore (Sip River) & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Rasbora daniconius</i> (Hamilton, 1822)	Sehore (Sip River) & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Tor khudree</i> (Sykes, 1839)	Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Tor tor</i> (Hamilton-Buchanan)	Sehore (Sip River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Harda (Anchinal River), Kolar (Kolar River)
Osteoglossiformes	Notopteridae	<i>Notopterus notopterus</i> (Pallas, 1769)	Sehore (Sip River) & Dewas (Jamner River), Sehore district, Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Choral (Choral River), Kolar (Kolar River), Pati (Goi River)
Perciformes	Badidae	<i>Badis badis</i> (Hamilton-Buchanan)	Sehore (Sip River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges

	Ambassidae	<i>Chanda nama</i> (Hamilton-Buchanan)	Kolar (Kolar River), Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Sehore (Sip River)
		<i>Chanda ranga</i> (Hamilton, 1822)	Sehore (Sip River)
	Channidae	<i>Channa gachua</i> (Hamilton, 1822)	Sehore (Sip River) & Dewas (Jamner River)
		<i>Channa marulius</i> (Hamilton-Buchanan)	Kolar (Kolar River) and Sehore (Sip River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Channa punctatus</i> (Bloch, 1793)	Harda (Anchinal River), Sehore (Sip River), Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Channa striatus</i> (Bloch)	Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Dewas (Jamner River), Sethani Ghat & Dongarwada, Sehore (Sip River)
	Nandidae	<i>Glossogobius giuris</i> (Hamilton, 1822)	Harda (Anchinal River), Pati (Goi River), Sehore (Sip River) & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Nandus nandus</i> (Hamilton-Buchanan)	Sehore (Sip River), Sehore district, Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Chakaldi (Kolar River), Choral (Choral River), Pati (Goi River)

Siluriformes	Amblycipitidae	<i>Amblyceps mangois</i> (Hamilton-Buchanan)	Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
	Heteropneustidae	<i>Heteropneustes fossilis</i> (Bloch, 1794)	Sehore (Sip River) & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
	Bagridae	<i>Mystus bleekari</i> (Day, 1877)	Sehore (Sip River) & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Choral (Choral River), Harda (Anchinal River), Kolar (Kolar River)
		<i>Mystus cavasius</i> (Hamilton-Buchanan)	Sehore (Sip River) & Dewas (Jamner River), Sehore district, Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges
		<i>Mystus tengara</i> (Hamilton-Buchanan)	Sehore (Sip River) & Dewas (Jamner River)
		<i>Mystus vittatus</i> (Bloch)	Sehore (Sip River)
	Siluridae	<i>Ompok bimaculatus</i> (Bloch)	Sehore (Sip River) & Dewas (Jamner River), Sehore district, Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Choral (Choral River), Harda (Anchinal River), Kolar (Kolar River)
		<i>Wallago attu</i> (Bloch & Schneider, 1801)	Sehore (Sip River) & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada – Hathed, Tawa and Palakmati from Satpura and Kolar, Bhagner, Gadaria, Juanri from Vindhya ranges, Kolar (Kolar River)

Synbranchiiformes	Mastacembelidae	<i>Mastacembelus armatus</i>	Choral (Choral River), Harda (Anchinal River), Kolar (Kolar River)
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Annexure 5.1.2.3 Checklist of the fisheries tributaries data of the current e zone of the Middle Narmada Basin.

Order	Family	Fish Taxa	Location
Beloniformes	Belonidae	<i>Xenentodon cancila</i> (Hamilton-Buchanan)	Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges, Harda (Anchinal River), Pati (Goi River)
Cypriniformes	Balitoridae	<i>Acanthocobitis botia</i> (Hamilton-Buchanan)	Choral (Choral River), Harda (Anchinal River), Pati (Goi River), Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Schistura denisoni denisoni</i> (Day)	Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
	Cyprinidae	<i>Amblypharyngodon microlepis</i> (Bleeker, 1853)	Dewas (Jamner River)
		<i>Amblypharyngodon mola</i> (Hamilton-Buchanan)	Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Aspidoparia jaya</i> (Hamilton, 1822)	Sehore (Sip River), & Dewas (Jamner River)

		<i>Barilius bendelisis</i> (Hamilton-Buchanan)	Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges, Sehore (Sip River), & Dewas (Jamner River)
		<i>Barilius vagra</i> (Hamilton-Buchanan)	Sehore (Sip River), & Dewas (Jamner River)
		<i>Brachydanio rerio</i> (Hamilton-Buchanan)	Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Catla catla</i> (Hamilton-Buchanan)	Kolar (Kolar River) , Sehore (Sip River), & Dewas (Jamner River)
		<i>Chela laubuca</i> (Hamilton-Buchanan)	Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Cirrhinus mrigala</i> (Hamilton, 1822)	Kolar (Kolar River) , Pati (Goi River), Sehore (Sip River)
		<i>Cirrhinus reba</i> (Hamilton, 1822)	Kolar (Kolar River) , Sehore (Sip River), & Dewas (Jamner River) Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Crossocheilus latius</i> <i>latius</i> (Hamilton-Buchanan)	Pati (Goi River)
		<i>Devario aequipinnatus</i> (McClell)	Sampling sites across seven tributaries of the river Narmada-Hathed

			Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges Sehore (Sip River)
		<i>Devario devario</i> (Hamilton-Buchanan)	Sehore (Sip River), & Dewas (Jamner River)
		<i>Devario malabaricus</i>	Choral (Choral River), Harda (Anchinal River), Pati (Goi River)
		<i>Esomus danricus</i> (Hamilton-Buchanan)	Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges Sehore (Sip River)
		<i>Garra gotyla</i> (Gray)	Sehore (Sip River), & Dewas (Jamner River)
		<i>Garra mullya</i> (Sykes)	Chakaldi (Kolar River) Choral (Choral River), Harda (Anchinal River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges, Sehore (Sip River),
		<i>Labeo bata</i> (Hamilton-Buchanan)	Kolar (Kolar River) , Sehore (Sip River), & Dewas (Jamner River)
		<i>Labeo boga</i> (Hamilton-Buchanan)	Sehore (Sip River), & Dewas (Jamner River)
		<i>Labeo boggut</i> (Sykes, 1839)	Chakaldi (Kolar River) Choral (Choral River), Harda (Anchinal River), Sehore (Sip River), & Dewas (Jamner River) Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner

		Gadaria Juanri from Vindhya ranges
	<i>Labeo calbasu</i> (Hamilton-Buchanan)	Kolar (Kolar River) , Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges,
	<i>Labeo dyocheilus</i> (McClell, 1839)	Kolar (Kolar River)
	<i>Labeo fimbriatus</i> (Bloch, 1795)	Kolar (Kolar River)
	<i>Labeo gonius</i> (Hamilton-Buchanan)	Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
	<i>Labeo pangusia</i> (Hamilton-Buchanan)	Sehore (Sip River), & Dewas (Jamner River), Sehore district) Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
	<i>Labeo rohita</i> (Hamilton-Buchanan)	Sehore (Sip River), & Dewas (Jamner River)
	<i>Nemacheilus botia</i>	Sehore (Sip River), Dewas (Jamner River)
	<i>Nemacheilus dayi</i>	Sehore (Sip River), & Dewas (Jamner River)
	<i>Osteobrama cotio</i> (Hamilton, 1822)	Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges, Harda

		(Anchinal River), Kolar (Kolar River) , Pati (Goi River)
	<i>Oxygaster bacaila</i> (Hamilton, 1822)	Sehore (Sip River)
	<i>Oxygaster gora</i> (Hamilton, 1822)	Sehore (Sip River)
	<i>Pethia conchoni</i> (Hamilton, 1822)	Sehore (Sip River), & Dewas (Jamner River)
	<i>Pethia ticto</i> (Hamilton, 1822)	Sehore (Sip River)
	<i>Puntius amphibious</i> (Valenciennes)	Sehore (Sip River), & Dewas (Jamner River)
	<i>Puntius chola</i> (Hamilton-Buchanan)	Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
	<i>Puntius chrysopterus</i>	Sehore (Sip River), & Dewas (Jamner River)
	<i>Puntius conchoni</i> (Hamilton-Buchanan)	Sehore (Sip River), Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
	<i>Puntius dorsalis</i> (Jerdon)	Sehore (Sip River)
	<i>Puntius filamentosus</i>	Dewas (Jamner River)
	<i>Puntius sarana</i> (Hamilton-Buchanan)	Bhamni (Banjar River) Chakaldi (Kolar River) Choral (Choral River), Harda (Anchinal River), Kolar (Kolar River) , Sehore (Sip River), / Dewas (Jamner River)
	<i>Puntius sarana orphoides</i> (Valenciennes)	Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner

			Gadaria Juanri from Vindhya ranges
		<i>Puntius sarana sarana</i> (Hamilton-Buchanan)	Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Puntius sophore</i> (Hamilton-Buchanan)	Choral (Choral River), Harda (Anchinal River), Pati (Goi River), Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Puntius ticto</i> (Hamilton-Buchanan)	Chakaldi (Kolar River) Choral (Choral River), Harda (Anchinal River), Pati (Goi River), Tawa (Tendwa River Sehore (Sip River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Rasbora daniconius</i> (Hamilton, 1822)	Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges,
		<i>Rasbora rasbora</i> (Hamilton, 1822)	Sehore (Sip River), & Dewas (Jamner River), Choral (Choral River)
		<i>Salmophasia boopis</i>	Harda (Anchinal River), Kolar (Kolar River)

		<i>Salmophasia horai</i> (Silas)	Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Salmostoma bacaila</i> (Hamilton-Buchanan)	Choral (Choral River), Harda (Anchinal River), Pati (Goi River), Sultanpur (Barna River), Sehore (Sip River),
		<i>Salmostoma orissaensis</i> (Banarescu, 1968)	Sehore (Sip River)
		<i>Securicula gora</i> (Hamilton-Buchanan)	Sehore (Sip River)
		<i>Systomus sarana</i> (Hamilton, 1822)	Sehore (Sip River), & Dewas (Jamner River)
		<i>Tor khudree</i> (Sykes, 1839)	Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Tor tor</i> (Hamilton-Buchanan)	Sehore (Sip River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges Harda (Anchinal River), Kolar (Kolar River),
	Homalopteridae	<i>Lepidocephalichthys guntea</i> (Hamilton-Buchanan)	Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges,
	Nemacheilidae	<i>Paracanthocobitis botia</i> (Hamilton, 1822)	Sehore (Sip River)

Osteoglossiformes	Notopteridae	<i>Notopterus notopterus</i> (Pallas, 1769)	Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges, Choral (Choral River), Kolar (Kolar River) , Pati (Goi River)
Perciformes	Badidae	<i>Badis badis</i> (Hamilton-Buchanan)	Sehore (Sip River), Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
	Mastacembelidae	<i>Chanda nama</i> (Hamilton-Buchanan)	Kolar (Kolar River) , Dewas (Jamner River) Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges Sehore (Sip River),
		<i>Chanda ranga</i> (Hamilton, 1822)	Sehore (Sip River)
	Belontiidae	<i>Colisa lalia</i> (Hamilton-Buchanan)	Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
	Channidae	<i>Channa gachua</i> (Hamilton, 1822)	Sehore (Sip River), & Dewas (Jamner River)
		<i>Channa marulius</i> (Hamilton-Buchanan)	Kolar (Kolar River) Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges

		<i>Channa orientalis</i> (Bloch & Schneider)	Choral (Choral River), Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Channa punctatus</i> (Bloch, 1793)	Harda (Anchinal River), Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Channa striatus</i> (Bloch)	Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges, Dewas (Jamner River), Sehore (Sip River)
	Cinidae	<i>Glossogobius giuris</i> (Hamilton, 1822)	Harda (Anchinal River), Pati (Goi River), Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Nandus nandus</i> (Hamilton- Buchanan)	Sehore (Sip River), Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges Chakaldi (Kolar River) Choral (Choral River), Pati (Goi River), Dewas

Siluriformes	Ambassidae	<i>Parambassis ranga</i> (Hamilton-Buchanan)	Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges, Sehore (Sip River)
		<i>Pseudambassis ranga</i>	Kolar (Kolar River) , Pati (Goi River)
	Amblycipitidae	<i>Amblyceps</i> (Blyth)	Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges,
		<i>Amblyceps mangois</i> (Hamilton-Buchanan)	Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
	Heteropneustidae	<i>Heteropneustes fossilis</i> (Bloch, 1794)	Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges,
	Bagridae	<i>Mystus bleekari</i> (Day, 1877)	Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges , Choral (Choral River), Harda (Anchinal River), Kolar (Kolar River)
		<i>Mystus cavasius</i> (Hamilton-Buchanan)	Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the

			river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Mystus tengara</i> (Hamilton- Buchanan)	Sehore (Sip River), & Dewas (Jamner River)
		<i>Mystus vittatus</i> (Bloch)	Sehore (Sip River)
		<i>Sperata aor</i> (Hamilton- Buchanan)	Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Sperata seenghala</i> (Sykes)	Sehore (Sip River), & Dewas (Jamner River), Kolar (Kolar River)
	Siluridae	<i>Wallago attu</i> (Bloch & Schneider, 1801)	Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges Kolar (Kolar River)
		<i>Ompok bimaculatus</i> (Bloch)	Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges, Choral (Choral River), Harda (Anchinal River), Kolar (Kolar River)
	Sisoridae	<i>Pseudolaguvia ribeiroi</i> (Hora, 1921)	Sampling sites across seven tributaries of the river Narmada- Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges

Synbranchiformes	Mastacembelidae	<i>Macrognaathus pancalus</i> (Hamilton-Buchanan)	Sehore (Sip River), & Dewas (Jamner River), Sampling sites across seven tributaries of the river Narmada-Hathed Tawa Palakmati from Satpura Kolar Bhagner Gadaria Juanri from Vindhya ranges
		<i>Mastacembelus armatus</i>	Choral (Choral River), Harda (Anchinal River), Kolar (Kolar River)

Annexure 5.2.1: Checklist of Macroinvertebrates in the Lower Narmada River Basin

S. No.	Phylum	Class	Species	Common Name/Type	Typical Habitat
1	Mollusca	Gastropoda	Thiara scabra	Freshwater Snail	Sandy-loam riverbeds (Kevadia)
2	Mollusca	Gastropoda	Thiara tuberculata	Red-rimmed Melania	Pollution-tolerant freshwater zones
3	Mollusca	Gastropoda	Bellamya bengalensis	Pond Snail	Lentic, organic-rich margins
4	Mollusca	Gastropoda	Lymnaea acuminata	Freshwater Snail	Weed-infested pools (Rajpipla)
5	Mollusca	Gastropoda	Gyraulus convexiculus	Ramshorn Snail	Shallow vegetation
6	Mollusca	Gastropoda	Digoniostoma sp.	Operculate Snail	Freshwater benthic zones
7	Mollusca	Gastropoda	Stenothyra sp.	Estuarine Snail	Brackish water muds (Bhadbhut)
8	Mollusca	Bivalvia	Corbicula striatella	Basket Clam	Clean sandy substrates (Poicha)
9	Mollusca	Bivalvia	Parreysia sp.	Freshwater Mussel	Riverine sand/silt
10	Mollusca	Bivalvia	Lamellidens marginalis	Marginal Mussel	Freshwater bottoms
11	Mollusca	Bivalvia	Sphaerium sp.	Fingernail Clam	Fine sediments
12	Annelida	Clitellata	Tubifex tubifex	Sludge Worm	High organic pollution (Bharuch)
13	Annelida	Clitellata	Limnodrilus hoffmeisteri	Oligochaete Worm	Sewage-impacted sediments
14	Annelida	Clitellata	Nais sp.	Naidid Worm	Algal mats/Detritus
15	Annelida	Polychaeta	Neanthes sp.	Ragworm	Estuarine mudflats (Ambetha)
16	Arthropoda	Insecta	Chironomus sp.	Bloodworm (Larva)	Hypoxic organic muds
17	Arthropoda	Insecta	Chaoborus sp.	Phantom Midge	Lentic water columns
18	Arthropoda	Insecta	Culicoides sp.	Biting Midge Larva	Marginal muds (Jhanor)
19	Arthropoda	Insecta	Tipula sp.	Crane Fly Larva	Moist banks/Shallow water
20	Arthropoda	Insecta	Psychoda sp.	Moth Fly Larva	Sewage trickling filters/outfalls
21	Arthropoda	Insecta	Pentaneura sp.	Chironomid	Freshwater sediments

S. No.	Phylum	Class	Species	Common Name/Type	Typical Habitat
22	Arthropoda	Malacostraca	Macrobrachium sp.	River Prawn	Estuarine/Freshwater interface

(Source: Compiled from CIFRI Bulletin No. 157 and Narmada Basin biological survey data)

Annexure 5.2.2: Seasonal Abundance of Macrobenthos in Lower Narmada Basin (*nos · per liter*)

Location	Summer (Pre-Monsoon)	Dominant Group (Summer)	Monsoon	Dominant Group (Monsoon)	Winter (Post-Monsoon)	Dominant Group (Winter)
Kevadia	222	Mollusca (46%)	1682	Mollusca (99%)	250	Mollusca (82%)
Poicha	418	Mollusca (78%)	83	Mollusca (100%)	55	Mollusca (84%)
Sisodara	985	Mollusca (87%)	232	Mollusca (72%)	1245	Annelida (57%)
Jhanor	5158	Mollusca (81%)	3550	Annelida (81%)	1124	Annelida (77%)
Bharuch	1311	Annelida (99%)	74	Annelida (76%)	186	Annelida (85%)
Bhadbhut	120	Annelida (62%)	9	Annelida (100%)	37	Malacostraca (76%)
Ambetha	Nil	Nil	9	Annelida (100%)	186	Annelida (100%)
Shakkarpura	Data Not Recorded	Data Not Recorded	46	Annelida (61%)	9	Mollusca (100%)

(Source: Derived from seasonal sampling data in Sardar Sarovar Dam impact assessment documents)

Annexure 5.2.3: Checklist of Fish Species in the Lower Narmada River Basin

S. No.	Family	Fish Taxa	Common Name	Habitat/Status
1	Cyprinidae	Catla catla	Catla	Freshwater / Commercial
2	Cyprinidae	Labeo rohita	Rohu	Freshwater / Commercial
3	Cyprinidae	Cirrhinus mrigala	Mrigal	Freshwater / Commercial
4	Cyprinidae	Labeo calbasu	Kalbasu	Freshwater
5	Cyprinidae	Tor tor	Mahseer	Freshwater / Endangered
6	Cyprinidae	Labeo bata	Bata	Freshwater
7	Cyprinidae	Puntius sarana	Olive Barb	Freshwater
8	Cyprinidae	Cirrhinus reba	Reba	Freshwater
9	Cyprinidae	Osteobrama cotio	Cotio	Freshwater
10	Cyprinidae	Salmostoma bacaila	Razorbelly Minnow	Freshwater
11	Cyprinidae	Rasbora daniconius	Slender Rasbora	Freshwater
12	Cyprinidae	Garra gotyla	Sucker Head	Hill stream / Tributaries
13	Bagridae	Rita rita	Rita	Freshwater / Benthic
14	Bagridae	Sperata seenghala	Giant River-catfish	Freshwater / Predator
15	Bagridae	Mystus cavasius	Gangetic Mystus	Freshwater
16	Bagridae	Sperata aor	Long-whiskered Catfish	Freshwater

S. No.	Family	Fish Taxa	Common Name	Habitat/Status
17	Bagridae	Rita pavementata	Gagra	Freshwater / Rocky beds
18	Siluridae	Wallago attu	Freshwater Shark	Freshwater / Predator
19	Siluridae	Ompok bimaculatus	Butter Catfish	Freshwater
20	Clupeidae	Tenulosa ilisha	Hilsa / Indian Shad	Anadromous / Migratory
21	Engraulidae	Setipinna phasa	Gangetic Hairfin Anchovy	Estuarine
22	Centropomidae	Lates calcarifer	Barramundi / Seabass	Estuarine / Catadromous
23	Mugilidae	Mugil cephalus	Flathead Grey Mullet	Estuarine
24	Mugilidae	Liza parsia	Goldspot Mullet	Estuarine
25	Gobiidae	Glossogobius giuris	Tank Goby	Fresh/Brackish
26	Gobiidae	Boleophthalmus sp.	Mudskipper	Intertidal Mudflats
27	Scatophagidae	Scatophagus argus	Spotted Scat	Estuarine
28	Polynemidae	Eleutheronema tetradactylum	Fourfinger Threadfin	Marine / Estuarine
29	Channidae	Channa striata	Striped Snakehead	Freshwater / Stagnant
30	Channidae	Channa punctata	Spotted Snakehead	Freshwater
31	Mastacembelidae	Mastacembelus armatus	Zig-zag Eel	Freshwater Benthic

S. No.	Family	Fish Taxa	Common Name	Habitat/Status
32	Notopteridae	Notopterus notopterus	Bronze Featherback	Freshwater
33	Belonidae	Xenentodon cancila	Freshwater Garfish	Freshwater Surface
34	Palaemonidae	Macrobrachium rosenbergii	Giant River Prawn	Amphidromous Crustacean
35	Penaeidae	Penaeus indicus	Indian White Prawn	Marine / Estuarine

(Source: Synthesized from CIFRI Bulletin 157, Inland Fishes of Gujarat Documentation, and Sardar Sarovar Impact Studies)